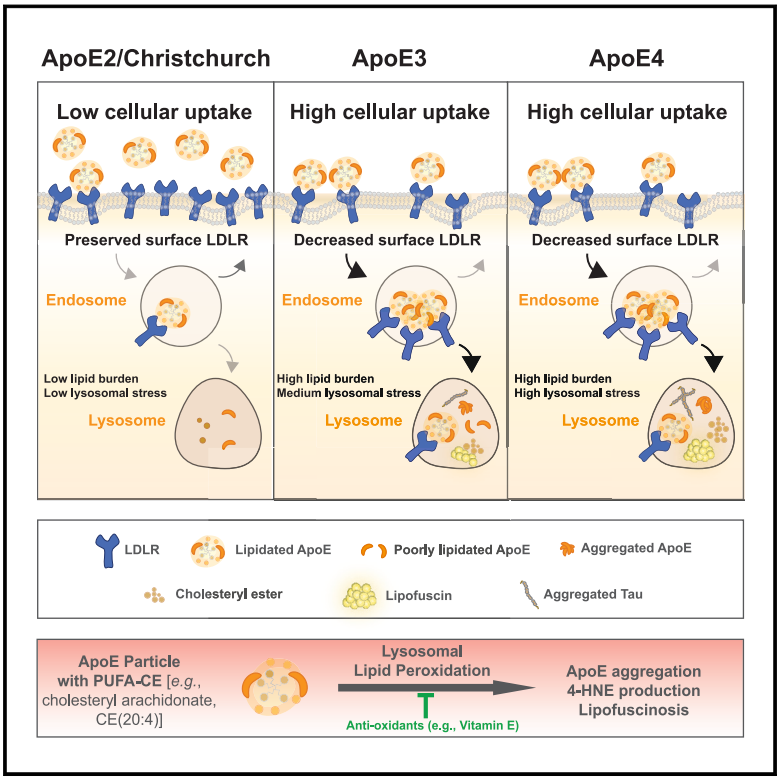
Cell

Decreased lipidated ApoE-receptor interactions confer protection against pathogenicity of ApoE and its lipid cargoes in lysosomes

Graphical abstract



Highlights

- Impaired LDLR binding by lipidated ApoE2 avoids lysosomal sequestration of LDLR
- Reduced ApoE-LDLR interaction decreases the uptake of ApoE and its lipid cargoes
- ApoE carrying PUFA-CE induces lipofuscin deposits *in vitro* and *in vivo*
- PUFA-CE-ApoE-induced lipofuscin involves lipid peroxidation and perturbs lysosomes

Authors

Jing L. Guo, Dylan Braun, Gabriel A. Fitzgerald, ..., David M. Holtzman, Christopher M. Koth, Gilbert Di Paolo

Correspondence

guo@tenvie.com (J.L.G.), dipaolo@dnli.com (G.D.P.)

In brief

ApoE2 and ApoE Christchurch variants are defective in LDLR binding, thus reducing the uptake of ApoE and associated lipid cargoes. LDLR-mediated internalization and endolysosomal delivery of ApoE lipoprotein particles carrying cholesterol esterified with polyunsaturated fatty acids result in lipofuscin deposition with the allelic series ApoE2 < ApoE3 < ApoE4.

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Article

Decreased lipidated ApoE-receptor interactions confer protection against pathogenicity of ApoE and its lipid cargoes in lysosomes

Jing L. Guo,[1,3,*] Dylan Braun,[1,3,5] Gabriel A. Fitzgerald,[1,3,5] Yun-Ting Hsieh,[1,3,5] Lionel Rougé,[1,5] Alexandra Litvinchuk,[2] Micah Steffek,[1,3] Nicholas E. Propson,[1] Catherine M. Heffner,[1] Claire Discenza,[1] Suk Ji Han,[1,3] Anil Rana,[1] Lukas L. Skuja,[1] Bi Qi Lin,[1] Elizabeth W. Sun,[1] Sonnet S. Davis,[1] Srijana Balasundar,[1] Isabel Becerra,[1] Jason C. Dugas,[1] Connie Ha,[1] Jennifer Hsiao-Nakamoto,[1] Fen Huang,[1,4] Shourya Jain,[1] Jennifer E. Kung,[1] Nicholas P.D. Liau,[1] Cathal S. Mahon,[1] Hoang N. Nguyen,[1] Nathan Nguyen,[1] Madhuja Samaddar,[1] Yajuan Shi,[1] David Tatarakis,[1] Yuxi Tian,[1] Yuda Zhu,[1] Jung H. Suh,[1] Thomas Sandmann,[1] Meredith E.K. Calvert,[1] Annie Arguello,[1] Lesley A. Kane,[1] Joseph W. Lewcock,[1] David M. Holtzman,[2] Christopher M. Koth,[1] and Gilbert Di Paolo[1,6,*]

[1]Denali Therapeutics Inc., South San Francisco, CA, USA
[2]Department of Neurology, Washington University School of Medicine, St. Louis, MO, USA
[3]Present address: Tenvie Therapeutics, South San Francisco, CA, USA
[4]Present address: Cajal Neuroscience, Seattle, WA, USA
[5]These authors contributed equally
[6]Lead contact
*Correspondence: guo@tenvie.com (J.L.G.), dipaolo@dnli.com (G.D.P.)
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SUMMARY

While apolipoprotein E (*APOE*) is the strongest genetic modifier for late-onset Alzheimer's disease (LOAD), the molecular mechanisms underlying isoform-dependent risk and the relevance of ApoE-associated lipids remain elusive. Here, we report that impaired low-density lipoprotein (LDL) receptor (LDLR) binding of lipidated ApoE2 (lipApoE2) avoids LDLR recycling defects observed with lipApoE3/E4 and decreases the uptake of cholesteryl esters (CEs), which are lipids linked to neurodegeneration. In human neurons, the addition of ApoE carrying polyunsaturated fatty acids (PUFAs)-CE revealed an allelic series (ApoE4 > ApoE3 > ApoE2) associated with lipofuscinosis, an age-related lysosomal pathology resulting from lipid peroxidation. Lipofuscin increased lysosomal accumulation of tau fibrils and was elevated in the *APOE4* mouse brain with exacerbation by tau pathology. Intrahippocampal injection of PUFA-CE-lipApoE4 was sufficient to induce lipofuscinosis in wild-type mice. Finally, the protective Christchurch mutation also reduced LDLR binding and phenocopied ApoE2. Collectively, our data strongly suggest decreased lipApoE-LDLR interactions minimize LOAD risk by reducing the deleterious effects of endolysosomal targeting of ApoE and associated pathogenic lipids.

INTRODUCTION

The apolipoprotein E (*APOE*) genotype is the most significant genetic modifier for late-onset Alzheimer's disease (LOAD), with *APOE4* and *APOE2* alleles greatly increasing and decreasing disease risk, respectively, relative to the more common *APOE3* allele.[1] ApoE has been implicated in various cellular pathways linked to AD pathogenesis, including amyloid-β (Aβ) clearance and aggregation, tau pathology, metabolism, endolysosomal trafficking, and neuroinflammation.[1–3] Reducing ApoE levels is generally protective in AD models.[1,4–6]

ApoE is the most abundant apolipoprotein in the brain, where it is primarily expressed by astrocytes and reactive microglia,[7] regulating cholesterol metabolism.[8,9] In disease models, ApoE4 perturbs cholesterol regulation, including cholesterol transport and esterification as well as expression of cholesterol metabolism-related genes.[10–13] Together with studies showing cholesterol dysregulation in a range of AD models and patients' brains, these studies suggest lipids are central to AD pathogenesis.[14–18] However, the precise molecular mechanisms underlying ApoE isoform-dependent lipid dyshomeostasis in LOAD are poorly understood.

One of the most striking ApoE isoform-specific biochemical features is that ApoE2 exhibits defective binding to low-density lipoprotein receptor (LDLR), based on ligand competition assays.[19,20] The amino acid substitution R158C, which distinguishes ApoE2 from ApoE3 and ApoE4, disrupts LDLR interaction in the lipidated state.[19,21] While this feature may relate to type III hyperlipoproteinemia observed in 1%–10% of *APOE2* homozygotes,[22] it has been underappreciated as a potential

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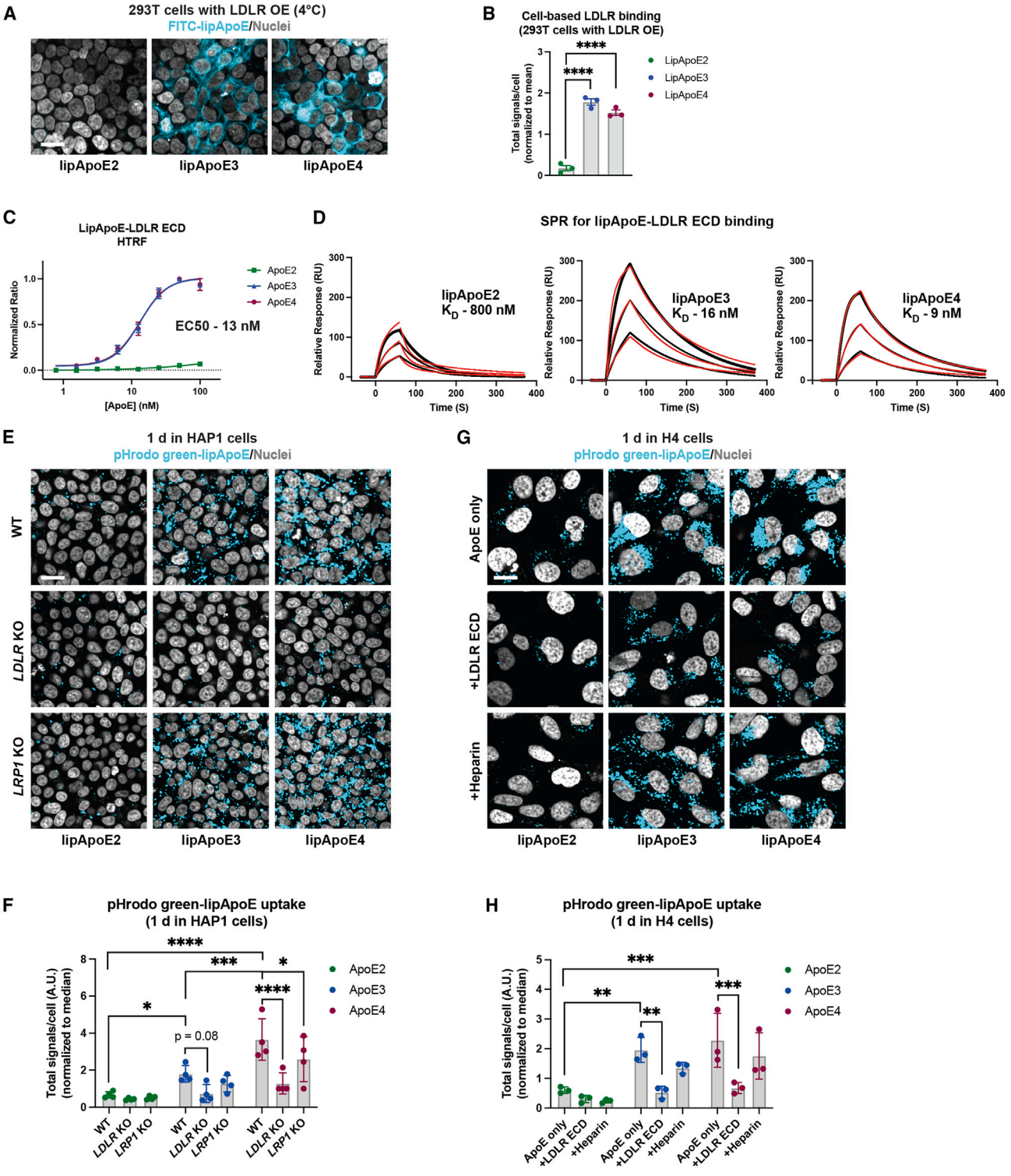


Figure 1. ApoE isoform-dependent LDLR binding results in differential cellular uptake

(A and B) Confocal images and quantification of FITC-labeled POPC-lipidated ApoE (10 μg/mL) bound to surface of 293T cells overexpressing LDLR after 1 h incubation at 4°C. Scale bar: 10 μm in (A).

(C) HTRF showing POPC-lipidated ApoE binding to LDLR ECD (3 independent experiments).

(D) SPR profiles for POPC-lipidated ApoE isoform binding to biotinylated LDLR ECD. Red: experimental data. Black: curve fit using a 1:1 kinetic binding model. Each curve represents binding at one concentration (0.5, 1, and 2 μM for lipApoE2; 0.0156, 0.03125, and 0.0625 μM for lipApoE3/E4).

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protective mechanism for LOAD, despite the fact LDLR is a key receptor for lipidated ApoE (lipApoE) in the brain.[23] Recently, a study reported a homozygous *APOE3* Christchurch mutation (R136S) carrier presenting a ~30-year delay in dementia onset caused by the *PSEN1* E280A mutation and limited tau pathology despite abundant amyloid burden.[24] A mildly protective effect of the heterozygous Christchurch mutation was also found in the same kindred.[25] The homozygous Christchurch mutation also suppresses AD pathologies in amyloid and non-amyloid mouse models.[26,27] While the protective mechanism of the R136S variant has been attributed to reduced binding to heparan sulfate proteoglycans (HSPGs),[24] it is unclear if it involves the lipidated state of ApoE. Additionally, this mutation was also reported to decrease LDLR binding by ~60%.[28] This commonality between the Christchurch mutation and ApoE2 led us to speculate that reduced lipApoE binding to LDLR could be a mechanism underlying decreased AD risk.

In this study, we demonstrate impaired LDLR binding of lipidated ApoE2 (lipApoE2) resulted in decreased LDLR-mediated uptake of ApoE and associated lipids, including cholesteryl ester (CE), thereby preventing excessive activation of microglia. Moreover, ApoE carrying polyunsaturated CE revealed an allelic series of neuronal lipofuscin accumulation (ApoE4 > ApoE3 > ApoE2) caused by lysosomal lipid peroxidation and accompanied by differential accumulation of tau fibrils in human induced pluripotent stem cell (iPSC)-derived neurons, with ApoE4 phenotypes rescuable by a mutation impairing receptor binding. ApoE4-potentiated lipofuscinosis was also observed in vulnerable brain regions of a tauopathy mouse model. Together with findings that Christchurch variant largely phenocopies ApoE2, our study suggests decreased lipidated ApoE-receptor interaction is a key protective mechanism for AD.

RESULTS

Cellular uptake of ApoE by LDLR is isoform-dependent

LDLR is a key receptor for lipidated ApoE (lipApoE). Analysis of single-nucleus RNA sequencing (RNA-seq) data from the Religious Orders Study and Memory and Aging Project (ROSMAP)[29] showed broad LDLR expression in various cell types in human brains, especially in excitatory neurons, where its mRNA levels were comparable to other ApoE receptors such as low-density lipoprotein receptor-related protein 1 (LRP1), LRP1B, and LRP8/APOER2 (Figure S1A). Since lipidation of ApoE is required to expose its LDLR-binding site,[30] we lipidated HEK293 cell-derived ApoE with 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC). Formation of homogeneous ApoE nanodiscs with a 10–20 nm diameter was confirmed by negative-stain transmission electron microscopy (TEM), with no appreciable isoform-specific differences (Figure S1B). Previously, impaired lipApoE2-LDLR binding was inferred from competitive LDL-binding assays, where LDLR-binding affinities of ApoE isoforms inversely correlated with LDL binding.[19,20] To directly assess ApoE-LDLR binding, we developed a cell-based binding assay in which lipApoE was labeled by fluorescein isothiocyanate (FITC) and incubated with 293T cells overexpressing human LDLR for 1 h at 4°C, a temperature that prevents endocytosis. Robust cell surface signals were observed for lipApoE3 and lipApoE4, while minimal signals were observed for lipApoE2, indicating reduced binding of lipApoE2 to LDLR (Figures 1A and 1B).

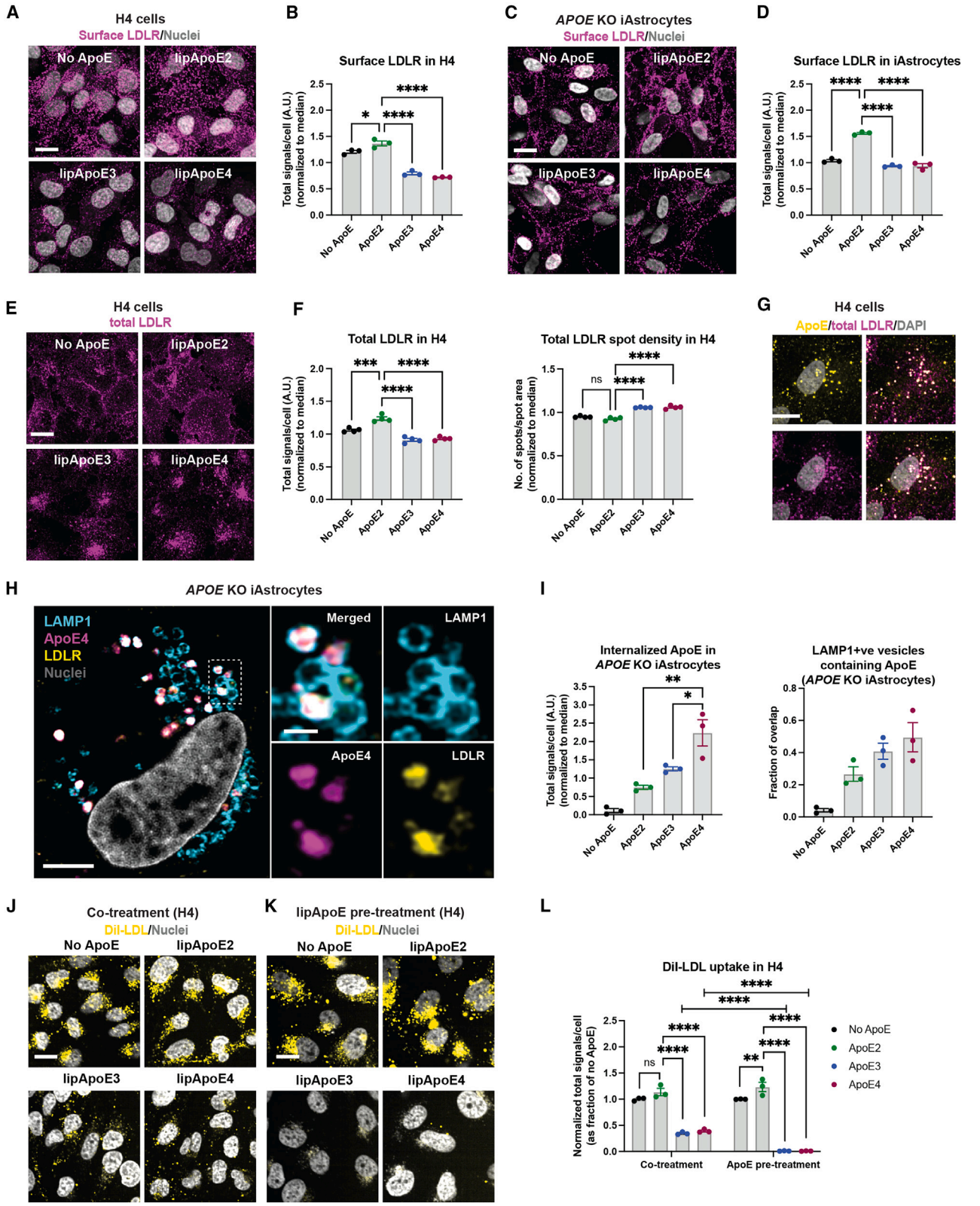
Next, homogeneous time-resolved fluorescence (HTRF) and surface plasmon resonance (SPR) were utilized to quantitatively measure binding of lipApoE to the extracellular domain (ECD) of LDLR. With HTRF, minimal LDLR binding was detected for lipApoE2 at concentrations up to 100 nM (Figure 1C). Conversely, lipApoE3 and lipApoE4 both exhibited high-affinity binding, demonstrating a half maximum effective concentration (EC_{50}) of ~13 nM. SPR similarly revealed a high-affinity LDLR binding for both lipApoE4 and lipApoE3 (K_D = 9 and 16 nM, respectively), whereas the lipApoE2-LDLR affinity was ~50-fold lower than lipApoE3 (K_D = 800 nM; Figure 1D).

To assess the impact of LDLR binding on lipApoE internalization, cells were treated with lipApoE labeled with pHrodo, a pH-sensitive dye that fluoresces in acidic endolysosomal compartments. Compared with lipApoE3 and lipApoE4, lipApoE2 showed significantly lower pHrodo signals after 5–6 h or 1-day incubation in wild-type (WT) HAP1 cells, indicating decreased cellular uptake (Figures 1E, 1F, S1C, and S1D). While both *LDLR* knockout (KO) and *LRP1* KO HAP1 cells demonstrated partially impaired uptake of lipApoE at 5–6 h, only *LDLR* KO cells showed dramatically reduced pHrodo signals for lipApoE3/E4 at 1 day. Reduced lipApoE2 internalization was also observed in H4 neuroglioma cells (Figures 1G, 1H, S1E, and S1F). Notably, co-incubation with LDLR ECD, which competes with cell surface LDLR for ApoE binding, suppressed internalization of lipApoE3/E4 both at 5–6 h and 1 day. Co-treatment with heparin, which competes with cell surface HSPGs for ApoE binding, only produced a significant inhibitory effect at 5–6 h. Co-incubation with the ECD of control receptor CD98hc did not affect lipApoE uptake (Figures S1G and S1H). Therefore, while multiple receptors mediate lipApoE internalization, LDLR plays a dominant role with longer-term treatment. Lastly, in human *APOE* KO iPSC-derived astrocytes (iAstrocytes), pHrodo-lipApoE uptake showed similar isoform trend that was abolished by blocking uptake of lipApoE3/E4 with LDLR ECD (Figures S1I and S1J), validating findings in a CNS cell type.

LipApoE3/E4 alter cellular distribution of LDLR and impair LDL uptake

To explore if LDLR-mediated uptake of lipApoE impacts LDLR recycling, we immunostained cell surface LDLR after 1-day lipApoE treatment. In H4, WT HAP1, and *APOE* KO iAstrocytes,

(E and G) Confocal images of pHrodo green-labeled POPC-lipidated ApoE (10 μg/mL) in WT, *LDLR* KO, or *LRP1* KO HAP1 cells (E) and H4 cells with or without co-treatment of 20 μg/mL LDLR ECD or 2.5 μg/mL heparin (G) after 1-day incubation. Scale bars: 10 μm.
(F and H) Quantification of pHrodo green signals in (E) and (G), respectively.
In all graphs, data shown are mean + SEM with each dot representing one independent experiment; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. For (B), one-way ANOVA was performed with Dunnett's multiple comparisons. For (F) and (H), two-way ANOVA was performed with Holm-Sidak's multiple comparisons.
See also Figure S1.



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incubation with lipApoE3/E4 significantly reduced surface LDLR relative to lipApoE2, revealing impaired LDLR recycling accompanying increased LDLR-mediated lipApoE uptake (Figures 2A–2D, S2A, and S2B). Interestingly, lipApoE2 raised surface LDLR in H4 and, more prominently, *APOE* KO iAstrocytes (Figures 2A–2D). Gene expression analysis of *APOE* KO iAstrocytes revealed upregulation of *LDLR* as well as enzymes involved in cholesterol synthesis (*HMGCR* and *DHCR7*) upon treatment with all lipApoE isoforms (Figure S2C), suggesting potential homeostatic responses to increased cholesterol efflux mediated by lipApoE. While all 3 ApoE isoforms impacted *LDLR* mRNA levels, lipApoE3/E4 abrogated the expected increase in surface LDLR by impairing its recycling. The compensatory upregulation of *LDLR* expression was most pronounced in astrocytes as this cell type is equipped with efficient cholesterol-efflux machinery.[31]

Immunostaining of total LDLR (cell surface + intracellular) in H4 cells after lipApoE3/E4 treatment showed a modest but significant reduction in signals, which also appeared slightly more clustered (Figures 2E and 2F). Cell-associated ApoE immunoreactivity extensively colocalized with LDLR puncta (Figure 2G) outside of transferrin-labeled recycling endosomes (Figure S2D), suggesting "trapping" of LDLR by ApoE in other endolysosomal compartments. By contrast, H4 cells treated with DiI-conjugated LDL (DiI-LDL) showed minimal colocalization of DiI-LDL with LDLR (Figure S2E) but significant overlap with lysosomal associated membrane protein 1 (LAMP1) immunoreactivity (Figure S2F), consistent with rapid endosomal dissociation of LDLR from LDL, with the latter targeted to lysosomes.

Super-resolution imaging on *APOE* KO iAstrocytes confirmed internalized lipApoE completely colocalized with endogenous LDLR (Mander's coefficient for ApoE puncta with colocalizing LDLR = 1 for all 3 isoforms), largely inside LAMP1 compartments (Figure 2H) (Mander's coefficients for ApoE puncta with colocalizing LAMP1 = 0.823 ± 0.061 for ApoE2; 0.903 ± 0.024 for ApoE3; 0.900 ± 0.008 for ApoE4). ApoE isoform differences were manifested in the differential levels of internalized lipApoE (ApoE4 > ApoE3 > ApoE2), giving rise to the same isoform trend for LAMP1-positive vesicles containing ApoE immunoreactivity (Mander's coefficients = 0.266 ± 0.045 for ApoE2, 0.410 ± 0.049 for ApoE3, 0.497 ± 0.090 for ApoE4) (Figure 2I).

Next, we performed DiI-LDL uptake in serum-starved H4 cells. When cells were co-treated with lipApoE and DiI-LDL for 3–4 h, both lipApoE3 and lipApoE4 led to 60%–65% inhibition of DiI-LDL uptake, likely via a direct competition with LDL for LDLR binding, whereas lipApoE2 had no effect (Figures 2J and 2L). When DiI-LDL was spiked into media after 1d pre-treatment with lipApoE3/E4, DiI-LDL uptake was nearly abolished (Figures 2K and 2L). By contrast, lipApoE2 pre-treatment modestly enhanced DiI-LDL uptake, consistent with slightly increased LDLR levels induced by ApoE2 (Figures 2B and 2F). Taken together, increased LDLR-mediated uptake of lipApoE3/E4, coupled with inefficient dissociation of LDLR from lipApoE upon internalization, results in redistribution of cell surface LDLR to endolysosomes, which exacerbates the suppression of LDL uptake by competitive binding.

ApoE isoform- and LDLR-dependent lipid uptake impacts lipid homeostasis

As CNS lipoproteins are high-density lipoprotein (HDL)-like,[16] we next tested if ApoE isoforms differentially mediate cellular uptake of pHrodo-labeled HDL pre-complexed with non-lipidated ApoE before addition to serum-starved H4 cells. With no ApoE, minimal uptake of pHrodo-HDL was observed for up to 1 day (Figures 3A and 3C) since HDL does not bind LDLR. Pre-complexing pHrodo-HDL with ApoE isoforms promoted its uptake in the order of ApoE4 ∼ ApoE3 > ApoE2 (Figures 3A and 3C). LDLR ECD suppressed pHrodo-HDL uptake with ApoE3 or ApoE4, indicating a key role of LDLR in ApoE-mediated HDL internalization. Blocking ApoE-HSPG interaction with heparin also attenuated pHrodo-HDL uptake with more pronounced effects at 6 h than at 1 day (Figures S3A–S3C), suggesting HSPGs may function as co-receptors to facilitate engagement with LDLR and/or LRP1, as shown for peripheral apolipoproteins.[32] Interestingly, the low pHrodo-HDL uptake mediated by ApoE2 was blocked by heparin while minimally inhibited by LDLR ECD, suggesting shifted dependency to HSPGs for ApoE2 due to impaired LDLR binding (Figures 3C, S3B, and S3C).

Elevated CE levels have been reported in AD brains as well as AD models.[12,14,15] To explore if ApoE isoforms can differentially modulate CE uptake, we pre-complexed BODIPY-conjugated CE (BODIPY-CE) with HDL in the presence or absence

Figure 2. Lipidated ApoE3 and ApoE4 alter cellular distribution of LDLR and impair LDL uptake

(A and C) Confocal images of cell surface LDLR staining in H4 cells and *APOE* KO iAstrocytes, respectively, after 1-day incubation with 10 μg/mL POPC-lipidated ApoE. Scale bars: 10 μm.

(B and D) Quantification of surface LDLR signals in (A) and (C), respectively.

(E and F) Confocal images of total LDLR staining for H4 cells after 1-day incubation with 10 μg/mL POPC-lipidated ApoE and quantification of total LDLR signals and spot density (i.e., spot number/spot area). Scale bar: 10 μm in (E).

(G) Confocal images showing colocalization of internalized ApoE4 with LDLR in H4 cells. Scale bar: 10 μm.

(H) Super-resolution confocal images showing co-immunostaining of LAMP1, ApoE, and LDLR in *APOE* KO iAstrocytes after 1-day treatment with 10 μg/mL POPC-lipidated ApoE4. Scale bar: 2.5 μm. Insets: higher-magnification images; scale bar: 0.5 μm.

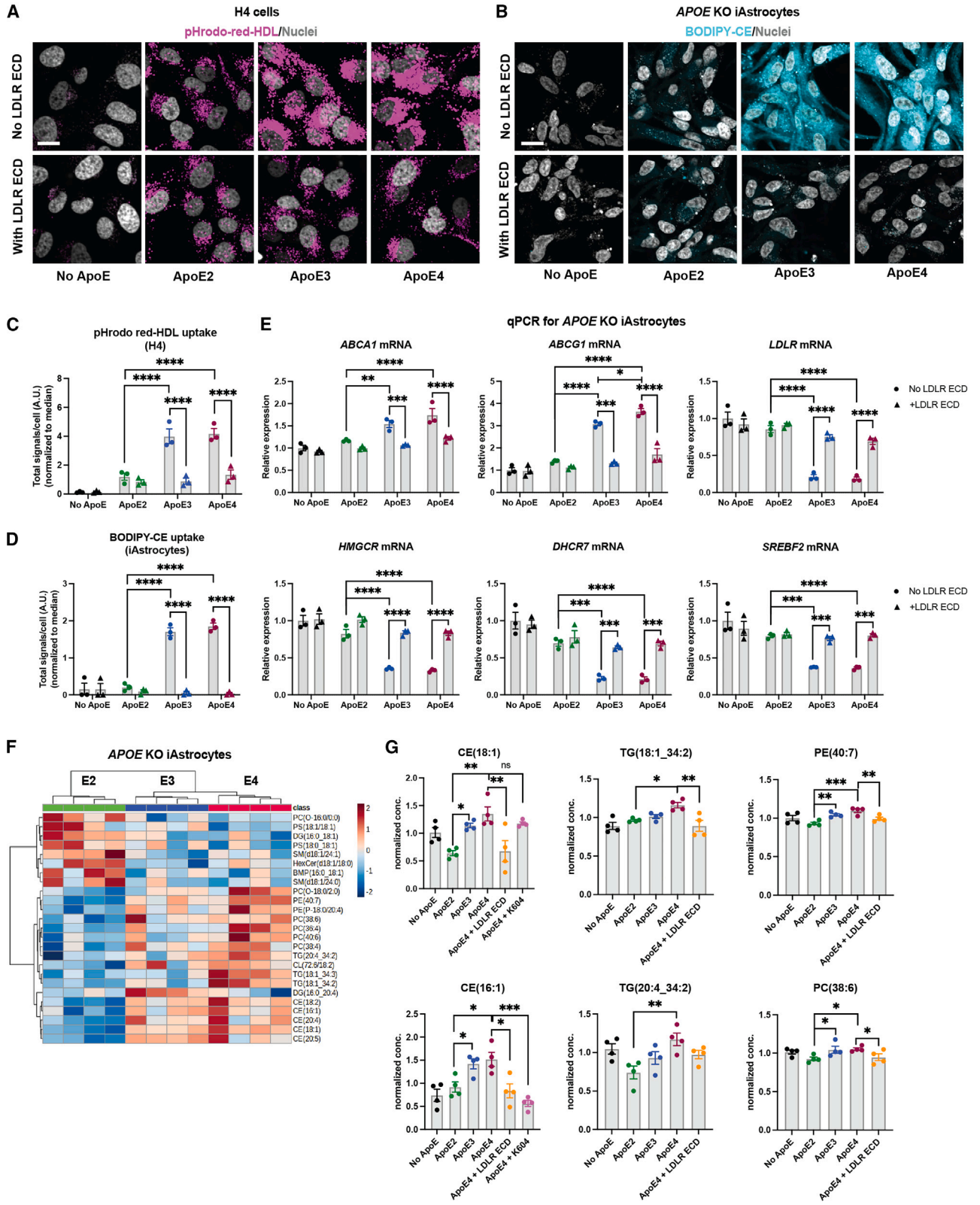
(I) Quantification of internalized lipApoE signals and LAMP1-positive vesicles containing ApoE immunoreactivity in (H).

(J and K) Confocal images of DiI-LDL signals in H4 cells after 3–4 h co-treatment of 10 μg/mL POPC-lipidated ApoE and 10 μg/mL DiI-LDL (J) or after 1 day pre-treatment with 10 μg/mL POPC-lipidated ApoE followed by 3–4 h incubation with DiI-LDL. Scale bars: 10 μm in (K).

(L) Quantification of DiI-LDL signals in H4 cells in (J) and (K).

In all graphs, data shown are mean + SEM with each dot representing one independent experiment; ns: $p > 0.05$, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. For (B), (D), (F), and (I), one-way ANOVA was performed with Sidak's multiple comparisons. For (L), two-way ANOVA was performed with Tukey's multiple comparisons.

See also Figure S2.



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of ApoE, then treated H4 cells with the complex. Without ApoE, HDL largely prevented cellular uptake of BODIPY-CE (Figures S3D and S3F). Pre-complexing BODIPY-CE/HDL with ApoE3 or ApoE4 but not ApoE2 produced higher intracellular BODIPY signals, suggesting ApoE3/E4 promotes internalization of BODIPY-CE via HDL, which otherwise sequesters BODIPY-CE extracellularly. The increased BODIPY-CE uptake mediated by ApoE3/E4 was abrogated by LDLR ECD, indicating LDLR dependency. Interestingly, a subset of BODIPY signals appeared as cytoplasmic puncta that colocalized with perilipin-2 (PLIN2), a lipid droplet marker (Figure S3E), and mirrored total BODIPY signals (Figure S3G). Since CE is hydrolyzed into cholesterol and fatty acid in lysosomes, and BODIPY is conjugated to the fatty acyl group of CE, BODIPY puncta likely represent BODIPY-fatty acids incorporated into lipid droplets. ApoE isoform- and LDLR-dependent BODIPY-CE uptake was similarly observed in *APOE* KO iAstrocytes (Figures 3B and 3D).

To determine how differential CE uptake impacts cholesterol homeostasis transcriptionally in H4 cells and *APOE* KO iAstrocytes, we used qPCR. In both cell types, ApoE3/E4 treatment, which caused higher CE uptake, upregulated expression of *ABCA1* and *ABCG1*, genes involved in cholesterol efflux, while genes mediating cholesterol synthesis (*HMGCR*, *SREBF2*, and *DHCR7*) were downregulated, as was LDLR (Figures 3E and S3H). These transcriptional changes likely reflect homeostatic responses to increased cholesterol burden. LDLR ECD attenuated ApoE3/E4-induced transcriptional alterations, as expected from blocked CE uptake.

Next, lipidomic analysis was performed on *APOE* KO iAstrocytes after 1-day treatment with CE (18:1)/HDL/ApoE complex (Figure 3F, 3G, and S3I). A heatmap for the top 25 lipid changes revealed segregation by ApoE isoforms, with ApoE2- and ApoE4-treated cells most distinct from each other while ApoE3-treated ones being intermediate (Figure 3F). The most prominent differences were observed for various CE species, phosphatidylcholine (PC) or phosphatidylethanolamine (PE) with long chains of polyunsaturated fatty acids (PUFAs), and multiple triglycerides (TGs), which exhibited a trend of ApoE4 $\geq$ ApoE3 > ApoE2. LDLR ECD co-treatment rescued ApoE4 phenotypes (Figures 3G and S3I). CE(18:1) was significantly elevated with ApoE3 or ApoE4 treatment compared with ApoE2, mirroring isoform-dependent uptake of exogenously supplied CE. Similar changes were observed in several other CE species, potentially from CE carried by HDL and/or CE generated from acetyl-coenzyme A acetyltransferase 1 (ACAT1)-mediated re-esterification of free cholesterol derived from lysosomal hydrolysis of CE(18:1), like CE(16:1). Notably, most CE species were lowered with ApoE2 treatment compared with the no-ApoE control, suggesting the cholesterol-efflux role of ApoE2 outweighs its lipid uptake function, leading to an overall reduction in cholesterol burden.

Differential lipid burden modulates inflammatory responses and transcription in microglia

Lipid challenges, including cholesterol, can induce pro-inflammatory responses in microglia.[8,9,33] To test if ApoE isoform-dependent lipid burden differentially impacts inflammatory responses in human iPSC-derived microglia (iMg), we applied the BODIPY-CE/HDL/ApoE treatment paradigm to *APOE* KO iMg in the presence of lipopolysaccharide (LPS), a pro-inflammatory stimulus. Interestingly, LPS upregulated LDLR expression (Figures 4C, LDLR panel and S4A), which increased baseline uptake of BODIPY-CE in the absence of ApoE (Figures 4A and 4B). Treatment of LPS-stimulated *APOE* KO iMg with ApoE3/E4 enhanced BODIPY-CE uptake relative to ApoE2, and LDLR ECD completely reversed the effect of ApoE4 (Figures 4A and 4B). Transcriptional changes in cholesterol homeostasis were consistent with those seen in the other cell types (Figure 4C, compared with Figures 3E and S3H). BODIPY-CE uptake results were replicated using *APOE* KO iMg from a different round of differentiation (Figures S4B and S4C). Cytokine/chemokine measurements on conditioned media collected at 1 day revealed a trend of ApoE4 > ApoE3 > ApoE2 $\sim$ "ApoE4 + LDLR ECD" for multiple cytokines, including interleukin (IL)-9, IL-13, IL-18, and interferon (IFN)-a2 (Figure 4D), suggesting increased lipid burden indeed promotes inflammatory responses. Despite a similar CE uptake induced by ApoE3 and ApoE4, the latter caused significantly higher secretion of the above-mentioned cytokines. A subset of cytokines showing ApoE4-specific changes were not rescuable by LDLR ECD, potentially implicating mechanisms other than lipid uptake (Figure S4D).

Next, we performed RNA-seq on the same cells, and differential expression analysis for individual genes revealed few differences between ApoE4 and ApoE3 treatment (Figure S4E). Moreover, top genes showing the most significant differences among ApoE4 vs. ApoE2, ApoE3 vs. ApoE2, and ApoE4 vs. ApoE4 + LDLR ECD comparisons were enriched with those involved in cholesterol homeostasis (Figures 4E and S4E).

Figure 3. ApoE isoform- and LDLR-dependent lipid uptake impacts lipid homeostasis

(A and C) Confocal images and quantification of pHrodo signals in H4 cells after 1-day incubation with 10 μg/mL pHrodo red-labeled HDL pre-complexed with 10 μg/mL ApoE with and without co-treatment of 20 μg/mL LDLR ECD. Scale bar: 10 μm in (A).

(B and D) Confocal images and quantification of BODIPY signals in *APOE* KO iAstrocytes after 1-day incubation with 10 μg/mL BODIPY-CE pre-complexed with 10 μg/mL HDL and 10 μg/mL ApoE with and without co-treatment of 20 μg/mL LDLR ECD. Scale bar: 10 μm.

(E) Relative expression of 6 genes quantified by qPCR for *APOE* KO iAstrocytes after 1-day treatment with 10 μg/mL CE(18:1) pre-complexed with 10 μg/mL HDL and 10 μg/mL ApoE, with and without co-treatment of 20 μg/mL LDLR ECD.

(F and G) Lipidomic analysis for *APOE* KO iAstrocytes after overnight treatment with 10 μg/mL E(18:1) pre-complexed with 10 μg/mL HDL and 10 μg/mL ApoE, with and without co-treatment of 20 μg/mL LDLR ECD or 500 nM ACAT1 inhibitor K604.

(F) A heatmap for the top 25 lipids showing the greatest differences among the 3 isoform treatment conditions.

(G) A subset of lipid species showing ApoE isoform-dependent changes.

In all graphs, data shown are mean + SEM with each dot representing one independent experiment; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. Two-way ANOVA was performed with Tukey's (C and D) or Sidak's (E) multiple comparisons; for (G), one-way ANOVA was performed with Sidak's multiple comparisons. See also Figure S3.

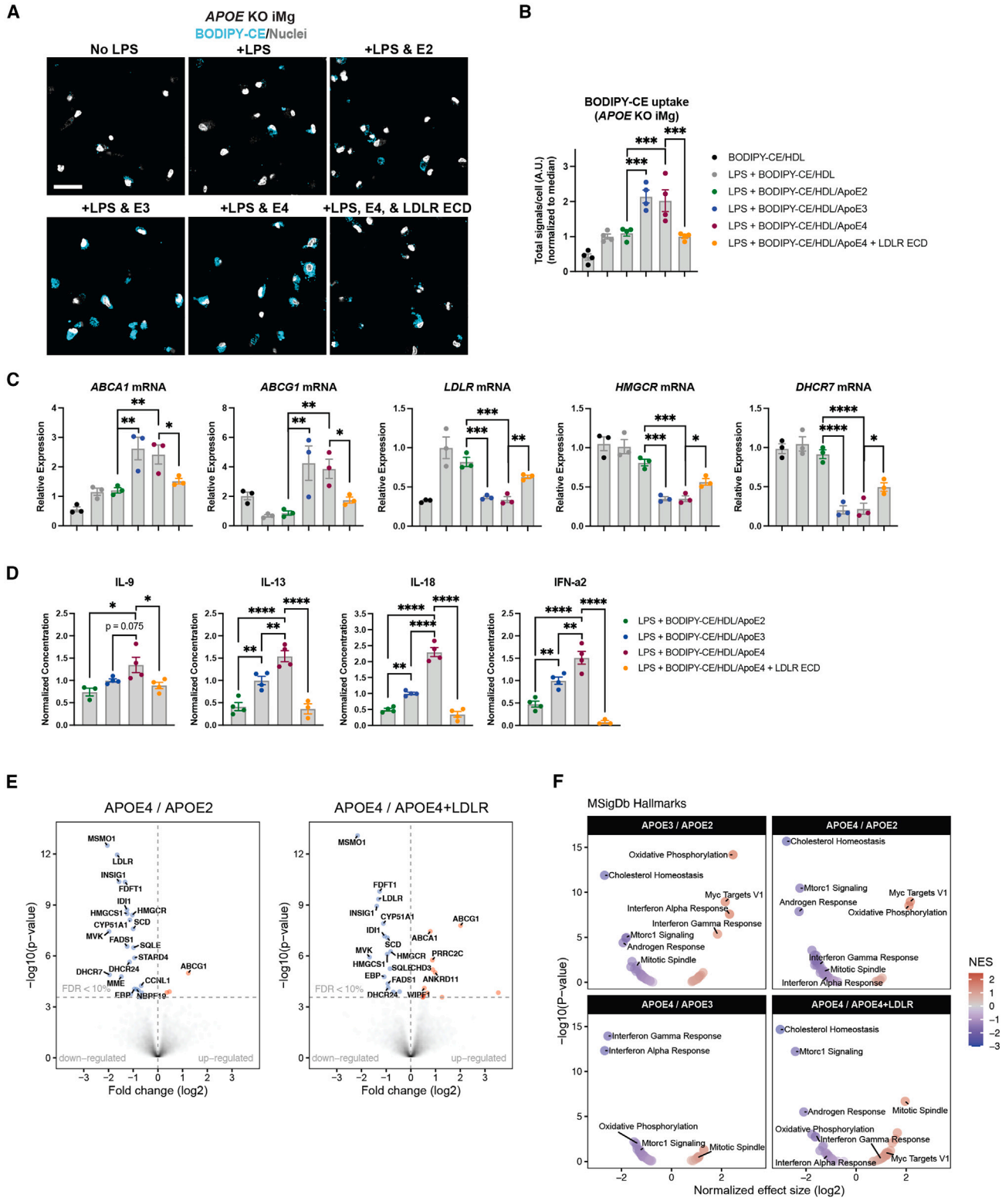


Figure 4. Differential lipid burden modulates inflammatory responses and transcription of microglia

(A and B) Confocal images and quantification of BODIPY signals in *APOE* KO iMg after 1-day incubation with 10 μg/mL BODIPY-CE pre-complexed with 10 μg/mL HDL and 10 μg/mL ApoE, with and without co-treatment of 20 μg/mL LDLR ECD, in medium containing 100 ng/mL LPS. Scale bar: 20 μm for (A). (C) Relative expression of 5 genes quantified by qPCR for APOE KO iMg after live imaging shown in (A).

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Additional pathways significantly modulated by ApoE isoforms included oxidative phosphorylation, eukaryotic translation elongation/initiation, and IFN response (Figures 4F and S4F). In general, ApoE3/E4 treatment increased gene expression in these pathways (Figures S4G and S4H), pointing to upregulation of protein synthesis and energy expenditure upon increased lipid burden. Compared with ApoE3 treatment, ApoE4 treatment more weakly upregulated the oxidative phosphorylation and the IFN response pathways, which is opposite to what would be expected from the cytokine protein data (Figures 4F and S4F). Lastly, LDLR ECD prevented the ApoE4-induced lipid burden and largely reverted gene expression changes observed with ApoE4, although several pathways showed exacerbated ApoE4 effects (e.g., oxidative phosphorylation and mitotic spindle in Figure 4F), possibly due to effects related to blocking ApoE-independent LDLR functions.

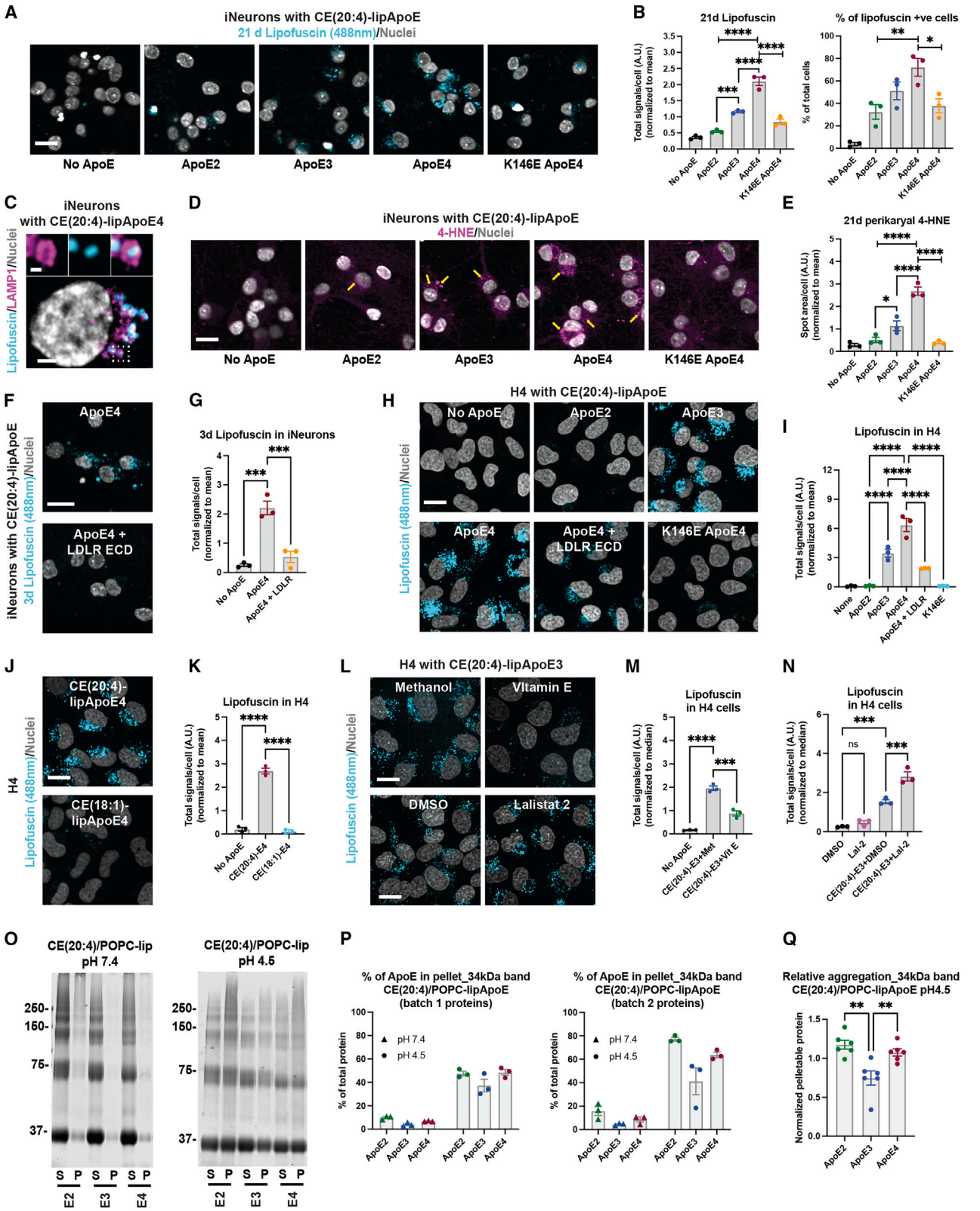
ApoE isoforms carrying CE(20:4) differentially induce lipofuscin in cultured cells

Our earlier study found PUFA-containing CE, including CE(20:4), prominently increased in the microglia of *APOE4* knockin (KI)/P301S *MAPT* mice.[10] Given that CE(20:4) is one of the most abundant CE species in human plasma,[34] we measured CE(20:4) levels in human cerebrospinal fluid (CSF) samples and detected elevated CE(20:4) in AD patients compared with age-matched control (Figure S5A). Since PUFAs are peroxidation-prone and may facilitate the formation of neuronal lipofuscin, a lysosomal pathology that is a hallmark of aging,[35] we generated ApoE particles consisting of POPC and CE(20:4) (CE(20:4)-lipApoE) (Figures S5B and S5C) and tested if ApoE isoforms can differentially modulate lipofuscin accumulation in human iPSC-derived neurons (iNeurons) (Figure S5D). Likely reflecting variable incorporation of CE(20:4) into the core, CE(20:4)-lipApoE particles appeared more heterogeneous than POPC-lipApoE, with diameters ranging from 10 to 40 nm, albeit with no appreciable isoform differences (Figure S5C, compared with Figure S1B). Longitudinal live imaging of CE(20:4)-lipApoE-treated iNeurons revealed both ApoE isoform-dependent and time-dependent accumulation of autofluorescence indicative of lipofuscin (Figures 5A, 5B, and S5E–S5G). At 7 days, lipofuscin induced by CE(20:4)-lipApoE3/E4 was weakly detectable, while signals for CE(20:4)-ApoE2 were minimal. After longer incubation (14 and 21 days), a progressively larger difference between ApoE4 and ApoE3 emerged (Figures 5B and S5G). Lipofuscin appeared mostly perikaryal and localized in LAMP1-positive lysosomes shown by super-resolution imaging (Figure 5C). Perikaryal immunoreactivities for 4-hydroxynonenal (4-HNE), a major peroxidation product of ω-6 fatty acids like arachidonic acid (20:4), also followed the order ApoE4 > ApoE3 > ApoE2 (Figures 5D and 5E), supporting the role of lipid peroxidation in lipofuscin biogenesis. To assess the impact of blocking receptor-mediated ApoE4 uptake, we exploited the K146E mutation, which impairs both LDLR and HSPG binding.[36] CE(20:4)-lipApoE4 carrying the K146E mutation greatly diminished lipofuscin and 4-HNE signals compared with ApoE4 (Figures 5A, 5B, 5D, 5E, and S5E–S5G), confirming lipofuscin formation requires CE(20:4)-lipApoE internalization. LDLR ECD co-treatment with CE(20:4)-lipApoE4 dramatically reduced lipofuscin (Figures 5F and 5G), confirming LDLR dependency. ApoE isoform-dependent lipofuscin accumulation was replicated in WT H4 and HAP1 cells, similarly blocked by LDLR ECD or KO of *LDLR* but not *LRP1* (Figures 5H, 5I, S5H, and S5I).

We next used H4 cells to further understand mechanisms underlying lipofuscinosis. Replacing CE(20:4) with CE(18:1) in the POPC-ApoE4 lipoproteins abolished lipofuscin induction (Figures 5J and 5K), denoting a PUFA requirement. Pre-complexing CE(20:4)-lipApoE3 with vitamin E, a lipophilic antioxidant, significantly decreased lipofuscin (Figures 5L and 5M), indicating lipid peroxidation requirement. Co-treatment of CE(20:4)-lipApoE3 with Lalistat-2, an inhibitor of lysosomal acid lipase that induced buildup of CE species, including CE(20:4) (Figure S5J), significantly enhanced lipofuscin formation (Figures 5L and 5N), suggesting lipofuscinosis involves peroxidation of PUFA in intact CE rather than free PUFA. Intriguingly, while Lalistat-2 treatment alone increased cellular levels of CE(20:4) to a similar extent as CE(20:4)-lipApoE3 (Figure S5J), Lalistat-2 alone did not induce lipofuscin (Figure 5N), revealing a synergistic effect between PUFA-CE and ApoE for lipofuscin formation.

Given similar LDLR binding by lipApoE3 and lipApoE4, other properties may contribute to the ApoE4 vs. ApoE3 difference in lipofuscinosis. To test if differential aggregation of CE(20:4)-lipApoE4 compared with CE(20:4)-lipApoE3 may play a role, we assessed aggregation status of different isoforms of CE(20:4)-lipApoE after 4-day incubation at 37°C with ultracentrifugation.[37] CE(20:4)-lipApoE remained largely soluble and mostly recovered in the supernatant fraction upon incubation at pH 7.4, but abundant protein was found in the pellet fraction indicative of aggregation after incubation at pH 4.5, which mimics lysosomal pH (Figures 5O and 5P). Interestingly, CE(20:4)-ApoE3 demonstrated the lowest aggregation at both pH 7.4 and pH 4.5, while CE(20:4)-ApoE2 exhibited the highest aggregation at pH 7.4 but comparable aggregation as CE(20:4)-ApoE4 at pH 4.5 (Figures 5O–5Q and S5K). Despite variability in the extent of aggregation across two independent batches of lipidated ApoE, the pH- and ApoE isoform-dependent aggregation of CE(20:4)-lipApoE was conserved. These results suggest internalization of CE(20:4)-lipApoE into acidic

(D) Normalized concentrations of 5 cytokines measured in conditioned media of *APOE* KO iMg after live imaging shown in Figure S4A.
(E and F) Results from RNA sequencing performed on *APOE* KO iMg after live imaging shown in Figure S4A.
(E) Volcano plots showing significantly different individual genes with adjusted $p < 0.1$, comparing ApoE4 vs. ApoE2 and ApoE4 vs. ApoE4 + LDLR ECD treatment conditions.
(F) Volcano plots showing Hallmark pathways that are the most altered for different pairs of comparisons.
In all graphs, data shown are mean + SEM with each dot representing one independent experiment; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. One-way ANOVA was performed with Holm-Sidak's multiple comparisons.
See also Figure S4.



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endolysosomal compartments likely promotes its aggregation, but the impaired cellular uptake of lipApoE2 protects against this deleterious event. The lower aggregation propensity of CE(20:4)-lipApoE3 may underlie its attenuated lipofuscin induction compared with CE(20:4)-lipApoE4 despite similar endolysosomal accumulation. In addition, POPC-lipApoE without CE(20:4) and CE(18:1)-lip ApoE4 that does not contain peroxidation-prone PUFA-CE showed minimal aggregation upon 37°C incubation at pH 4.5 (Figures S5L and S5M), suggesting PUFA-CE increases the propensity of ApoE particles to aggregate.

PUFA-CE-ApoE4 promotes lipofuscin deposits *in vivo*

To explore if ApoE isoform-dependent lipofuscinosis occurs *in vivo*, we examined autofluorescent signals in the brains of ~9.5-month-old *APOE* KO (EKO), *APOE3* KI (E3), and *APOE4* KI (E4) mice with and without P301S *MAPT* background. Overall, lipofuscin was most frequently detected in the hippocampus and entorhinal cortex, the most severely degenerated brain regions in P301S *MAPT/APOE* KI mice (Figures 6A, 6B, S6A, and S6B). Human ApoE4 expression alone significantly enhanced lipofuscin buildup in the hippocampus relative to ApoE3 (Figures 6A and 6B). While P301S *MAPT* expression generally promoted lipofuscinosis across *APOE* genotypes, P301S *MAPT/APOE4* KI (Tau/E4) mice showed the highest lipofuscin in both brain regions, suggesting ApoE4 and mutant tau together exacerbate lipofuscinosis. Co-labeling with neuronal and microglial markers (NeuN and Iba1, respectively) revealed predominant localization of lipofuscin in neurons with occasional detection in Tau/E4 microglia (Figure 6C). These findings demonstrate ApoE4-potentiated lipofuscinosis *in vivo*.

Our recent study demonstrated liver X receptor (LXR) agonist dosing significantly decreased PUFA-CE species (CE(18:2), CE(20:4), CE(20:5), and CE(22:6)) that accumulated in Tau/E4 mouse brain, accompanied by reduced tau pathology and neurodegeneration.[10] LXR agonist also significantly reduced lipofuscin in Tau/E4 hippocampi in the same cohort (Figures 6D and 6E), supporting a correlation between PUFA-CE accumulation and lipofuscinosis. To assess the sufficiency of PUFA-CE-containing ApoE4 lipoproteins in causing lipofuscinosis *in vivo*, we performed stereotaxic injection of PBS or CE(20:4)-lipApoE4 into each side of the hippocampus of 6-month-old WT mice (Figures 6F and 6G). Although lipofuscin was observed along the injection path, likely reflecting parenchymal damage, more lipofuscin was detected in regions throughout the hemisphere receiving CE(20:4)-lipApoE4 inoculation than that receiving PBS. This difference appeared most prominent in cortex, hippocampus, and fimbria-fornix, away from injection paths. Furthermore, co-labeling with neuronal and microglial markers near the injection site in the cortex and hippocampus revealed lipofuscin in both cell types but significantly more in microglia than in neurons (Figures S6C and S6D), perhaps reflecting microglial activation caused by injection-induced physical damage. These results demonstrate PUFA-CE-lipApoE4 is sufficient to promote lipofuscinosis in living WT animals.

PUFA-CE-ApoE-induced lipofuscin perturbs lipid metabolism and lysosomal function

Given robust lipidomic and lipofuscin phenotypes in Tau/E4 mice, we investigated if CE(20:4)-lipApoE4 impacts lipids in iNeurons overexpressing P301S tau. Lipidomics uncovered significant alterations in numerous lipid species after 14-day treatment with CE(20:4)-lipApoE4 compared with the other isoforms (Figures S6E and S6F). Apart from the dramatic increase in CE(20:4) itself, multiple PUFA-containing TGs (e.g.,TG(20:4_34:2)) and plasmalogens (e.g., PE(P-16:0/20:4)) along with PE species, the precursor to plasmalogens, were significantly elevated. By contrast, several diglycerides (DGs) and subgroups of sphingolipids, including sphingomyelins (SMs), ceramides (Cer), and hexosylceramides (HexCer), showed significant reduction, demonstrating broad lipid dysregulation accompanies ApoE4-potentiated lipofuscinosis.

Increasing evidence suggests ApoE4 promotes spreading of tau pathology in AD brains.[38,39] To explore if ApoE4-potentiated lipofuscin influences the neuronal accumulation of preformed tau fibrils (tau pffs), we added pHrodo-labeled tau pffs to iNeurons pre-treated with CE(20:4)-lipApoE for 14 days. After another 13–14 day incubation, total pHrodo signals from tau pffs that

Figure 5. ApoE isoforms carrying PUFA-CE differentially induce lipofuscin in cultured cells

(A and B) Confocal images and quantification of lipofuscin (488 nm channel) in NGN2 iNeurons after 21-day incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE isoforms. Scale bar: 10 μm in (A).

(C) Super-resolution confocal images demonstrating predominant localization of lipofuscin in LAMP1-positive vesicles in NGN2 iNeurons after 14-day treatment with 20 μg/mL CE(20:4)/POPC-lipidated ApoE4. Scale bar: 2.5 μm. Insets: higher-magnification images for single or merged channels; scale bar: 0.5 μm.

(D and E) Confocal images and quantification of 4-HNE immunoreactivities in NGN2 iNeurons after the same treatment conditions as in (A). Scale bar: 10 μm in (D). Yellow arrows point to examples of punctate perikaryal 4-HNE signals being quantified.

(F, H, and J) Confocal images of lipofuscin (488 nm channel) in NGN2 iNeurons (F) or in H4 cells (H and J) after 3-day incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE with or without 20 μg/mL LDLR ECD, or with 20 μg/mL CE(18:1)/POPC-lipidated ApoE4. Scale bars: 10 μm.

(G, I, and K) Quantifications of lipofuscin in (F), (H), and (K), respectively.

(L–N) Confocal images and quantification of lipofuscin (488 nm channel) in H4 cells after 1-day treatment of 20 μg/mL CE(20:4)/POPC-lipidated ApoE3 with 25 μM vitamin E (in methanol) or 25 μM Lalistat-2 (Lal-2, in DMSO) compared with respective vehicle control. Scale bars: 10 μm.

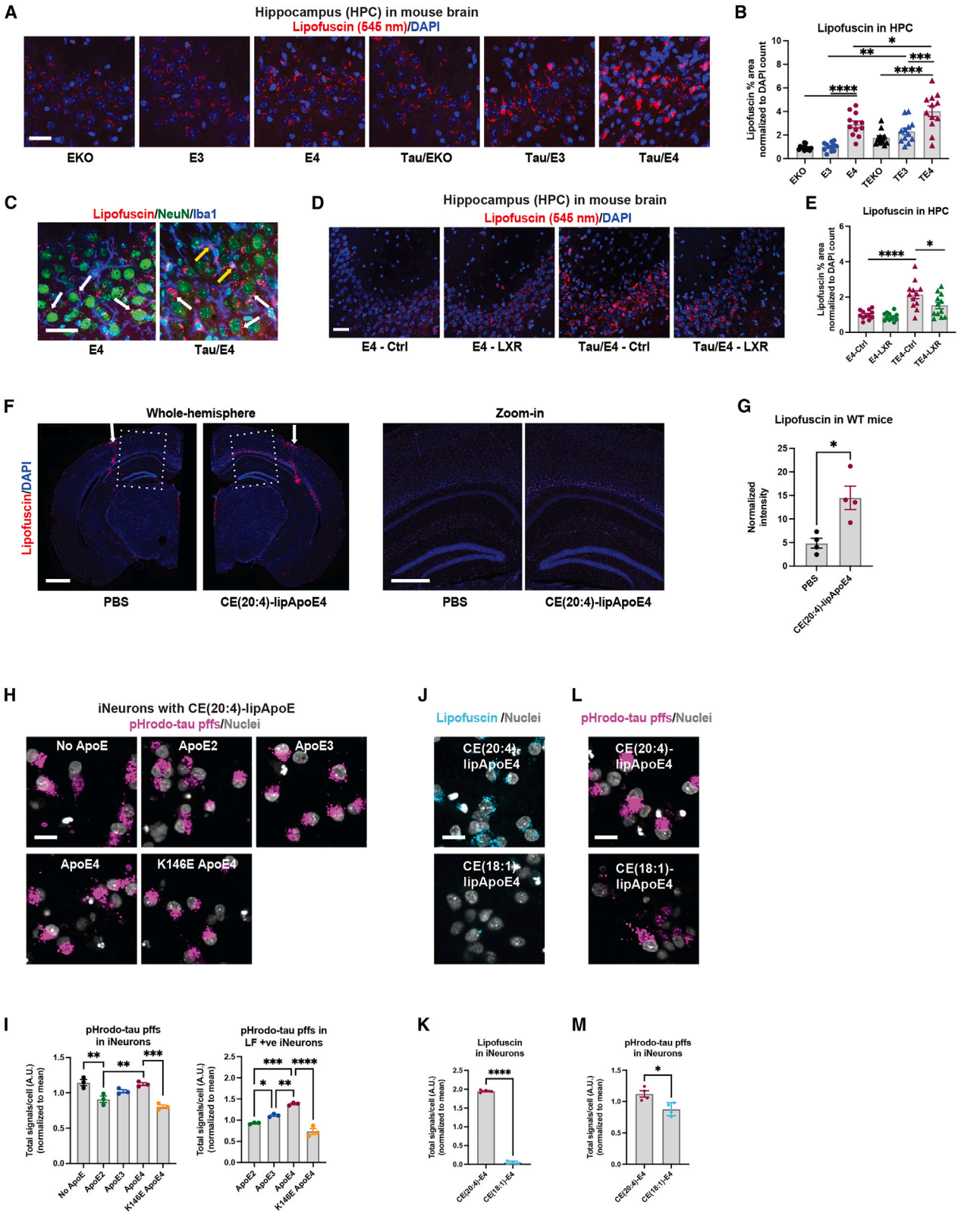
(O) Non-reduced SDS-PAGE analysis for supernatant (S) and pellet (P) fractions from ultracentrifugation of CE(20:4)/POPC-lipidated ApoE after 4-day incubation at 37°C at pH 7.4 or pH 4.5.

(P) Quantification of percentage of ApoE recovered in the pellet fraction for experiments shown in (O) using 2 different batches of CE(20:4)/POPC-lipidated ApoE (densitometry of the 34 kDa band is shown).

(Q) Quantification of relative aggregation for CE(20:4)/POPC-lipidated ApoE isoforms after 4-day incubation at 37°C at pH 4.5, with percentage of pelletable ApoE for each isoform normalized by mean pelletable ApoE in each experiment (combined results from 3 independent experiments for 2 batches of preparations).

In all graphs, data shown are mean + SEM with each dot representing one independent experiment; ns: $p > 0.05$, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. Except for (P), one-way ANOVA was performed with Holm-Sidak's multiple comparisons.

See also Figure S5.



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were internalized and accumulated in the endolysosomes demonstrated a trend of ApoE4 > ApoE3 > ApoE2 > K146E ApoE4, with the isoform-dependent difference being more pronounced among lipofuscin-positive neurons (Figures 6H and 6I), suggesting neuronal accumulation of tau pffs is augmented by a higher lipofuscin load. To confirm the effect of PUFA-CE-lipApoE4 on tau pff accumulation is causally linked to lipofuscin, we directly compared CE(20:4)-lipApoE4 with CE(18:1)-lipApoE4, which did not induce lipofuscin as CE(18:1) is not prone to peroxidation (Figures 6J and 6K). Indeed, CE(18:1)-lipApoE4 produced significantly lower accumulation of pHrodo-tau pffs than CE(20:4)-lipApoE4 (Figures 6L and 6M), consistent with lipofuscin-potentiated tau pff accumulation.

Next, we tested if lipofuscin produces lysosomal abnormalities in H4 cells. Cathepsin activity measured by the 6QC probe[40] showed a modest but significant reduction with CE(20:4)-lipApoE3 or CE(20:4)-lipApoE4 treatment compared with other treatments that minimally induced lipofuscin (Figures S6G and S6H). LAMP1 immunostaining revealed no treatment effect on the total lysosomal area, but CE(20:4)-lipApoE3/E4 treatment resulted in prominent lysosomal clustering (Figures S6I–S6K), a phenotype previously observed in association with lysosomal impairment.[41,42] Therefore, acute lipofuscin induction by CE(20:4)-ApoE3/4 is sufficient to cause lysosomal defects.

Christchurch mutation R136S reduces LDLR binding and decreases lipid burden

Similar to ApoE2, Christchurch (R136S) mutation is protective in autosomal dominant AD cases[24,25] and AD mouse models.[26,27] To understand if the R136S mutation exerts its protection partly via an LDLR-dependent manner, we introduced it into both ApoE3 and ApoE4, including the K146E ApoE4 variant as a positive control for receptor binding deficiency. Consistent with the literature,[24] high-performance liquid chromatography (HPLC)-based heparin binding assay performed on non-lipidated ApoE demonstrated impaired heparin binding by the Christchurch mutation on both ApoE3 and ApoE4 background with the overall ranking of ApoE4 > ApoE3 > ApoE4ch > ApoE2 > ApoE3ch > K146E ApoE4 (Figure S7A). POPC-lipidated ApoE particles were generated for both ApoE3ch and ApoE4ch, with the size and morphology of POPC-lipApoE3ch nanodiscs confirmed to be similar to POPC-lipApoE2/E3/E4 by negative-stain EM (Figure S7B, compared with Figure S1B). A competitive HTRF assay measuring binding between His-tagged lipApoE4 and biotinylated LDLR ECD in the presence of different concentrations of untagged lipApoE variants revealed a complete lack of inhibition by lipK146E ApoE4, our positive control, and greatly right-shifted inhibition curves of lipApoE3ch and lipApoE4ch compared with lipApoE3 and lipApoE4, respectively (Figure 7A), confirming decreased LDLR binding affinity by R136S mutation. Additionally, ApoE3ch and ApoE4ch showed significant reduction in pHrodo-lipApoE, pHrodo-HDL, and BODIPY-CE uptake assays compared with their respective WT counterparts (Figures 7B–7E, S7C, and S7D). Lastly, CE(20:4)-lipidated Christchurch mutants showed significantly attenuated lipofuscinosis (Figures 7F and 7G). Taken together, decreased LDLR-dependent cellular uptake of ApoE and its lipid cargoes may be a common mechanism underlying the protective effects of ApoE2 and ApoE3ch.

DISCUSSION

Our study suggests impaired binding of lipApoE2 and Christchurch variant to LDLR may be protective in AD. Pertinent to ApoE's physiological function of lipid transport, reduced LDLR binding by lipApoE2 lowered its cellular uptake and that of its lipid cargoes, preventing excessive intracellular lipid burden observed with lipApoE3/E4. Specifically, when ApoE carries PUFA-CE, LDLR-dependent internalization caused lysosomal deposition of lipofuscin, a pathological autofluorescent lipopigment of obscure origin since its discovery in 1840 (reviewed in Seehafer and Pearce[43]) that relates to neuronal aging[35,43] and vulnerability in AD brain.[44] PUFA-CE-lipApoE-induced lipofuscinosis mirrored the ApoE2 < ApoE3 < ApoE4 allelic series and involved lipid peroxidation of PUFA-CE and

Figure 6. ApoE4 induces lipofuscin in mouse brain and perturbs tau pff degradation in neurons

(A and B) Confocal images of lipofuscin (545 nm channel) in the hippocampus of 9.5-month-old *APOE* KO (EKO), *APOE3* KI (E3), *APOE4* KI (E4), P301S MAPT/APOE KO (Tau/EKO), P301S *MAPT/APOE3* KI (Tau/E3), and P301S *MAPT/APOE4* KI (Tau/E4) mice with quantification (12 mice/group). Scale bar: 20 μm in (A).

(C) Confocal images of lipofuscin co-labeled with neuronal (NeuN) and microglial (Iba1) markers in the hippocampus of 9.5-month-old E4 and Tau/E4 mice. White arrows point to examples of neuronal lipofuscin adjacent to NeuN staining. Yellow arrows point to examples of lipofuscin that appears inside Iba1-positive microglia.

(D and E) Confocal images of lipofuscin (545 nm channel) in the hippocampus of 9.5-month-old E4 or Tau/E4 mice after 3.5-month treatment with vehicle (Ctrl) or LXR agonist GW3965 (LXR) with quantification (12 mice/group). Scale bar: 20 μm in (D).

(F) Images of thresholded lipofuscin in the brain of 6-month-old WT mice at 2 weeks after injection of PBS or CE(20:4)/POPC-lipidated ApoE4 into each side of the hippocampus. Arrows denote the approximate injection sites. Scale bars: 1 mm for the whole-hemisphere images; 500 μm for the zoom-in images.

(G) Quantification of thresholded lipofuscin around the coronal plane of injection but with the injection tract excluded, for the experiment shown in (F) (4 mice, one-tailed paired t test).

(H) Confocal images of pHrodo signals detected in NGN2 iNeurons after 14-day treatment with 20 μg/mL CE(20:4)/POPC-lipidated ApoE followed by 13-day incubation with 2.5 μg/mL pHrodo red-labeled tau pffs. Scale bar: 10 μm.

(I) Quantification of pHrodo signals in (H) for all neurons or only among neurons classified as lipofuscin-positive (LF +ve).

(J–M) Confocal images and quantification of lipofuscin (J and K) and pHrodo signals (L and M) detected in NGN2 iNeurons after 14-day treatment with 20 μg/mL CE(20:4)/POPC-lipidated ApoE4 or CE(18:1)/POPC-lipidated ApoE4 followed by 14-day incubation with 2.5 μg/mL pHrodo red-labeled tau pffs. Scale bars: 10 μm.

In all graphs, data shown are mean + SEM with each dot representing one independent experiment or one animal; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. For (B) and (E), one-way ANOVA was performed with Sidak's multiple comparisons. For (I), one-way ANOVA was performed with Holm-Sidak's multiple comparisons. For (K) and (M), two-tailed unpaired t test was performed.

See also Figure S6.

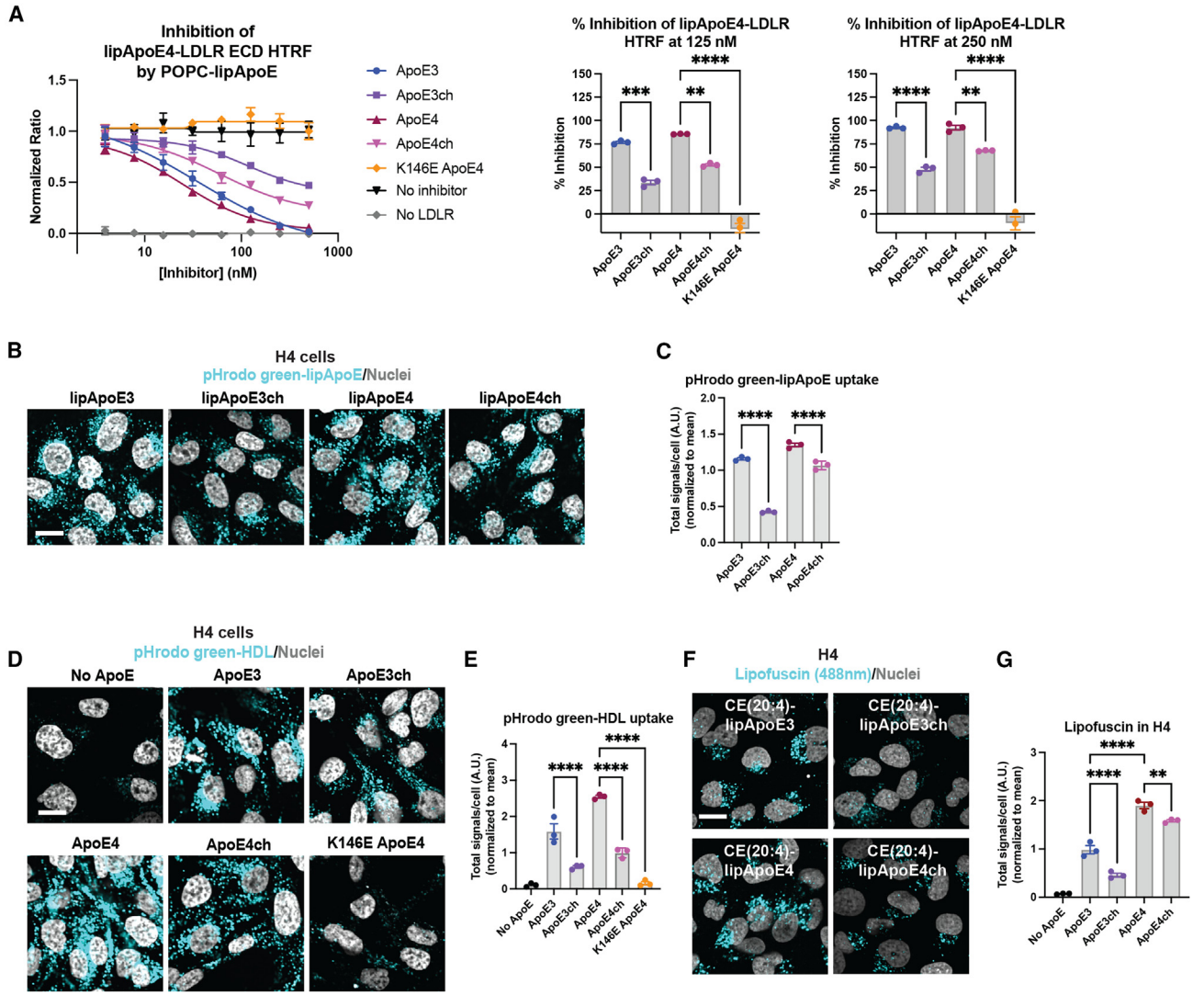


Figure 7. Christchurch mutation reduces LDLR binding, lipid uptake, and lipofuscin

(A) Competitive HTRF demonstrating inhibition of tagged lipApoE4-LDLR ECD binding in the presence of different concentrations of untagged lipidated ApoE variants ($n = 3$ independent experiments). No inhibitor: no addition of untagged lipidated ApoE. The first panel shows the complete dose response. The second and third panels show the % inhibition at 125 and 250 nM, respectively.

(B and C) Confocal images and quantification of pHrodo green-labeled POPC-lipidated ApoE isoforms (10 μg/mL) in H4 cells after 1-day incubation. Scale bar: 10 μm in (B).

(D and E) Confocal images and quantification of pHrodo signals detected in H4 cells after 1-day incubation with 10 μg/mL pHrodo green-labeled HDL pre-complexed with 10 μg/mL ApoE isoforms. Scale bar: 10 μm in (D).

(F and G) Confocal images and quantification of lipofuscin in H4 cells after 3-day incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE isoforms. Scale bar: 10 μm in (F).

In all bar graphs, data shown are mean + SEM with each dot representing one independent experiment; $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. One-way ANOVA was performed with Holm-Sidak's multiple comparisons.

See also Figure S7.

perturbed lysosomes, with an impact on lysosomal accumulation of tau pffs. Lipofuscinosis was also observed with endogenous expression of *APOE* in *APOE4* KI mouse brains, where it was worsened by P301S *MAPT* expression and ameliorated by an LXR agonist previously shown to reduce brain PUFA-CE.[10] PUFA-CE also enhanced ApoE aggregation in an isoform-dependent manner, especially at acidic/lysosomal pH. Our study provides a mechanistic framework potentially clarifying ApoE isoform-dependent impact on LOAD risk, implicating receptor-mediated internalization of ApoE and associated pathogenic lipid cargoes.

Past studies have historically focused more on pathogenic mechanisms driven by ApoE4 than protective mechanisms mediated by ApoE2,[1,21] which is also associated with greater

longevity independently of AD.[45,46] To assess if reduced LDLR binding of lipApoE2 could be protective, we adopted a simple paradigm of treating cultured cells with lipid-complexed HEK293-purified ApoE isoforms. Using quantitative, direct binding assays, we demonstrated a ~50- to 100-fold decrease in LDLR-binding affinity for ApoE2 vs. ApoE3/E4 and a comparable binding affinity for ApoE3 and ApoE4. Human CSF ApoE concentrations were reported to be ~7–9 μg/mL (~206–265 nM), which are below the LDLR-binding K_D of lipApoE2, i.e., ~800 nM,[47,48] suggesting lipApoE2 minimally engages LDLR in the brain, consistent with *Ldlr* KO mouse studies.[23]

Previous work showed ApoE isoforms differentially perturb endosomal recycling of receptors, such as LRP8/ApoER2, with a deleterious impact of ApoE4 on signaling by reelin, a key ligand of this receptor.[49] Our study now demonstrates lipApoE isoform-dependent impact on the endolysosomal trafficking of a distinct ApoE receptor, LDLR, which primarily mediates lipid metabolism.[50] We first showed LDLR is the dominant receptor mediating uptake and endolysosomal accumulation of lipApoE in multiple human cellular models. LDLR-mediated uptake of lipApoE3 and lipApoE4 led to endolysosomal trapping of LDLR and depletion of cell surface LDLR, resulting in further reduced LDL uptake. LipApoE2 did not show these effects due to defective LDLR binding. Our findings thus potentially explain the reported *APOE* genotype effects on human plasma LDL levels, with ApoE2 and Christchurch variant associated with lower LDL[51] or ApoB levels,[52] a profile that correlates with lower cardiovascular disease risk.[53] While LDL biology is mostly relevant for peripheral diseases,[32,54,55] the CNS primarily contains HDL-like particles, many of which harbor ApoE and can thus engage LDLR.[56] LDLR's ligands in the CNS, besides ApoE, are not well defined. However, important roles were shown for LDLR in modulating AD pathologies,[57–59] in part via uptake and clearance of Aβ by astrocytes.[60] Therefore, reduced LDLR binding of ApoE2 may free up LDLR for binding and clearance of other ligands, including Aβ, thus decreasing amyloid pathology.

Growing evidence suggests ApoE's role in lipid transport is connected to LOAD pathogenesis. ApoE-mediated transport of lipids from neurons to glia in flies[61] and mammalian cells is neuroprotective.[62] Additionally, *APOE4* expression in human iMg was shown to induce secretion of lipids, which promoted tau phosphorylation in neurons.[63] CE, the main transport and storage form of cholesterol, is of particular interest as a candidate disease-relevant ApoE cargo, based on its upregulation in AD patient brain and various AD models, along with the fact that its downregulation is protective in various AD models.[10,12,15,64] Our study showed in multiple cell types greatly enhanced CE uptake by HDL-complexed ApoE3 or ApoE4 compared with ApoE2 due to differential LDLR binding. While LDLR-dependent lipApoE uptake is a normal physiological process, we hypothesized the disease-relevant protective mechanism for ApoE2 involves decreased endolysosomal delivery of ApoE and pathogenic lipids into neurons. Of note, neurons are vulnerable to toxic lipids, such as oxidized lipids,[65,66] particularly when lysosomes are compromised.[67] Therefore, we assessed the pathogenic potential of peroxidation-prone PUFA-CE by treating iNeurons with CE(20:4)-lipApoE isoforms. We observed accumulation of lipofuscin along with an established toxic product of arachidonic acid peroxidation, 4-HNE, following the allelic series of ApoE4 > ApoE3 > ApoE2. Lipofuscin is believed to arise from oxidative damage and crosslinking of macromolecules induced by PUFA peroxidation.[35,43] Consistently, we found reduced lipofuscin accumulation with antioxidant vitamin E and a complete lack of lipofuscinosis when CE(20:4) is replaced by peroxidation-resistant CE(18:1), supporting the role of lipid peroxidation in lipofuscinosis. Inhibition of lysosomal acid lipase, which broadly increased levels of CE, including CE(20:4), exacerbated lipofuscinosis but only in the presence of lipApoE. Importantly, lipofuscin reduced lysosomal proteolysis and caused abnormal lysosomal clustering, suggesting deleterious impact on lysosomes. Reducing ApoE4-receptor interactions with the K146E mutation rescued lipofuscin, 4-HNE, and lysosomal phenotypes to levels observed with ApoE2.

While lipofuscin is a key pathology in neuronal ceroid lipofuscinosis and *GRN*-linked frontotemporal dementia, it may be more broadly involved in neurodegeneration.[68–71] Indeed, long and poorly myelinated cortical neurons with higher lipofuscin burden in the transentorhinal cortex are the most vulnerable to neurofibrillary tangle formation, and lipofuscin-free neurons are generally more resilient in AD.[44] Our findings linking ApoE isoform-dependent lipofuscinosis to tau pff accumulation thus offer a potential mechanism for the ApoE4 effect on enhancing spreading of tau pathology.[38,39] ApoE4-potentiated lipofuscin buildup was validated in *APOE4* KI/P301S *MAPT* mouse brains, where it was observed in both neurons and microglia of vulnerable regions, namely the hippocampus and entorhinal cortex. Critically, chronic dosing of these mice with an LXR agonist previously shown to decrease PUFA-CE, including CE(20:4),[10] significantly diminished lipofuscinosis. Moreover, intrahippocampal injection of CE(20:4)-lipApoE4 was sufficient to induce lipofuscin in WT mouse brain. Therefore, PUFA-CE species, together with ApoE, may be key drivers of brain lipofuscinosis. However, other lipids and proteins, such as ApoE receptors themselves, given their sensitivity to peroxidation-derived reactive species,[72] may also be involved.

Besides characterizing the common protective variant ApoE2, we studied the rare ApoE R136S (Christchurch) variant, which is protective in the *PSEN1* E280A kindred.[24,25] Not only did we confirm the R136S mutation impairs LDLR binding,[28] besides HSPG binding, but we also showed reduced LDLR-mediated uptake of ApoE-associated lipid cargoes and decreased lipofuscinosis. Decreased LDLR binding may thus also contribute to the protective effects of this mutation.[26,27]

Studies suggest that poorly lipidated ApoE may seed Aβ due to its propensity to aggregate and that lipidated ApoE is protective in AD models.[1] Our work suggests that an enrichment of PUFAs, such as PUFA-CEs, in the ApoE particle may render lipidated ApoE pathological, particularly in the aging brain where oxidative stress is greater and the degradative capacity of lysosomes is diminished. Since PUFA-CE is strongly upregulated in the brain of amyloid mouse models,[14] Aβ elevation may significantly amplify this pathogenic process as part of the amyloid cascade. Although ApoE isoform-dependent receptor binding could only explain protective mechanism of ApoE2, with ApoE4 minimally differentiated from ApoE3 in most uptake assays, higher aggregation propensity of lipApoE4 compared

with lipApoE3 shown in our study may contribute to its exacerbated pathogenicity over ApoE3 upon internalization. Interestingly, while lipApoE2 is aggregation-prone at acidic pH in the cell-free assay, its reduced internalization ability may protect it against this potentially deleterious structural property, which has been revealed in other settings.[73] This may explain why *APOE2* is a risk factor for age-related macular degeneration, since drusen, the autofluorescent lipid particles that are prominent in the diseased eye, accumulate and likely oxidize extracellularly.[74] Collectively, our data suggest the ApoE allelic series, in the lipidated form, may modify LOAD risk via two distinct but related mechanisms: (1) a protective mechanism mediated by decreased lipApoE-receptor interactions, as shown for ApoE2 and Christchurch; and (2) a deleterious mechanism mediated by increased aggregation of PUFA-containing lipoproteins upon reaching the acidic lysosomes, as exhibited by lipApoE4.

Our study suggests unchecked endolysosomal delivery of ApoE and its lipid cargoes, including PUFA-CE, is pathogenic and that inhibiting lipApoE-receptor interactions in the CNS may reduce LOAD risk or disease progression, in part by decreasing the formation of lipofuscin and promoting lysosomal health. Rescue of lipApoE4-induced phenotypes by mutations impairing receptor interactions indicates ApoE4 can be modified to phenotypically resemble protective variants, potentially paving the way for development of novel therapeutics.

Limitations of the study

In this study, we used artificially lipidated ApoE containing only 1–2 lipid species, which does not capture the complexity of brain ApoE lipoproteins. Our study focuses on ApoE isoform-dependent effects downstream of ApoE-LDLR interaction without addressing the source of lipApoE in the brain (e.g., astrocytes or microglia). While our study showed a key role of LDLR in the uptake of ApoE and associated lipids, we cannot rule out the contributions of other ApoE receptors to isoform-dependent phenotypes. Lastly, our study suggests lipofuscin induced by PUFA-CE lipApoE may promote pathological tau accumulation in cell culture, but it did not directly test this *in vivo*.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Gilbert Di Paolo (dipaolo@dnli.com).

Materials availability

Information and requests on recombinant ApoE proteins should be directed to the lead contact, Gilbert Di Paolo (dipaolo@dnli.com).

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request. RNA-seq data have been deposited at NCBI GEO under accession GSE250249 and are publicly available as of the date of publication.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

J.L.G. and G.D.P. conceived the study and designed experiments. J.L.G., C.M.K., G.A.F., L.R., M. Steffek, and C.S.M. designed biochemical approaches. J.L.G., D.B., G.A.F., Y.-T.H., L.R., M. Steffek, N.E.P., C.M.H., S.J.H., A.R., L.L.S., B.Q.L., E.W.S., S.B., C.H., S.J., N.P.D.L., N.N., M. Samaddar, and Y.S. performed cellular and/or biochemical studies. J.L.G., D.B., G.A.F., L.R., Y.-T.H., M. Steffek, C.M.H., S.J.H., A.R., S.S.D., J.E.K., J.H.S., and T.S. analyzed data. A.L. and D.M.H. contributed lipofuscin data in *APOE4* mice. I.B., C.D., H.N.N., J.C.D., M.E.K.C., and A.A. performed *in vivo* study and/or analyzed data. J.H.-N., F.H., Y.T., Y.Z., and D.T. analyzed human CSF or single-nucleus RNA-seq (snRNA-seq) data. J.L.G. and G.D.P. wrote the manuscript. C.M.K., L.A.K., J.W.L., and D.M.H. provided feedback on data and edited the manuscript.

DECLARATION OF INTERESTS

All authors, except N.N., A.L., and D.M.H., are full-time employees and/or shareholders of Denali Therapeutics. J.L.G., D.B., G.A.F., Y.-T.H., M. Steffek, and S.J.H. are currently full-time employees and shareholders of Tenvie Therapeutics. D.M.H. is on the scientific advisory board of Denali Therapeutics, Genentech, Cajal Neuroscience, C2N Diagnostics, and *Cell*. D.M.H. consults for Asteroid.

STAR★METHODS

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REFERENCES

1. Martens, Y.A., Zhao, N., Liu, C.C., Kanekiyo, T., Yang, A.J., Goate, A.M., Holtzman, D.M., and Bu, G. (2022). ApoE Cascade Hypothesis in the pathogenesis of Alzheimer's disease and related dementias. Neuron *110*, 1304–1317. https://doi.org/10.1016/j.neuron.2022.03.004.
2. Johnson, L.A. (2020). APOE and metabolic dysfunction in Alzheimer's disease. Int. Rev. Neurobiol. *154*, 131–151. https://doi.org/10.1016/bs.irn.2020.02.002.
3. Windham, I.A., and Cohen, S. (2024). The cell biology of APOE in the brain. Trends Cell Biol. *34*, 338–348. https://doi.org/10.1016/j.tcb.2023.09.004.
4. Bales, K.R., Verina, T., Dodel, R.C., Du, Y., Altstiel, L., Bender, M., Hyslop, P., Johnstone, E.M., Little, S.P., Cummins, D.J., et al. (1997). Lack of apolipoprotein E dramatically reduces amyloid beta-peptide deposition. Nat. Genet. *17*, 263–264. https://doi.org/10.1038/ng1197-263.
5. Huynh, T.V., Liao, F., Francis, C.M., Robinson, G.O., Serrano, J.R., Jiang, H., Roh, J., Finn, M.B., Sullivan, P.M., Esparza, T.J., et al. (2017). Age-Dependent Effects of apoE Reduction Using Antisense Oligonucleotides in a Model of beta-amyloidosis. Neuron *96*, 1013–1023.e4. https://doi.org/10.1016/j.neuron.2017.11.014.
6. Shi, Y., Yamada, K., Liddelow, S.A., Smith, S.T., Zhao, L., Luo, W., Tsai, R.M., Spina, S., Grinberg, L.T., Rojas, J.C., et al. (2017). ApoE4 markedly exacerbates tau-mediated neurodegeneration in a mouse model of tauopathy. Nature *549*, 523–527. https://doi.org/10.1038/nature24016.
7. Fernandez, C.G., Hamby, M.E., McReynolds, M.L., and Ray, W.J. (2019). The role of APOE4 in disrupting the homeostatic functions of astrocytes and microglia in aging and Alzheimer's disease. Front. Aging Neurosci. *11*, 14. https://doi.org/10.3389/fnagi.2019.00014.
8. Cantuti-Castelvetri, L., Fitzner, D., Bosch-Queralt, M., Weil, M.T., Su, M., Sen, P., Ruhwedel, T., Mitkovski, M., Trendelenburg, G., Lütjohann, D., et al. (2018). Defective cholesterol clearance limits remyelination in the aged central nervous system. Science *359*, 684–688. https://doi.org/10.1126/science.aan4183.
9. Nugent, A.A., Lin, K., van Lengerich, B., Lianoglou, S., Przybyla, L., Davis, S.S., Llapashtica, C., Wang, J., Kim, D.J., Xia, D., et al. (2020). TREM2 regulates microglial cholesterol metabolism upon chronic phagocytic challenge. Neuron *105*, 837–854.e9. https://doi.org/10.1016/j.neuron.2019.12.007.
10. Litvinchuk, A., Suh, J.H., Guo, J.L., Lin, K., Davis, S.S., Bien-Ly, N., Tycksen, E., Tabor, G.T., Remolina Serrano, J., Manis, M., et al. (2024). Amelioration of Tau and ApoE4-linked glial lipid accumulation and neurodegeneration with an LXR agonist. Neuron *112*, 384–403.e8. https://doi.org/10.1016/j.neuron.2023.10.023.
11. Lin, Y.T., Seo, J., Gao, F., Feldman, H.M., Wen, H.L., Penney, J., Cam, H.P., Gjoneska, E., Raja, W.K., Cheng, J., et al. (2018). APOE4 causes widespread molecular and cellular alterations associated with Alzheimer's disease phenotypes in human iPSC-derived brain cell types. Neuron *98*, 1141–1154.e7. https://doi.org/10.1016/j.neuron.2018.05.008.
12. Blanchard, J.W., Akay, L.A., Davila-Velderrain, J., von Maydell, D., Mathys, H., Davidson, S.M., Effenberger, A., Chen, C.Y., Maner-Smith, K., Hajjar, I., et al. (2022). APOE4 impairs myelination via cholesterol dysregulation in oligodendrocytes. Nature *611*, 769–779. https://doi.org/10.1038/s41586-022-05439-w.
13. Tcw, J., Qian, L., Pipalia, N.H., Chao, M.J., Liang, S.A., Shi, Y., Jain, B.R., Bertelsen, S.E., Kapoor, M., Marcora, E., et al. (2022). Cholesterol and matrisome pathways dysregulated in astrocytes and microglia. Cell *185*, 2213–2233.e25. https://doi.org/10.1016/j.cell.2022.05.017.
14. Chan, R.B., Oliveira, T.G., Cortes, E.P., Honig, L.S., Duff, K.E., Small, S.A., Wenk, M.R., Shui, G., and Di Paolo, G. (2012). Comparative lipidomic analysis of mouse and human brain with Alzheimer disease. J. Biol. Chem. *287*, 2678–2688. https://doi.org/10.1074/jbc.M111.274142.
15. van der Kant, R., Langness, V.F., Herrera, C.M., Williams, D.A., Fong, L.K., Leestemaker, Y., Steenvoorden, E., Rynearson, K.D., Brouwers, J.F., Helms, J.B., et al. (2019). Cholesterol metabolism is a druggable axis that independently regulates tau and amyloid-beta in iPSC-derived Alzheimer's disease neurons. Cell Stem Cell *24*, 363–375.e9. https://doi.org/10.1016/j.stem.2018.12.013.
16. Feringa, F.M., and van der Kant, R. (2021). Cholesterol and Alzheimer's disease; from risk genes to pathological effects. Front. Aging Neurosci. *13*, 690372. https://doi.org/10.3389/fnagi.2021.690372.
17. Romero-Molina, C., Garretti, F., Andrews, S.J., Marcora, E., and Goate, A.M. (2022). Microglial efferocytosis: diving into the Alzheimer's disease gene pool. Neuron *110*, 3513–3533. https://doi.org/10.1016/j.neuron.2022.10.015.
18. Miranda, A.M., Ashok, A., Chan, R.B., Zhou, B., Xu, Y., McIntire, L.B., Area-Gomez, E., Di Paolo, G., Duff, K.E., Oliveira, T.G., et al. (2022). Effects of APOE4 allelic dosage on lipidomic signatures in the entorhinal cortex of aged mice. Transl. Psychiatry *12*, 129. https://doi.org/10.1038/s41398-022-01881-6.
19. Weisgraber, K.H., Innerarity, T.L., and Mahley, R.W. (1982). Abnormal lipoprotein receptor-binding activity of the human E apoprotein due to cysteine-arginine interchange at a single site. J. Biol. Chem. *257*, 2518–2521. https://doi.org/10.1016/S0021-9258(18)34954-8.
20. Yamamoto, T., Choi, H.W., and Ryan, R.O. (2008). Apolipoprotein E isoform-specific binding to the low-density lipoprotein receptor. Anal. Biochem. *372*, 222–226. https://doi.org/10.1016/j.ab.2007.09.005.
21. Li, Z., Shue, F., Zhao, N., Shinohara, M., and Bu, G. (2020). APOE2: protective mechanism and therapeutic implications for Alzheimer's disease. Mol. Neurodegener. *15*, 63. https://doi.org/10.1186/s13024-020-00413-4.
22. Khalil, Y.A., Rabès, J.P., Boileau, C., and Varret, M. (2021). APOE gene variants in primary dyslipidemia. Atherosclerosis *328*, 11–22. https://doi.org/10.1016/j.atherosclerosis.2021.05.007.
23. Fryer, J.D., Demattos, R.B., McCormick, L.M., O'Dell, M.A., Spinner, M.L., Bales, K.R., Paul, S.M., Sullivan, P.M., Parsadanian, M., Bu, G., et al. (2005). The low density lipoprotein receptor regulates the level of central nervous system human and murine apolipoprotein E but does not modify amyloid plaque pathology in PDAPP mice. J. Biol. Chem. *280*, 25754–25759. https://doi.org/10.1074/jbc.M502143200.
24. Arboleda-Velasquez, J.F., Lopera, F., O'Hare, M., Delgado-Tirado, S., Marino, C., Chmielewska, N., Saez-Torres, K.L., Amarnani, D., Schultz, A.P., Sperling, R.A., et al. (2019). Resistance to autosomal dominant Alzheimer's disease in an APOE3 Christchurch homozygote: a case report. Nat. Med. *25*, 1680–1683. https://doi.org/10.1038/s41591-019-0611-3.
25. Quiroz, Y.T., Aguillon, D., Aguirre-Acevedo, D.C., Vasquez, D., Zuluaga, Y., Baena, A.Y., Madrigal, L., Hincapié, L., Sanchez, J.S., Langella, S., et al. (2024). APOE3 Christchurch heterozygosity and autosomal

dominant Alzheimer's disease. N. Engl. J. Med. *390*, 2156–2164. https://doi.org/10.1056/NEJMoa2308583.

26. Chen, Y., Song, S., Parhizkar, S., Lord, J., Zhu, Y., Strickland, M.R., Wang, C., Park, J., Tabor, G.T., Jiang, H., et al. (2024). APOE3ch alters microglial response and suppresses Abeta-induced tau seeding and spread. Cell *187*, 428–445.e20. https://doi.org/10.1016/j.cell.2023.11.029.
27. Nelson, M.R., Liu, P., Agrawal, A., Yip, O., Blumenfeld, J., Traglia, M., Kim, M.J., Koutsodendris, N., Rao, A., Grone, B., et al. (2023). The APOE-R136S mutation protects against APOE4-driven Tau pathology, neurodegeneration and neuroinflammation. Nat. Neurosci. *26*, 2104–2121. https://doi.org/10.1038/s41593-023-01480-8.
28. Lalazar, A., Weisgraber, K.H., Rall, S.C., Jr., Giladi, H., Innerarity, T.L., Levanon, A.Z., Boyles, J.K., Amit, B., Gorecki, M., and Mahley, R.W. (1988). Site-specific mutagenesis of human apolipoprotein E. Receptor binding activity of variants with single amino acid substitutions. J. Biol. Chem. *263*, 3542–3545. https://doi.org/10.1016/S0021-9258(18)68957-4.
29. Mathys, H., Peng, Z., Boix, C.A., Victor, M.B., Leary, N., Babu, S., Abdelhady, G., Jiang, X., Ng, A.P., Ghafari, K., et al. (2023). Single-cell atlas reveals correlates of high cognitive function, dementia, and resilience to Alzheimer's disease pathology. Cell *186*, 4365–4385.e27. https://doi.org/10.1016/j.cell.2023.08.039.
30. Chen, J., Li, Q., and Wang, J. (2011). Topology of human apolipoprotein E3 uniquely regulates its diverse biological functions. Proc. Natl. Acad. Sci. USA *108*, 14813–14818. https://doi.org/10.1073/pnas.1106420108.
31. Chen, J., Zhang, X., Kusumo, H., Costa, L.G., and Guizzetti, M. (2013). Cholesterol efflux is differentially regulated in neurons and astrocytes: implications for brain cholesterol homeostasis. Biochim. Biophys. Acta *1831*, 263–275. https://doi.org/10.1016/j.bbalip.2012.09.007.
32. Mahley, R.W., Huang, Y., and Rall, S.C., Jr. (1999). Pathogenesis of type III hyperlipoproteinemia (dysbetalipoproteinemia). Questions, quandaries, and paradoxes. J. Lipid Res. *40*, 1933–1949. https://doi.org/10.1016/S0022-2275(20)32417-2.
33. Andreone, B.J., Przybyla, L., Llapashtica, C., Rana, A., Davis, S.S., van Lengerich, B., Lin, K., Shi, J., Mei, Y., Astarita, G., et al. (2020). Alzheimer's-associated PLCgamma2 is a signaling node required for both TREM2 function and the inflammatory response in human microglia. Nat. Neurosci. *23*, 927–938. https://doi.org/10.1038/s41593-020-0650-6.
34. Godoy-Corchuelo, J.M., Fernández-Beltrán, L.C., Ali, Z., Gil-Moreno, M.J., López-Carbonero, J.I., Guerrero-Sola, A., Larrad-Sainz, A., Matias-Guiu, J., Matias-Guiu, J.A., Cunningham, T.J., et al. (2022). Lipid metabolic alterations in the ALS-FTD spectrum of disorders. Biomedicines *10*, 1105. https://doi.org/10.3390/biomedicines10051105.
35. Moreno-García, A., Kun, A., Calero, O., Medina, M., and Calero, M. (2018). An overview of the role of lipofuscin in age-related neurodegeneration. Front. Neurosci. *12*, 464. https://doi.org/10.3389/fnins.2018.00464.
36. Mann, W.A., Lohse, P., Gregg, R.E., Ronan, R., Hoeg, J.M., Zech, L.A., and Brewer, H.B., Jr. (1995). Dominant expression of type III hyperlipoproteinemia. Pathophysiological insights derived from the structural and kinetic characteristics of ApoE-1 (Lys146–>Glu). J. Clin. Invest. *96*, 1100–1107. https://doi.org/10.1172/JCI118096.
37. Liao, F., Li, A., Xiong, M., Bien-Ly, N., Jiang, H., Zhang, Y., Finn, M.B., Hoyle, R., Keyser, J., Lefton, K.B., et al. (2018). Targeting of nonlipidated, aggregated apoE with antibodies inhibits amyloid accumulation. J. Clin. Invest. *128*, 2144–2155. https://doi.org/10.1172/JCI96429.
38. Ferrari-Souza, J.P., Bellaver, B., Ferreira, P.C.L., Benedet, A.L., Povala, G., Lussier, F.Z., Leffa, D.T., Therriault, J., Tissot, C., Soares, C., et al. (2023). APOEepsilon4 potentiates amyloid beta effects on longitudinal tau pathology. Nat Aging *3*, 1210–1218. https://doi.org/10.1038/s43587-023-00490-2.
39. Steward, A., Biel, D., Dewenter, A., Roemer, S., Wagner, F., Dehsarvi, A., Rathore, S., Otero Svaldi, D., Higgins, I., Brendel, M., et al. (2023). ApoE4 and connectivity-mediated spreading of tau pathology at lower amyloid levels. JAMA Neurol. *80*, 1295–1306. https://doi.org/10.1001/jamaneurol.2023.4038.
40. Widen, J.C., Tholen, M., Yim, J.J., Antaris, A., Casey, K.M., Rogalla, S., Klaassen, A., Sorger, J., and Bogyo, M. (2021). AND-gate contrast agents for enhanced fluorescence-guided surgery. Nat. Biomed. Eng. *5*, 264–277. https://doi.org/10.1038/s41551-020-00616-6.
41. Henry, A.G., Aghamohammadzadeh, S., Samaroo, H., Chen, Y., Mou, K., Needle, E., and Hirst, W.D. (2015). Pathogenic LRRK2 mutations, through increased kinase activity, produce enlarged lysosomes with reduced degradative capacity and increase ATP13A2 expression. Hum. Mol. Genet. *24*, 6013–6028. https://doi.org/10.1093/hmg/ddv314.
42. Li, X., Rydzewski, N., Hider, A., Zhang, X., Yang, J., Wang, W., Gao, Q., Cheng, X., and Xu, H. (2016). A molecular mechanism to regulate lysosome motility for lysosome positioning and tubulation. Nat. Cell Biol. *18*, 404–417. https://doi.org/10.1038/ncb3324.
43. Seehafer, S.S., and Pearce, D.A. (2006). You say lipofuscin, we say ceroid: defining autofluorescent storage material. Neurobiol. Aging *27*, 576–588. https://doi.org/10.1016/j.neurobiolaging.2005.12.006.
44. Braak, H., Del Tredici, K., Schultz, C., and Braak, E. (2000). Vulnerability of select neuronal types to Alzheimer's disease. Ann. N. Y. Acad. Sci. *924*, 53–61. https://doi.org/10.1111/j.1749-6632.2000.tb05560.x.
45. Shinohara, M., Kanekiyo, T., Tachibana, M., Kurti, A., Shinohara, M., Fu, Y., Zhao, J., Han, X., Sullivan, P.M., Rebeck, G.W., et al. (2020). APOE2 is associated with longevity independent of Alzheimer's disease. eLife *9*, e62199. https://doi.org/10.7554/eLife.62199.
46. Tesi, N., van der Lee, S., Hulsman, M., van Schoor, N.M., Huisman, M., Pijnenburg, Y., van der Flier, W.M., Reinders, M., and Holstege, H. (2024). Cognitively healthy centenarians are genetically protected against Alzheimer's disease. Alzheimers Dement. *20*, 3864–3875. https://doi.org/10.1002/alz.13810.
47. Wahrle, S.E., Shah, A.R., Fagan, A.M., Smemo, S., Kauwe, J.S.K., Grupe, A., Hinrichs, A., Mayo, K., Jiang, H., Thal, L.J., et al. (2007). Apolipoprotein E levels in cerebrospinal fluid and the effects of ABCA1 polymorphisms. Mol. Neurodegener. *2*, 7. https://doi.org/10.1186/1750-1326-2-7.
48. Koch, M., Furtado, J.D., Falk, K., Leypoldt, F., Mukamal, K.J., and Jensen, M.K. (2017). Apolipoproteins and their subspecies in human cerebrospinal fluid and plasma. Alzheimers Dement. (Amst) *6*, 182–187. https://doi.org/10.1016/j.dadm.2017.01.007.
49. Lane-Donovan, C., and Herz, J. (2017). The ApoE receptors Vldlr and Apoer2 in central nervous system function and disease. J. Lipid Res. *58*, 1036–1043. https://doi.org/10.1194/jlr.R075507.
50. Beglova, N., and Blacklow, S.C. (2005). The LDL receptor: how acid pulls the trigger. Trends Biochem. Sci. *30*, 309–317. https://doi.org/10.1016/j.tibs.2005.03.007.
51. Rasmussen, K.L., Luo, J., Nordestgaard, B.G., Tybjærg-Hansen, A., and Frikke-Schmidt, R. (2023). APOE and vascular disease: sequencing and genotyping in general population cohorts. Atherosclerosis *385*, 117218. https://doi.org/10.1016/j.atherosclerosis.2023.117218.
52. He, K.Y., Khramtsova, E.A., Cabrera-Socorro, A., Zhang, Y., Li, S., Sarver, B.A.J., Smets, B., Li, Q.S., De Muynck, L., Parrado, A.R., et al. (2023). Characterization of APOE Christchurch carriers in 455,306 UK Biobank participants. Mol. Neurodegener. *18*, 92. https://doi.org/10.1186/s13024-023-00684-7.
53. Wadhera, R.K., Steen, D.L., Khan, I., Giugliano, R.P., and Foody, J.M. (2016). A review of low-density lipoprotein cholesterol, treatment strategies, and its impact on cardiovascular disease morbidity and mortality. J. Clin. Lipidol. *10*, 472–489. https://doi.org/10.1016/j.jacl.2015.11.010.

54. Brown, M.S., and Goldstein, J.L. (1983). Lipoprotein receptors in the liver. Control signals for plasma cholesterol traffic. J. Clin. Invest. *72*, 743–747. https://doi.org/10.1172/JCI111044.

55. Herz, J., and Willnow, T.E. (1995). Lipoprotein and receptor interactions in vivo. Curr. Opin. Lipidol. *6*, 97–103. https://doi.org/10.1097/00041433-199504000-00007.

56. Merrill, N.J., Davidson, W.S., He, Y., Díaz Ludovico, I., Sarkar, S., Berger, M.R., McDermott, J.E., Van Eldik, L.J., Wilcock, D.M., Monroe, M.E., et al. (2023). Human cerebrospinal fluid contains diverse lipoprotein subspecies enriched in proteins implicated in central nervous system health. Sci. Adv. *9*, eadi5571. https://doi.org/10.1126/sciadv.adi5571.

57. Kim, J., Castellano, J.M., Jiang, H., Basak, J.M., Parsadanian, M., Pham, V., Mason, S.M., Paul, S.M., and Holtzman, D.M. (2009). Overexpression of low-density lipoprotein receptor in the brain markedly inhibits amyloid deposition and increases extracellular A beta clearance. Neuron *64*, 632–644. https://doi.org/10.1016/j.neuron.2009.11.013.

58. Katsouri, L., and Georgopoulos, S. (2011). Lack of LDL receptor enhances amyloid deposition and decreases glial response in an Alzheimer's disease mouse model. PLoS One *6*, e21880. https://doi.org/10.1371/journal.pone.0021880.

59. Shi, Y., Andhey, P.S., Ising, C., Wang, K., Snipes, L.L., Boyer, K., Lawson, S., Yamada, K., Qin, W., Manis, M., et al. (2021). Overexpressing low-density lipoprotein receptor reduces tau-associated neurodegeneration in relation to apoE-linked mechanisms. Neuron *109*, 2413–2426.e7. https://doi.org/10.1016/j.neuron.2021.05.034.

60. Basak, J.M., Verghese, P.B., Yoon, H., Kim, J., and Holtzman, D.M. (2012). Low-density lipoprotein receptor represents an apolipoprotein E-independent pathway of Abeta uptake and degradation by astrocytes. J. Biol. Chem. *287*, 13959–13971. https://doi.org/10.1074/jbc.M111.288746.

61. Liu, L., MacKenzie, K.R., Putluri, N., Maletić-Savatić, M., and Bellen, H.J. (2017). The glia-neuron lactate shuttle and elevated ROS Promote Lipid synthesis in neurons and lipid droplet accumulation in glia via APOE/D. Cell Metab. *26*, 719–737.e6. https://doi.org/10.1016/j.cmet.2017.08.024.

62. Ioannou, M.S., Jackson, J., Sheu, S.H., Chang, C.L., Weigel, A.V., Liu, H., Pasolli, H.A., Xu, C.S., Pang, S., Matthies, D., et al. (2019). Neuron-astrocyte metabolic coupling protects against activity-induced fatty acid toxicity. Cell *177*, 1522–1535.e14. https://doi.org/10.1016/j.cell.2019.04.001.

63. Haney, M.S., Pálovics, R., Munson, C.N., Long, C., Johansson, P.K., Yip, O., Dong, W., Rawat, E., West, E., Schlachetzki, J.C.M., et al. (2024). APOE4/4 is linked to damaging lipid droplets in Alzheimer's disease microglia. Nature *628*, 154–161. https://doi.org/10.1038/s41586-024-07185-7.

64. Shibuya, Y., Chang, C.C., and Chang, T.Y. (2015). ACAT1/SOAT1 as a therapeutic target for Alzheimer's disease. Future Med. Chem. *7*, 2451–2467. https://doi.org/10.4155/fmc.15.161.

65. Liu, L., Zhang, K., Sandoval, H., Yamamoto, S., Jaiswal, M., Sanz, E., Li, Z., Hui, J., Graham, B.H., Quintana, A., et al. (2015). Glial lipid droplets and ROS induced by mitochondrial defects promote neurodegeneration. Cell *160*, 177–190. https://doi.org/10.1016/j.cell.2014.12.019.

66. Dong, Y., D'Mello, C., Pinsky, W., Lozinski, B.M., Kaushik, D.K., Ghorbani, S., Moezzi, D., Brown, D., Melo, F.C., Zandee, S., et al. (2021). Oxidized phosphatidylcholines found in multiple sclerosis lesions mediate neurodegeneration and are neutralized by microglia. Nat. Neurosci. *24*, 489–503. https://doi.org/10.1038/s41593-021-00801-z.

67. Tian, R., Abarientos, A., Hong, J., Hashemi, S.H., Yan, R., Dräger, N., Leng, K., Nalls, M.A., Singleton, A.B., Xu, K., et al. (2021). Genome-wide CRISPRi/a screens in human neurons link lysosomal failure to ferroptosis. Nat. Neurosci. *24*, 1020–1034. https://doi.org/10.1038/s41593-021-00862-0.

68. Ward, M.E., Chen, R., Huang, H.Y., Ludwig, C., Telpoukhovskaia, M., Taubes, A., Boudin, H., Minami, S.S., Reichert, M., Albrecht, P., et al. (2017). Individuals with progranulin haploinsufficiency exhibit features of neuronal ceroid lipofuscinosis. Sci. Transl. Med. *9*, eaah5642. https://doi.org/10.1126/scitranslmed.aah5642.

69. Radke, J., Stenzel, W., and Goebel, H.H. (2015). Human NCL neuropathology. Biochim. Biophys. Acta *1852*, 2262–2266. https://doi.org/10.1016/j.bbadis.2015.05.007.

70. Simon, M.J., Logan, T., DeVos, S.L., and Di Paolo, G. (2023). Lysosomal functions of progranulin and implications for treatment of frontotemporal dementia. Trends Cell Biol. *33*, 324–339. https://doi.org/10.1016/j.tcb.2022.09.006.

71. Klein, M., and Hermey, G. (2023). Converging links between adult-onset neurodegenerative Alzheimer's disease and early life neurodegenerative neuronal ceroid lipofuscinosis? Neural Regen. Res. *18*, 1463–1471. https://doi.org/10.4103/1673-5374.361544.

72. Ramsden, C.E., Keyes, G.S., Calzada, E., Horowitz, M.S., Zamora, D., Jahanipour, J., Sedlock, A., Indig, F.E., Moaddel, R., Kapogiannis, D., et al. (2022). Lipid peroxidation induced ApoE receptor-ligand disruption as a unifying hypothesis underlying sporadic Alzheimer's disease in humans. J. Alzheimers Dis. *87*, 1251–1290. https://doi.org/10.3233/JAD-220071.

73. La Cunza, N., Tan, L.X., Thamban, T., Germer, C.J., Rathnasamy, G., Toops, K.A., and Lakkaraju, A. (2021). Mitochondria-dependent phase separation of disease-relevant proteins drives pathological features of age-related macular degeneration. JCI Insight *6*, e142254. https://doi.org/10.1172/jci.insight.142254.

74. Toops, K.A., Tan, L.X., and Lakkaraju, A. (2016). Apolipoprotein E isoforms and AMD. Adv. Exp. Med. Biol. *854*, 3–9. https://doi.org/10.1007/978-3-319-17121-0_1.

75. Huynh, T.V., Wang, C., Tran, A.C., Tabor, G.T., Mahan, T.E., Francis, C.M., Finn, M.B., Spellman, R., Manis, M., Tanzi, R.E., et al. (2019). Lack of hepatic apoE does not influence early Abeta deposition: observations from a new APOE knock-in model. Mol. Neurodegener. *14*, 37. https://doi.org/10.1186/s13024-019-0337-1.

76. Yoshiyama, Y., Higuchi, M., Zhang, B., Huang, S.M., Iwata, N., Saido, T.C., Maeda, J., Suhara, T., Trojanowski, J.Q., and Lee, V.M.Y. (2007). Synapse loss and microglial activation precede tangles in a P301S tauopathy mouse model. Neuron *53*, 337–351. https://doi.org/10.1016/j.neuron.2007.01.010.

77. Guttikonda, S.R., Sikkema, L., Tchieu, J., Saurat, N., Walsh, R.M., Harschnitz, O., Ciceri, G., Sneeboer, M., Mazutis, L., Setty, M., et al. (2021). Fully defined human pluripotent stem cell-derived microglia and tri-culture system model C3 production in Alzheimer's disease. Nat. Neurosci. *24*, 343–354. https://doi.org/10.1038/s41593-020-00796-z.

78. Zhang, Y., Pak, C., Han, Y., Ahlenius, H., Zhang, Z., Chanda, S., Marro, S., Patzke, C., Acuna, C., Covy, J., et al. (2013). Rapid single-step induction of functional neurons from human pluripotent stem cells. Neuron *78*, 785–798. https://doi.org/10.1016/j.neuron.2013.05.029.

79. Lin, H.C., He, Z., Ebert, S., Schörnig, M., Santel, M., Nikolova, M.T., Weigert, A., Hevers, W., Kasri, N.N., Taverna, E., et al. (2021). NGN2 induces diverse neuron types from human pluripotency. Stem Cell Rep. *16*, 2118–2127. https://doi.org/10.1016/j.stemcr.2021.07.006.

80. Newhouse, Y., Peters-Libeu, C., and Weisgraber, K.H. (2005). Crystallization and preliminary X-ray diffraction analysis of apolipoprotein E-containing lipoprotein particles. Acta Crystallogr. Sect. F Struct. Biol. Cryst. Commun. *61*, 981–984. https://doi.org/10.1107/S1744309105032410.

81. Punjani, A., Rubinstein, J.L., Fleet, D.J., and Brubaker, M.A. (2017). cryoSPARC: algorithms for rapid unsupervised cryo-EM structure determination. Nat. Methods *14*, 290–296. https://doi.org/10.1038/nmeth.4169.

82. Rohou, A., and Grigorieff, N. (2015). CTFFIND4: fast and accurate defocus estimation from electron micrographs. J. Struct. Biol. *192*, 216–221. https://doi.org/10.1016/j.jsb.2015.08.008.

83. Degorce, F., Card, A., Soh, S., Trinquet, E., Knapik, G.P., and Xie, B. (2009). HTRF: A technology tailored for drug discovery - a review of theoretical aspects and recent applications. Curr. Chem. Genomics *3*, 22–32. https://doi.org/10.2174/1875397300903010022.

84. Chew, K.S., Wells, R.C., Moshkforoush, A., Chan, D., Lechtenberg, K.J., Tran, H.L., Chow, J., Kim, D.J., Robles-Colmenares, Y., Srivastava, D.B., et al. (2023). CD98hc is a target for brain delivery of biotherapeutics. Nat. Commun. *14*, 5053. https://doi.org/10.1038/s41467-023-40681-4.

85. Ikenoya, M., Yoshinaka, Y., Kobayashi, H., Kawamine, K., Shibuya, K., Sato, F., Sawanobori, K., Watanabe, T., and Miyazaki, A. (2007). A selective ACAT-1 inhibitor, K-604, suppresses fatty streak lesions in fat-fed hamsters without affecting plasma cholesterol levels. Atherosclerosis *191*, 290–297. https://doi.org/10.1016/j.atherosclerosis.2006.05.048.

86. Muffat, J., Li, Y., Yuan, B., Mitalipova, M., Omer, A., Corcoran, S., Bakiasi, G., Tsai, L.H., Aubourg, P., Ransohoff, R.M., et al. (2016). Efficient derivation of microglia-like cells from human pluripotent stem cells. Nat. Med. *22*, 1358–1367. https://doi.org/10.1038/nm.4189.

87. Li, W., and Lee, V.M.Y. (2006). Characterization of two VQIXXK motifs for tau fibrillization in vitro. Biochemistry *45*, 15692–15701. https://doi.org/10.1021/bi061422+.

88. Ewels, P.A., Peltzer, A., Fillinger, S., Patel, H., Alneberg, J., Wilm, A., Garcia, M.U., Di Tommaso, P., and Nahnsen, S. (2020). The nf-core framework for community-curated bioinformatics pipelines. Nat. Biotechnol. *38*, 276–278. https://doi.org/10.1038/s41587-020-0439-x.

89. Dobin, A., Davis, C.A., Schlesinger, F., Drenkow, J., Zaleski, C., Jha, S., Batut, P., Chaisson, M., and Gingeras, T.R. (2013). STAR: ultrafast universal RNA-seq aligner. Bioinformatics *29*, 15–21. https://doi.org/10.1093/bioinformatics/bts635.

90. Patro, R., Duggal, G., Love, M.I., Irizarry, R.A., and Kingsford, C. (2017). Salmon provides fast and bias-aware quantification of transcript expression. Nat. Methods *14*, 417–419. https://doi.org/10.1038/nmeth.4197.

91. Robinson, M.D., McCarthy, D.J., and Smyth, G.K. (2010). edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. Bioinformatics *26*, 139–140. https://doi.org/10.1093/bioinformatics/btp616.

92. Ritchie, M.E., Phipson, B., Wu, D., Hu, Y., Law, C.W., Shi, W., and Smyth, G.K. (2015). limma powers differential expression analyses for RNA-sequencing and microarray studies. Nucleic Acids Res. *43*, e47. https://doi.org/10.1093/nar/gkv007.

93. Robinson, M.D., and Oshlack, A. (2010). A scaling normalization method for differential expression analysis of RNA-seq data. Genome Biol. *11*, R25. https://doi.org/10.1186/gb-2010-11-3-r25.

94. Benjamini, Y., and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. J. R. Stat. Soc. *57*, 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

95. Korotkevich, G., Sukhov, V., Budin, N., Shpak, B., Artyomov, M.N., and Sergushichev, A. (2021). Fast gene set enrichment analysis. Preprint at bioRxiv. https://doi.org/10.1101/060012.

96. Logan, T., Simon, M.J., Rana, A., Cherf, G.M., Srivastava, A., Davis, S.S., Low, R.L.Y., Chiu, C.L., Fang, M., Huang, F., et al. (2021). Rescue of a lysosomal storage disorder caused by Grn loss of function with a brain penetrant progranulin biologic. Cell *184*, 4651–4668.e25. https://doi.org/10.1016/j.cell.2021.08.002.

97. Fujita, M., Gao, Z., Zeng, L., McCabe, C., White, C.C., Ng, B., Green, G.S., Rozenblatt-Rosen, O., Phillips, D., Amir-Zilberstein, L., et al. (2024). Cell subtype-specific effects of genetic variation in the Alzheimer's disease brain. Nat. Genet. *56*, 605–614. https://doi.org/10.1038/s41588-024-01685-y.

98. Hao, Y., Hao, S., Andersen-Nissen, E., Mauck, W.M., 3rd, Zheng, S., Butler, A., Lee, M.J., Wilk, A.J., Darby, C., Zager, M., et al. (2021). Integrated analysis of multimodal single-cell data. Cell *184*, 3573–3587.e29. https://doi.org/10.1016/j.cell.2021.04.048.

99. Lun, A.T.L., McCarthy, D.J., and Marioni, J.C. (2016). A step-by-step workflow for low-level analysis of single-cell RNA-seq data with Bioconductor. F1000Res *5*, 2122. https://doi.org/10.12688/f1000research.9501.2.

100. Germain, P.L., Lun, A., Garcia Meixide, C., Macnair, W., and Robinson, M.D. (2021). Doublet identification in single-cell sequencing data using scDblFinder. F1000Res *10*, 979. https://doi.org/10.12688/f1000research.73600.2.

101. Siletti, K., Hodge, R., Mossi Albiach, A., Lee, K.W., Ding, S.L., Hu, L., Lönnerberg, P., Bakken, T., Casper, T., Clark, M., et al. (2023). Transcriptomic diversity of cell types across the adult human brain. Science *382*, eadd7046. https://doi.org/10.1126/science.add7046.

102. McCarthy, D.J., Campbell, K.R., Lun, A.T.L., and Wills, Q.F. (2017). Scater: pre-processing, quality control, normalization and visualization of single-cell RNA-seq data in R. Bioinformatics *33*, 1179–1186. https://doi.org/10.1093/bioinformatics/btw777.

103. Pang, Z., Zhou, G., Ewald, J., Chang, L., Hacariz, O., Basu, N., and Xia, J. (2022). Using MetaboAnalyst 5.0 for LC-HRMS spectra processing, multi-omics integration and covariate adjustment of global metabolomics data. Nat. Protoc. *17*, 1735–1761. https://doi.org/10.1038/s41596-022-00710-w.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Goat anti-human LDLR	R&D Systems	Cat: AF2148; RRID:AB_2135126
Rabbit anti-human ApoE	Abcam	Cat: ab52607; RRID:AB_867704
Mouse pan anti-human ApoE	Cell Signaling	Cat: 74417; RRID:AB_3094529
Mouse anti-LAMP1	DSHB	Clone H4A3; RRID:AB_2296838
Rabbit anti-PLIN2 (ADFP)	ThermoFisher Scientific	Cat: MA5-32664; RRID:AB_2809941
Mouse anti-TuJ1 (β3-tubulin)	Cell Signaling	Cat: 4466S; RRID:AB_1904176
Chicken anti-MAP2 (microtubule-associated protein 2)	Abcam	Cat: ab5392; RRID:AB_2138153
Mouse anti-4-HNE	Abcam	Cat: ab48506; RRID:AB_867452
AT8 (Mouse anti-pS202/T205 tau)	ThermoFisher Scientific	Cat: MN1020; RRID:AB_223647
Goat anti-Iba1	Novus	Cat: NB100-1028; RRID:AB_3148646 or AB_521594
Rabbit anti-NeuN	Abcam	Cat: ab177487; RRID:AB_2532109
Donkey anti-mouse 488nm	ThermoFisher Scientific	Cat: A-21202
Donkey anti-rabbit 488nm	ThermoFisher Scientific	Cat: A-21206
Donkey anti-rabbit 488nm PLUS	ThermoFisher Scientific	Cat: A-32790
Donkey anti-rabbit 568nm	ThermoFisher Scientific	Cat: A-10042
Donkey anti-mouse 647 nm	ThermoFisher Scientific	Cat: A-31571
Donkey anti-rabbit 647nm	ThermoFisher Scientific	Cat: A-31573
Donkey anti-goat 647nm	ThermoFisher Scientific	Cat: A-21447
Donkey anti-chicken 647nm	ThermoFisher Scientific	Cat: A-78952
Biological samples		
Human CSF samples	Precision Med	7005, 8009, and 8200
Chemicals, peptides, and recombinant proteins		
ApoE2, ApoE3, R136S ApoE3, ApoE4, R136S ApoE4, K146E ApoE4	Generated in-house	N/A
LDLR ECD	R&D Systems; or generated in-house	Cat: 2148-LD/CF-MTO
Heparin	Fisher Scientific	Cat: BP2524
DiI-LDL	ThermoFisher Scientific	Cat: L3482
HDL	EMD Millipore	Cat: LP3-5MG
BODIPY-CE	ThermoFisher Scientific	Cat: C3927MP
CE(18:1)	Avanti Polar Lipids	Cat: 700269
CE(20:4)	Cayman Chem	Cat: 22595
POPC	Avanti Polar Lipids	Cat: 850457
PBS	Gibco	Cat: 10010023
Bio-Beads SM-2 Resin	Bio-Rad	Cat: 1523920
Lipofectamine LTX-Plus	Invitrogen	Cat: A12621
Bovine Serum Albumin	Sigma-Aldrich	Cat: A2153-500G
NucBlue Live ReadyProbes	ThermoFisher Scientific	Cat: R37605
DAPI	Invitrogen	Cat: D1306
Fluorescein-transferrin	ThermoFisher Scientific	Cat: T2871
LysoTracker Deep Red	Invitrogen	Cat: L12492
InstantBlue	Abcam	Cat: ab119211
ProLong Gold Antifade Mountant	ThermoFisher Scientific	Cat: P36930

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Neutravidin	ThermoFisher Scientific	Cat: 31000
KnockOut DMEM/F-12	ThermoFisher Scientific	Cat: 12660012
DMEM/F12 DMEM/F-12, HEPES	ThermoFisher Scientific	Cat: 11330032
MEM Non-Essential Amino Acids Solution (100X)	ThermoFisher Scientific	Cat: 11140050
N-2 Supplement (100X)	ThermoFisher Scientific	Cat: 17502048
Recombinant Human NT-3	Peprotech	Cat: 4500350UG
Recombinant Human/Murine/Rat BDNF	Peprotech	Cat: 450-02-50UG
Laminin Mouse Protein, Natural	ThermoFisher Scientific	Cat: 23017015
Neurobasal-A Medium	ThermoFisher Scientific	Cat: 10888022
GlutaMAX Supplement	ThermoFisher Scientific	Cat: 35050061
B-27 Supplement (50X), minus vitamin A	ThermoFisher Scientific	Cat: 12587010
Neural Induction Medium	STEMCELL Technologies	Cat: 05839
Neural Rosette Selection Reagent	STEMCELL Technologies	Cat: 05832
Astrocyte Differentiation Medium	STEMCELL Technologies	Cat: 100-0014
Astrocyte Maturation Medium	STEMCELL Technologies	Cat:100-0035
Microglia Differentiation Medium	STEMCELL Technologies	Cat:100-0019
Microglia Maturation Medium	STEMCELL Technologies	Cat:100-0020
Cryostor	STEMCELL Technologies	Cat: 07930
Y27632	STEMCELL Technologies	Cat: 72302
mTeSR Plus Medium	STEMCELL Technologies	Cat: 100-0276
Dulbecco's Modified Eagle Medium (DMEM)	ThermoFisher Scientific	Cat: 11965118
1% Penicillin/Streptomycin	Gibco	Cat: 15140122
10% Fetal Bovine Serum	VWR	Cat: 97068-085
Iscove's Modified Dulbecco's Medium (IMDM)	ThermoFisher Scientific	Cat: 12440053
Opti-MEM	Gibco	Cat: 31985062
Hank's Balanced Salt Solution (HBSS)	Gibco	Cat: 14025092
Live Cell Imaging Solution	Invitrogen	Cat: A14291DJ
Vitronectin	ThermoFisher Scientific	Cat: A31804
hm-CSF	Peprotech	Cat: 300–25
hIL34	Peprotech	Cat: 200–34
Critical commercial assays		
pHrodo iFL Green Microscale Protein Labeling Kit	ThermoFisher Scientific	Cat: P36015
pHrodo iFL Red Microscale Protein Labeling Kit	ThermoFisher Scientific	Cat: P36014
RNAeasy Plus Micro Kit	Qiagen	Cat: 74034
Superscript IV VILO Master Mix	ThermoFisher Scientific	Cat: 11756050
TaqMan Fast Advanced Master Mix	ThermoFisher Scientific	Cat: 444456
BirA Biotin-protein ligase biotinylation kit	Avidity LLC	Cat: EC 6.3.4.15
FluoReporter FITC protein labeling kit	Invitrogen	Cat: F6434
QuantSeq 3' mRNA-seq Library Prep Kit	Lexogen	Cat: A01173
4-12% Bis-Tris Gel	Invitrogen	Cat: NP0322BOX
Deposited data		
RNA-seq data	This paper	NCBI GEO accession: GSE250249
Experimental models: Cell lines		
WT HAP1 Cells	Horizon Discovery	C631
LDLR KO HAP1 Cells	Horizon Discovery	HZGHC003978c019
LRP1 KO HAP1 Cells	Horizon Discovery	HZGHC001773c001
H4 Cells	ATCC	HTB-148

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
293T Cells	ATCC	CRL-3216
iPSC	ThermoFisher Scientific	A18945
NGN2 iPSC	Sigma	66540561
Experimental models: Organisms/strains		
Mouse: ApoE KO line (B6.129P2-Apoetm1Unc/J)	Jackson Laboratory	Cat: #002052 RRID: IMSR_JAX:002052
Mouse: *APOE3*$^{flox/flox}$	Laboratory of D. Holtzman	Huynh et al.[75]
Mouse: *APOE4*$^{flox/flox}$	Laboratory of D. Holtzman	Huynh et al.[75]
Mouse: P301S MAPT/APOE KO	Laboratory of D. Holtzman	Shi et al.[6]
Mouse: P301S *MAPT/APOE3* KI	Laboratory of D. Holtzman	Shi et al.[6]
Mouse: P301S *MAPT/APOE4* KI	Laboratory of D. Holtzman	Shi et al.[6]
Oligonucleotides		
Human LDLR Taqman Probe	ThermoFisher Scientific	Cat: Hs01023087_m1
Human HMGCR Taqman Probe	ThermoFisher Scientific	Cat: Hs00168352_m1
Human DHCR7 Taqman Probe	ThermoFisher Scientific	Cat: Hs01023087_m1
Human ABCA1 Taqman Probe	ThermoFisher Scientific	Cat: Hs01059137_m1
Human ABCG1 Taqman Probe	ThermoFisher Scientific	Cat: Hs00245154_m1
Human SREBF2 Taqman Probe	ThermoFisher Scientific	Cat: Hs01081784_m1
Human ACTB Taqman Probe	ThermoFisher Scientific	Cat: Hs01060665_g1
Recombinant DNA		
pcDNA3.1-ApoE2	This paper	N/A
pcDNA3.1-ApoE3	This paper	N/A
pcDNA3.1-ApoE4	This paper	N/A
pcDNA3.1- R136S ApoE3	This paper	N/A
pcDNA3.1- R136S ApoE4	This paper	N/A
pcDNA3.1- K146E ApoE4	This paper	N/A
pRK5- LDLR ECD	This paper	N/A
Software and algorithms		
Harmony	Revvity	HCA
Image Studio	Li-Cor	Version 5.2
ImageJ	National Institutes of Health	Version 2.14.0
MetaboAnalyst	Xia Lab at McGill	Version 5.0, https://www.metaboanalyst.ca
Prism	GraphPad	Version 10
R Studio	R Studio	URL: https://www.rstudio.com/
SnapGene	SnapGene	Version 6.0.5
Biacore Insight Evaluation	Cytiva	Version 4.0.8.20368
Other		
PDL-coated 96-well plate	Corning Biocoat	Cat: 354640
96-well PhenoPlate	Perkin Elmer	Cat: 6055700
Alpha Plate, 384-well	Perkin Elmer	Cat: 6005350
Slide-A-Lyzer MINI dialysis device 10kDa MWCO	ThermoFisher Scientific	Cat: 69570
HiTrap Heparin HP column	Cytiva	Cat: 17040701
HiLoad Superdex S200 pg	Cytiva	Cat: 2898335
C1 Chip	Cytiva	Cat: BR100540
CF2/1 300 grid	Electron Microscopy Sciences	Cat: CF312100
High sensitivity RNA ScreenTape	Agilent	5067-5579
High Sensitivity RNA ScreenTape Ladder	Agilent	5067-5581
High Sensitivity D100 ScreenTape	Agilent	5067-5584

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human CSF samples

Cerebrospinal fluid (CSF) samples from Alzheimer's disease (AD) (n=23) or mild cognitive impairment (MCI) patients (n = 7) and controls (n = 19) procured from Precision Med Inc (Solana Beach, Ca, USA) were used in this study. The samples were obtained with informed consent under WIRB® protocol #20081324 and #20100229 as well as protocols 7005, 8009, and 8200, reviewed and approved by their Institutional Review Board. Repeated CSF collection was taken for some subjects over time and a total of n=92 CSF samples were used in the lipidomic analysis. The demographics of the study subjects are described in the following table.

Demographics and characterization of human subjects used for CSF collection

Measurement	Control	MCI	AD
n	19	7	23
Age (mean (SD))	61.98 (7.88)	71.60 (6.37)	67.19 (8.67)
Gender = M(%)	14 (73.7)	1 (14.3)	12 (52.2)
Ethnicity= Caucasian (%)	19 (100.0)	7 (100.0)	23 (100.0)
MMSE_mean (mean (SD))	29.68 (0.48)	25.19 (0.71)	21.68 (3.33)
ADAS_mean (mean (SD))	NaN (NA)	19.03 (6.08)	24.67 (6.57)
CDR_MEAN (mean (SD))	NaN (NA)	0.63 (0.19)	0.80 (0.37)
Number of APOE4 allele	–	–	–
0	16 (84.2%)	4 (57.1%)	11 (47.8%)
1	2 (10.5%)	1 (14.3%)	8 (34.8%)
2	0 (0%)	2 (28.6%)	3 (13.0%)
Missing information	1 (5.3%)	0 (0%)	1 (4.3%)
CSF Abeta42 (pg/mL, mean (SD))	57.03 (31.51)	18.55 (18.75)	26.70 (22.10)
CSF Total Tau (pg/mL, mean (SD))	417.72 (216.43)	1296.66 (875.73)	1232.31 (926.85)
Plasma pTau181 (pg/mL, mean (SD))	1.80 (0.77)	3.64 (0.90)	3.37 (1.47)
Plasma pTau217 (pg/mL, mean (SD))	0.23 (0.14)	0.75 (0.19)	0.73 (0.48)

Animals

Human *APOE3*$^{flox/flox}$, *APOE4*$^{flox/flox}$ and *APOE* KO mice were previously described.[75] These mice were further crossed to P301S *MAPT* tau transgenic mice overexpressing human 1N4R tau harboring the P301S mutation under PrnP promoter (Stock No. 008169, Jackson Laboratories)[76] to generate *APOE* KI/P301S *MAPT* or *APOE* KO/ P301S *MAPT* mice on murine *Apoe* knockout background.[6] Lipofuscin imaging was performed on the same cohort of 9.5-month-old mice used in a previous study,[10] with 6 male and 6 female animals from each of the 6 genotypes: *APOE* KO, *APOE3* KI, *APOE4* KI, P301S MAPT/APOE KO, P301S *MAPT*/*APOE3* KI, P301S *MAPT*/*APOE4* KI. The LXR agonist impact on lipofuscin accumulation was also tested on a published cohort of 9.5-month-old mice,[10] with 12 male animals from each of the 4 groups: *APOE4* KI, *APOE4* KI with LXR agonist GW3965 treatment, P301S *MAPT*/*APOE4* KI, P301S *MAPT*/*APOE4* KI with LXR agonist GW3965 treatment. Animals received a rodent chow AIN-76 (Research Diets Inc.) with or without supplement of 10 mg/kg LXR agonist GW3965 (HY-10627, MCE Inhibitors) from 6 to 9.5 months.

For hippocampal injection of CE(20:4)/POPC-lipidated ApoE4 or PBS into WT mice, four 6-month-old C57BL/6J male mice (JAX strain #00664) were used. All procedures in animals were performed in adherence to ethical regulations and protocols approved by the Denali Therapeutics Inc. Institutional Animal Care and Use Committee. Mice were housed under a 12-hour light / 12-hour dark cycle and had access to water and standard rodent diet *ad libitum*.

Cell lines

H4 cells (ATCC, HTB-148) and 293T cells (ATCC, CRL-3216) were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (VMR), 1% GlutaMax (GIBCO) and 1% Penicillin/Streptomycin (GIBCO). WT, *LDLR* KO, and *LRP1* KO HAP1 cells (Horizon Discovery, WT: C631; *LDLR* KO: HZGHC003978c019; *LRP1* KO: HZGHC001773c001) were cultured in Iscove's Modified Dulbecco's Medium (IMDM) supplemented with 10% fetal bovine serum (VMR), 1% GlutaMax (GIBCO) and 1% Penicillin/Streptomycin (GIBCO). All cells are grown at 37°C with 5% CO2.

iPSC maintenance and generation of *APOE* KO iPSC

iPSCs were maintained using Matrigel-coated 6-well plates and mTeSR-plus from STEMCELL Technologies. CRISPR/Cas9 genome editing of iPSCs was carried out by Synthego using vials of iPSCs cryopreserved from ThermoFisher cell line A18945, which is

homozygous for APOE3 allele. *APOE* knockout (KO) clones were identified from a pooled knock-in approach to create *APOE2/2* KI lines. iPSCs from this pool were single cell cloned into 96-well Matrigel-coated plates using single cell sorting on a BD FACS Aria. Clones were then screened for small insertions and deletions leading to effective KO of *APOE* by Sanger sequencing (performed by Genewiz) and confirmed in genome viewer SnapGene. Further validation of *APOE* KO was performed using Western blotting. Confirmed clones were cryopreserved and thawed periodically to perform routine differentiations into iPSC-derived CNS cell types outlined below.

iPSC differentiation into astrocytes

Neural progenitor cells (NPCs) were first generated from iPSCs plated into an Aggrewell 400 plate for embryoid body (EB) formation and cultured in STEMCELL Technologies neural induction medium (Cat# 08581). Medium changes and replating of EBs were followed as specified in the protocol. Rosette selection was performed using a truncated 30-minute incubation in rosette selection medium. This was done to reduce the amount of unwanted contaminating cells in the eventual rosettes that were selected and replated. Following selection, rosettes were maintained for an additional week in neural induction medium and cryopreserved at 2.5-5x10^6 cells/mL in Cryostor from STEMCELL Technologies. Astrocytes were then differentiated from cryopreserved NPCs using the reagents and following the protocol from STEMCELL Technologies. NPCs were thawed into Matrigel coated 6-well plates containing neural induction medium + 5 μM Y27632 and allowed to recover for 24 hours. NPCs were expanded for 3-5 more days in neural induction medium. Once confluent, NPCs were singularized using accutase and seeded according to manufacturer's protocol (STEMCELL Technologies) and maintained in astrocyte differentiation medium for the allotted time indicated in the protocol. All medium changes, splitting ratios and time frames were followed as indicated in the astrocyte StemDiff protocol. Astrocyte progenitors were then transitioned to the STEMCELL Technologies astrocyte maturation medium and expanded for ~2 weeks prior to cryopreservation in Cryostor. Partially matured astrocytes were then expanded for an additional 7-10 days in maturation medium prior to plating for assays.

iPSC differentiation into microglia

Hematopoietic progenitor cells (HPCs) were generated using partial reagents from STEMCELL Technologies and a previously published protocol[77] with modifications. To confirm quality of progenitors, FACS analysis was performed as indicated by STEMCELL Technologies to ensure at least 90% purity for CD43. To differentiate HPCs into microglia, frozen vials of qualified HPCs were thawed in 37°C water bath until a small frozen crystal was still observed. Cryoprotectant was washed away by titrating cells into DMEM-F12 and centrifuging at 500 x *g* for 5 min. Cells were then resuspended in Microglia Differentiation Medium (STEMCELL Technologies, Catalog No. 100-0019) and differentiated according to the manufacturer's protocol with suggested medium changes and splitting. After 24 d of differentiation, cells were switched into Microglia Maturation Medium (STEMCELL Technologies, Catalog No. 100-0020). iMg were qualified as greater than 90% double positive for CD45 and CD11b and plated for experiments between 31 d and 39 d.

NGN2 iNeuron differentiation

Differentiation of Doxycycline-inducible Neurogenin-2 (NGN2) iPSCs (Sigma # 66540561) into human neurons was performed as previously described.[78,79] Briefly, NGN2 iPSCs were cultured on a Matrigel-coated plate or flask and maintained in STEMCell Technologies' mTeSR plus medium (Catalog No. 100-0276) following standard non-enzymatic passaging techniques. To begin differentiation into neurons, NGN2 iPSCs were detached by accutase, pelleted by centrifugation at 300 x *g* for 5 min and resuspended in pre-differentiation medium (refer to the table below) containing 10 μM Y27632 and 2 μg/mL doxycycline at approximately 7.8e^4 cells/cm^2 and plate onto a Matrigel-coated plate or flask. This is termed day -3 of the differentiation. For the next two days, day -2 and -1 of the differentiation, a complete medium change with the pre-differentiation medium containing 2 μg/mL of doxycycline was done. On day 0, cells were again lifted by accutase and plated at 20,000 cells/well on Corning BD Biocoat 96-well plates (Catalog No. 354640) in differentiation medium (refer to the table below) containing 1 μg/mL laminin and 2 μg/mL doxycycline. After that, 50-70% medium change was performed every 7 days.

NGN2 Pre-Differentiation Media Components	NGN2 Differentiation Media Components
KnockOut DMEM/F-12	DMEM/F12 DMEM/F-12, HEPES
MEM Non-Essential Amino Acids Solution (100X)	Neurobasal-A Medium
N-2 Supplement (100X)	MEM Non-Essential Amino Acids Solution (100X)
Recombinant Human NT-3	GlutaMAX Supplement
Recombinant Human/Murine/Rat BDNF	B-27 Supplement (50X), minus vitamin A
Laminin Mouse Protein, Natural	N-2 Supplement (100X)
–	Recombinant Human NT-3
–	Recombinant Human/Murine/Rat BDNF
–	Laminin Mouse Protein, Natural

METHOD DETAILS

Expression and purification of HEK cell-derived ApoE

A DNA fragment encoding for full length (including signal sequence) human ApoE3 protein was synthesized with codon optimization for mammalian expression and cloned into a pcDNA3.1 vector (GenScript). Mutagenesis of the resulting construct was performed to generate the following constructs: ApoE2, R136S ApoE3, ApoE4, R136S ApoE4, and K146E ApoE4. Proteins were expressed by transient transfection in HEK293 cells at 37°C for 5 days. Cells were harvested by centrifugation at 3000 rpm for 10 min, after which the supernatant was filtered through a 0.22 μm sterile filter and loaded onto a HiTrap heparin HP column (Cytiva) pre-equilibrated with 25 mM Tris-HCl pH 7.5, 10 mM NaCl, 1 mM DTT and 10 mM sodium cholate (Buffer A). The column was then washed with 18 column volumes (CV) of 25 mM Tris-HCl pH 7.5, 10 mM NaCl, 1 mM DTT, 0.1% Triton X-100, followed by an additional 18 CV of Buffer A. Bound protein was eluted with a 20 CV gradient to 25 mM Tris-HCl pH 7.5, 500 mM NaCl, 1 mM DTT, 10 mM sodium cholate. Eluted fractions containing purified ApoE were pooled, concentrated to >2 mg/ml and loaded onto a HiLoad Superdex S200 pg (Cytiva) pre-equilibrated with 25 mM Tris-HCl pH 7.5, 150 mM NaCl and 1 mM TCEP. Fractions containing purified ApoE were again pooled and concentrated to >2 mg/ml. All ApoE preparations used in iMg experiments had endotoxin levels < 1 EU/mg.

To generate materials for the homogeneous time-resolved fluorescence (HTRF)-based ApoE-LDLR binding assay, constructs encoding for ApoE2, ApoE3 and ApoE4 were used as template. DNA fragments lacking the native signal sequence were subcloned into a derived pRK5 containing a mouse kappa light chain signal sequence followed by a 6x histidine (His) tag, a $(GGGS)_2$ linker, a BirA ligase recognition sequence (Avi tag, GLNDIFEAQKIEWHE) and another $(GGGS)_2$ linker. Proteins were expressed by transient transfection in HEK293 cells at 37°C for 5 days. Cells were harvested by centrifugation at 3000 rpm for 10 min and the resulting supernatant was filtered through a 0.22 μm sterile filter and loaded onto a Ni-NTA Excel resin pre-equilibrated with 25 mM Tris pH7.5, 150 mM NaCl, 10 mM imidazole, 1 mM DTT and 10 mM sodium cholate (Buffer B). The column was washed with 18 CV of 25 mM Tris-HCl pH7.5, 150 mM NaCl, 0.1% Triton X-100, 1 mM DTT, followed by 18 CV of Buffer B. Protein was eluted using a gradient of 25 mM Tris pH7.5, 150 mM NaCl, 500mM imidazole, 1 mM DTT and 10 mM sodium cholate. Elution fractions containing purified ApoE were pooled, concentrated to >2 mg/ml and resolved on a Superdex S200 16/60 column pre-equilibrated with 25 mM Tris pH 7.5, 150 mM NaCl and 1 mM DTT. Fractions containing purified ApoE were pooled and concentrated to >2 mg/ml.

ApoE lipidation

Purified ApoE was lipidated with 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC) using a previously described procedure,[80] with some modifications. A solution of POPC (Avanti Polar Lipids, #850457) in chloroform was dried under vacuum for 30 minutes using a Rotavap and further dried overnight under vacuum. The resulting film was resuspended to a final concentration of 10 mg/mL using a solution of 25 mM Tris pH 7.5, 150 mM NaCl supplemented with 100mM sodium cholate (molar ratio of POPC: cholate = 1:8). Purified ApoE was mixed with the solution to the molar ratio of 1:130 for ApoE:POPC and incubated under constant rotation at RT. After 2 h, the reaction mixture was transferred to 4°C and an equal volume of bio-beads (biobeads SM2 resin, BioRad) were added to remove excess lipids, with the beads replaced 3 times over the course of a 48 h period. Finally, the bio-beads were filtered out using 0.22 μm filter Eppendorf units and the resulting lipid-reconstituted ApoE nanodiscs were polished over a HiLoad Superdex S200 pg (Cytiva) pre-equilibrated with PBS and 1 mM DTT, under low endotoxin conditions.

For CE(20:4)/POPC lipidation of ApoE for use in the lipofuscin assay, an aliquot of CE(20:4) (Cayman Chem, 22595) was freshly resuspended into chloroform and mixed with POPC at a molar ratio of 1:4 for CE(20:4):POPC. Lipid preparation and lipidation were carried out as described above for POPC lipidation. The earliest eluting peak from SEC purification (Figure S5B) was found to be the most efficient in lipofuscin induction on NGN2 iNeurons in pilot experiments and was thus selected for experiments shown in this study. CE(18:1)/POPC-lipidated ApoE4 was generated following the same protocol, except for switching CE(20:4) with CE(18:1) (Avanti Polar Lipids, #700269).

The concentrations of lipidated ApoE preparations were measured by NanoDrop Spectrophotometer and adjusted to the same concentrations for different isoforms. Aliquots of lipidated ApoE were snap frozen and stored at -80°C. Thawed aliquots were temporarily stored at 4°C and used within 3 weeks. CE(20:4)/POPC-lipidated ApoE4 used for mouse brain injection had an endotoxin level < 10 EU/mg.

Negative stain TEM for lipidated ApoE

Samples were diluted to a final concentration of 2.5 to 5 μg/mL in 25 mM HEPES pH 7.5, 150 mM NaCl; 3 μL of the solution was placed onto a CF2/1 300 grid from Electron Microscopy Sciences (EMS) previously cleaned using a PELCO EasiGlow glow system discharge (Ted Pella). After 60 seconds incubation, sample was blotted with Whatman paper and stained three times into 20 μL of a filtered solution of 1% uranyl acetate (30 seconds with last drop). Grids were imaged using a Hitachi7800 TEM with a NanoSprint15 Mk-II AMT camera (AMT Imaging). All image processing was performed using CryoSPARC.[81] Micrographs were contrast transfer function (CTF) corrected using CTFFIND4.[82] Particles were then selected using the blob picker and subjected to several rounds of 2D classification.

Purification of recombinant human LDLR extracellular domain

The full-length human low density lipoprotein receptor (LDLR) extracellular domain (ECD, residues A1-R767) with its native signal peptide sequence was synthesized (Genscript) with codon optimization for mammalian expression and cloned into a derived

pRK5 vector upstream of a C-terminal Avi tag, followed by a TEV cleavage site (ENLYFQG) and a 6x His tag. Protein was expressed by transient transfection in HEK293 cells at 37°C for 5 days. Cells were harvested by centrifugation at 3000 rpm for 10 min and the resulting supernatant was filtered through a 0.22 μm sterile filter and loaded onto Ni-NTA Excel resin pre-equilibrated with 25 mM HEPES pH 7.5, 150 mM NaCl, 1 mM $CaCl_2$ (Buffer C) at 4°C. The resin was washed with Buffer C supplemented with 35 mM imidazole and protein was then step eluted with Buffer C containing 300 mM imidazole. For cell-based assays, the elution fractions containing purified LDLR ECD were pooled, concentrated to >2 mg/ml and resolved on a HiLoad Superdex S200 column pre-equilibrated with Buffer C, under low endotoxin conditions. Fractions containing purified LDLR ECD were pooled and concentrated to >2 mg/ml. In cellular experiments, human LDLR ECD purchased from R&D Systems (2148-LD/CF-MTO) was used interchangeably with internally generated materials.

For HTRF experiments, elution fractions from the Ni-NTA affinity purification were pooled and incubated with TEV protease overnight while dialyzing against Buffer C at 4°C. Cleaved LDLR-ECD was then reloaded onto a Ni-NTA Excel column, washed with Buffer C supplemented with 30mM and eluted with Buffer C containing 300 mM imidazole. The flowthrough and wash fractions containing tag-cleaved LDLR-ECD were pooled and in-vitro biotinylated using a BirA Biotin-protein ligase biotinylation kit, according to the manufacturer's instructions (Avidity LLC). The reaction was incubated at RT for 3 h and the biotionlyated LDLR-ECD was separated from free biotin by resolving on a Superdex S200 16/60 column pre-equilibrated with Buffer C. Fractions containing purified, biotinylated LDLR ECD were pooled and concentrated to >2 mg/ml.

HTRF for lipApoE-LDLR binding assessment

Homogeneous Time Resolved Fluorescence (HTRF) measurements[83] were carried out using an anti-6xHis-XL665 mAb acceptor and Streptavidin-Eu cryptate donor according to the manufacturer's instructions. The assay was performed in a buffer consisting of 100 mM Tris HCl, pH 7.5, 150 mM NaCl, 2 mM $CaCl_2$, and 0.1% BSA. LDLR ECD-Biotin was dispensed into a 384-well assay plate (AlphaPlate 384-well, Perkin Elmer), in triplicate. Then, 8-point, 2-fold serial dilutions of His-tagged ApoE were prepared in assay buffer and dispensed into the 384-well assay plate in triplicate, followed immediately by sequential additions of 1X acceptor and donor fluorophores. Each well therefore contained 10 nM LDLR ECD-biotin, 0.3X each of acceptor and donor fluorophores, and the indicated concentrations of ApoE, in a total volume of 20 μL. The assay plate was then covered in adhesive foil to protect it from light and incubated for 1 hour at RT while shaking at 1050 rpm on an orbital plate shaker. After 1 hour, the plate was read on BMG Labtech PHERAstar, dual time-resolved fluorescence on endpoint mode with top optic. Measurements were made with Read A at 337/665 and Read B at 337/620. The assay signal was calculated with Read A over Read B, multiplied by a factor of 10,000 on the MARS software. EC50s were calculated on Graph Pad Prism 9 and curves were fitted with a 4-parameter sigmoidal curve. Three separate titrations were performed in triplicate for each ApoE isoform with points representing mean +/- standard deviation of the average of the triplicates from each titration. In the competitive HTRF assay to assess inhibition of lipApoE-LDLR binding by different ApoE variants, untagged, lipidated ApoE was mixed with 30 nM LDLR ECD-Biotin for 30 min at RT and then combined with 30 nM His-tagged lipApoE4, along with donor and acceptor fluorophores described above. The total reaction volume remained at 20 uL and all the other assay parameters remained constant. Three separate titrations were performed in triplicate for each lipApoE variant.

SPR for lipApoE-LDLR binding measurement

For surface plasmon resonance (SPR) experiments, a C1 chip (Cytiva BR100540) was used in the Biacore 8k (Cytiva) that had been amine coupled with neutravidin (Thermo 31000) to a density of approximately 700 RUs. The biotinylated LDLR ECD (2 μg/mL) was immobilized onto the neutravidin surface to a density from 330 to 360 RUs. Lipidated ApoE2, E3, E4 were injected over the SPR surface at a flow rate of 50 μL per min with association time of 60 seconds and a final dissociation time of 300 seconds. Concentrations tested were as follows: 0.5, 1, and 2 μM for lipidated ApoE2; 0.0156, 0.03125, and 0.0625 μM for lipidated ApoE3 and ApoE4. All data were analyzed in the Biacore Insight Evaluation software (version 4.0.8.20368) and the curves were fit using a 1:1 kinetic model.

HPLC-based heparin binding assay

HEK-derived ApoE was diluted to 1.85 mg/mL in 20 mM Tris-HCl pH 7.5, after which 50 μg of the sample was injected onto a POROS™ Heparin 50 μm Column, 2.1 x 30 mm, 0.1 mL (Thermo Scientific™, 4333411) pre-equilibrated with 20 mM Tris-HCl pH 7.5 (Buffer A). The column was washed with 5 CV of Buffer A and eluted with 20 mM Tris-HCl pH 7.5 containing 2 M NaCl (Buffer B) at 0.08 M/min NaCl gradient.

Cell-based LDLR binding assay

POPC/cholesterol-lipidated ApoE was labeled using FluoReporter FITC protein labeling kit (Invitrogen, F6434) with excess dye removed by dialysis into PBS using Slide-A-Lyzer MINI dialysis device with 10kDa MWCO (ThermoFisher, #69570). Concentrations of the FITC-labeled different ApoE isoforms were equalized based on densitometry of protein bands on a 4-12% Bis-Tris gel (Invitrogen) stained by InstantBlue (Abcam, ab119211). After adding 1mM DTT, the FITC-lipApoE preparations were stored at 4°C for up to 2 weeks. HEK293T cells were plated at 25,000 cells/well in a PDL-coated 96-well plate (Corning Biocoat, 354640). One day after plating, cells were transfected with Flag tagged-human LDLR on pcDNA3.1 (GenScript) using Lipofectamine LTX-Plus (Invitrogen, A12621) in serum-free Opti-MEM (GIBCO) and switched back into complete medium at 6 h after transfection. Two days

after transfection, cells were stained with NucBlue (Invitrogen, R37605) for 15 min at 37°C and then incubated with 10 μg/mL FITC-labeled lipidated ApoE (~294 nM) diluted in cold medium for 1 h at 4°C. After removing the treatment medium, cells were washed once with Hanks' Balanced Salt Solution (GIBCO, 14025092) and placed on cold Live Cell Imaging Solution (Invitrogen, A14291DJ) before live imaging. After live imaging of cell-bound FITC-lipApoE, cells were fixed and immunostained to confirm equivalent LDLR overexpression across treatment conditions.

pHrodo-lipApoE uptake assay

For assessing cellular uptake, lipidated ApoE was labeled using pHrodo-iFL green microscale labeling kit (ThermoFisher, P36015) at a dye:protein ratio of 7:1. Excess dye removal and concentration adjustment were performed as that for FITC-labeled lipApoE described above with 1 mM DTT added. Aliquots of pHrodo green-lipApoE were snap frozen and stored at -80°C. Thawed aliquots were kept at 4°C for up to 2 weeks. Cells were plated at appropriate densities (8,000-10,000 cells/well for H4, ~20,000 cells/well for HAP1 lines, 15,000-20,000 cells/well for *APOE* KO iAstrocytes) in a PDL-coated 96-well plate (PerkinElmer, or Corning Biocoat). The next day, cells were treated with 10 μg/mL pHrodo green-labeled lipidated ApoE (~294 nM) in serum-free Opti-MEM. To test how uptake of lipApoE is mediated by cell surface receptors, pHrodo green-lipApoE was pre-incubated with LDLR ECD (R&D Systems #2148-LD/CF-MTO, or generated as described above), low-molecular-weight heparin (Fisher Scientific, BP2524), or CD98hc ECD[84] at 37°C for 30–40 min before being added to cells. Note that the final concentrations of the molecules in treatment medium were 20 μg/mL for LDLR ECD (~224 nM) and CD98hc ECD (~434 nM) and 2.5 μg/mL for heparin (~400 nM), but pre-incubation with pHrodo green-lipApoE was done at 2x or 4x concentrations. For all live imaging experiments, nuclear staining was done by incubation with NucBlue. Live imaging for pHrodo green signals was performed at 5-6 h or 1 d after treatment. The exact uptake efficiency of ApoE4 *vs.* ApoE3 varies across different lipidation batches, but an overall trend of ApoE4 >= ApoE3 > ApoE2 was observed.

pHrodo-HDL/BODIPY-CE uptake assays and CE(18:1)/HDL/ApoE treatment

High-density lipoprotein (HDL, EMD Milipore, LP3-5MG) was stored at 4°C for up to 2 months in Nitrogen gas-purged glass vials as small aliquots to minimize oxidation. BODIPY-CE (CholEsteryl BODIPY™ FL C12, ThermoFisher C3927MP) and CE(18:1) (Avanti Polar Lipids #700269) were reconstituted in chloroform to 1 mg/mL, aliquoted into glass vials at 500 μL or 1 mL per vial and dried under Nitrogen gas. Dried CE was kept at -20°C or -80°C for long-term storage or re-dissolved in ethanol to reach 1 mg/mL with the aid of a heat gun accompanied by vigorous vortexing. Small aliquots of CE in ethanol were kept at -20°C in Nitrogen gas-purged glass vials for short-term use within 1-2 months. Right before cell treatment, an aliquot of CE was warmed on a 37°C heat block for ~5 min followed by vigorous vortexing to ensure the powder was fully dissolved again.

For assessing ApoE-mediated HDL uptake, HDL was labeled using pHrodo-iFL red microscale labeling kit (ThermoFisher, P36014) or pHrodo-iFL green microscale labeling kit (ThermoFisher, P36015) following the same protocol as that for lipApoE labeling described above with 1 mM DTT added. Aliquots of pHrodo-labeled HDL were stored at 4°C and used within 2 weeks. One day after plating cells to a PDL-coated 96-well plate, pHrodo-labeled HDL was pre-complexed with non-lipidated ApoE with and without the addition of LDLR ECD or heparin at 37°C for ~45 min before being added to cells in serum-free Opti-MEM. The final concentrations in the treatment medium were 10 μg/mL for pHrodo-HDL and ApoE, 20 μg/mL for LDLR ECD, and 2.5 μg/mL for heparin, but the pre-complexing was typically done at 2x or 4x concentrations.

BODIPY-CE uptake assay or CE(18:1)/HDL/ApoE treatment was performed following a similar protocol as pHrodo-HDL uptake except with the use of unlabeled HDL and the inclusion of BODIPY-CE or CE(18:1) during pre-complexing. The final concentrations in the treatment medium were 10 μg/mL for ApoE, HDL and BODIPY-CE or CE(18:1). Note that extracellular BODIPY-CE fluorescence signals are extensively self-quenched, thus imaging for BODIPY-CE uptake was performed without medium removal as BODIPY signals only show up after cellular uptake of BODIPY-CE. For lipidomic study on CE(18:1)/HDL/ApoE-treated *APOE* KO iAstrocytes, ApoE4 treatment was done with and without 20 μg/mL LDLR ECD or 500nM ACAT1 inhibitor K604.[85]

For BODIPY-CE/HDL/ApoE treatment of *APOE* KO iMg, cells were plated onto a PDL-coated 96-well plate with additional coating of vitronectin (ThermoFisher, A31804) and switched from Microglia Maturation Medium into NGD medium[86] supplemented with 20 ng/mL of hM-CSF (Peprotech,300-25) and 20 ng/mL of hIL34 (Peprotech, 200-34), i.e., NGD++ medium. The plating density was at 8,000-12,000 cells/well for the first round of differentiation and 16,000-18,000 cells/well for the second round of differentiation. One day after plating, iMg were treated with 10 μg/mL ApoE, HDL and BODIPY-CE pre-complexed at 37°C for ~45 min at 2x concentrations in NGD++ medium with the addition of 100 ng/mL LPS. Live imaging of BODIPY signals was performed after 1 d incubation.

DiI-LDL uptake assay

Two paradigms were used to assess uptake of DiI-conjugated LDL (ThermoFisher, L3482) in H4 cells. In the co-treatment paradigm, cells were serum-starved in Opti-MEM for 1.5 - 2 h before being treated with 10 μg/mL DiI-LDL and 10 μg/mL POPC-lipidated ApoE simultaneously in Opti-MEM, with live imaging performed at 3-4 h after treatment. In the second paradigm, cells were pre-treated with 10 μg/mL POPC-lipidated ApoE in Opti-MEM for 1 d, followed by spike-in of DiI-LDL to reach 10 μg/mL and further incubation for 3-4 h before live imaging.

Lipofuscin induction in NGN2 iNeurons

To monitor lipofuscin formation over time, NGN2 iNeurons were treated with 20 μg/mL CE(20:4)/POPC-lipidated ApoE in 50% fresh medium on day 7 and replenished with 10 μg/mL CE(20:4)/POPC-lipidated ApoE in 50% fresh medium on days 14 and 21. Live imaging for autofluorescent lipofuscin was performed on days 14, 21 and 28 (*i.e.*, after 7 d, 14 d and 21 d treatment, respectively). Lipofuscin was detected in a broad spectrum but appeared the most intense in the 488 nm channel, which was used for image capture and quantification. To confirm the role of LDLR on lipofuscin induction in iNeurons, live imaging for lipofuscin was performed after 3 d treatment of 20 μg/mL CE(20:4)/POPC-lipidated ApoE4 with or without 20 μg/m LDLR ECD that was pre-incubated for 30 min at 37°C with ApoE. For lipidomic study, NGN2 iNeurons were infected with lentivirus expressing P301S 4R2N tau (gene synthesis into pLVX-puro vector by GenScript, lentiviral production by SignaGen) at MOI of 5 on day 11, treated with 20 μg/mL CE(20:4)/POPC-lipidated ApoE in 50% fresh medium on day 14 and replenished with 10 μg/mL CE(20:4)/POPC-lipidated ApoE in 50% fresh medium on day 21. Lipid extraction was done on day 28.

pHrodo-tau pff accumulation assay in NGN2 iNeurons

4R2N full-length tau carrying P301S mutation was expressed and purified by CepterBio as previously described.[87] For fibrillization, 40 μM P301S tau was incubated with 40 μM low-molecular-weight heparin and 2 mM DTT in PBS for 5 d at 37°C with constant agitation at 1,000 rpm in a ThermoMixer fitted with a heated lid. At the end of the incubation, ultracentrifugation was performed on the fibrillization reaction at 100,000 x *g* for 30 min at 4°C, following which the supernatant was removed and the pellet fraction containing preformed tau fibrils (tau pffs) was resuspended in PBS. Proteins in the supernatant and the pellet fractions were run on a 4-12% Bis-Tris gel and stained with InstantBlue for estimating the concentration of tau in the pellet fraction. For assessing neuronal uptake and accumulation of tau pffs, the P301S tau pffs were diluted to 1 mg/mL and labeled using pHrodo-iFL red microscale labeling kit (ThermoFisher, P36014) at a dye:protein ratio of 8:1. Excess dye was removed by 2 rounds of ultracentrifugation with the resulting pellet resuspended in PBS with sonication, aliquoted and stored at -80°C. pHrodo-tau pff treatment was done on NGN2 iNeurons after 14 d incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE or CE(18:1)/POPC-lipidated ApoE, whereby 50% of conditioned medium was removed followed by the addition of sonicated pHrodo-tau pffs to a final concentration of 2.5 μg/mL. Live imaging of pHrodo-tau signals was performed after 13-14 d incubation, during which cells were replenished with 10 μg/mL CE(20:4)/POPC-lipidated ApoE or CE(18:1)/POPC-lipidated ApoE once at 7 d after tau pff addition.

Lipofuscin induction in H4 and HAP1 cells

Lipofuscin induction on H4 and various lines of HAP1 cells (WT, *LDLR* KO, *LRP1* KO) was monitored by live imaging after 3 d incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE or CE(18:1)/POPC-lipidated ApoE in serum-free Opti-MEM. To validate the role of LDLR in lipofuscin induction in H4 cells, 20 μg/mL CE(20:4)/POPC-lipidated ApoE4 was added to cells with or without 20 μg/mL LDLR ECD that was pre-incubated for 30 min at 37°C with ApoE. To test modulation of lipofuscin accumulation by Vitamin E, 20 μg/mL CE(20:4)/POPC-lipidated ApoE3 was pre-incubated with 25 μM Vitamin E (Sigma,V-082-1ML, in methanol) or volume-matched methanol (vehicle control) for 30 min at 37°C before adding the mixture to H4 cells for 1 d incubation. To test the effect of Lalistat 2 (Cayman, 25347-5) on lipofuscin accumulation, H4 cells were pre-treated with 25 μM Lalistat 2 (in DMSO) or volume-matched DMSO (vehicle control) for 1 d before the spike-in of CE(20:4)/POPC-lipidated ApoE3 to a final concentration of 20 μg/mL followed by 1 d further incubation.

Cathepsin activity assay

6QC probe was synthesized as previously described[40] and reconstituted by DMSO into 5 mM stock solution. For measuring cathepsin activity in H4 cells pre-treated with ApoE, 6QC probe was diluted into 5 μM in serum-free Opti-MEM together with NucBlue and incubated with H4 cells for 2 h at 37°C before live imaging. Upon cleavage by lysosomal cathepsins, the 6QC probe becomes de-quenched and emits fluorescence in the 647nm channel.

Immunocytochemistry

Immunostaining for cell surface LDLR was performed by incubating live cells with an anti-human LDLR antibody (R&D Systems, AF2148) diluted 1:200 in live staining medium (cold Opti-MEM supplemented with 25 mM HEPES) for 1 h at 4°C. After three washes with cold live staining medium, cells were fixed with 4% paraformaldehyde (PFA) for 15 min at RT. After two PBS washes, cells with blocked with 5% bovine serum albumin (BSA) for 1 h at RT, followed by incubation with donkey anti-goat Alexa 647nm secondary antibody (Invitrogen) diluted 1:1000 in 5% BSA for 2 h at RT. Following two PBS washes, cells were incubation with DAPI (Invitrogen, D1306) diluted 1:5000 in PBS for at least 15 min. Imaging was done after one last PBS wash. For immunostaining of intracellular proteins, including internalized lipApoE, cells were fixed with 4% PFA for 15 min at RT, washed twice with PBS, blocked with 5% BSA containing 0.05% saponin for 1 h at RT and incubated with primary antibodies (see key resources table) diluted in 5% BSA containing 0.05% saponin for 2–3 h at RT or overnight at 4°C. After removal of primary antibody staining solution, cells were washed 3x in PBS and incubated with Alexa-labeled secondary antibodies diluted in 5% BSA containing 0.05% saponin for 1-2 h at RT. Post-secondary PBS washes and DAPI staining were done as that for cell surface LDLR staining described above. For labeling of recycling endosomes, cells were incubated with Alexa 488nm-conjugated transferrin for 15 min before fixing by 4% PFA. To confirm perfect colocalization of internalized ApoE with LDLR was not due to bleed-through, ApoE staining was done with and without addition of the

primary antibody for human LDLR but with both secondary antibodies, which showed completely no signals in the channel corresponding to LDLR in the absence of primary antibody for LDLR.

High-content imaging and image analysis

Live imaging or imaging of fixed and immunostained cells was performed using Opera Phenix high-content imaging system (Revvity) with 40x or 63x water objective at the confocal mode. Fluorescent signals were automatically quantified using the spot analysis algorithm within Harmony HCS software (Revvity) with appropriate thresholding. Key parameters that were quantified include cell numbers, corrected spot intensity mean, total spot area and total number of spots. For most experiments, phenotypic comparisons were made based on the calculation of corrected spot intensity mean x total spot area / cell number (i.e., total signals/cell). For analysis of pHrodo-tau pff signals in subpopulation of iNeurons, perinuclear signals were quantified for neurons with > 3 lipofuscin spots per cell (arbitrarily defined as "lipofuscin-positive"). To make sure 4-HNE immunoreactivities were not contaminated by autofluorescent signals from lipofuscin, no primary antibody control was performed, which confirmed no detectable autofluorescent signals at the contrast set to visualize 4-HNE puncta which are much brighter. To analyze lysosomal clustering, LAMP1 spot analysis was performed separately with the splitting sensitivity set at 0 (minimum) or 1 (maximum). At the minimal splitting sensitivity, LAMP1 spots that are clustered together are automatically counted as a single spot. Subsequently, LAMP1 spot area covered by large single spots reaching a defined size threshold ("clustered" lysosomes) was expressed as a percentage of total LAMP1 spot area quantified using maximal splitting sensitivity (total lysosomes). The percentage of clustered/total lysosomes for each treatment condition was further normalized to the mean percentage across all conditions in each independent experiment and termed "LAMP1 clustering index". For cell count estimation of *APOE* KO iMg before harvesting media for cytokine/chemokine assay, Digital Phase Contrast and Brightfield images were taken at 10x.

Super-resolution imaging and analysis

Fixed and stained cells in 96-well plate were rinsed twice with PBS prior to imaging on a Leica SP8 microscope using a 63x/1.4 N.A. oil immersion objective with a pixel size of ~30 nm at 63X magnification in lightning acquisition mode. For acquisition of super-resolution images, lightning processing was performed in the adaptive mode and images were then visualized and processed in LAS X and ImageJ. For colocalization analysis, ImageJ's Coloc2 tool was used for calculating the Mander's overlap coefficients.

Cell-free aggregation of lipidated ApoE

POPC-lipidated ApoE, CE(20:4)/POPC-lipidated ApoE, and CE(18:1)/POPC-lipidated ApoE4 in PBS (pH 7.4) were diluted into 0.5 mg/mL using PBS (pH 7.4) and incubated at 37°C for 4 d. To adjust the pH to 4.5, a pre-determined volume of diluted hydrochloric acid was spiked in at 1/10 of the total volume prior to 37°C incubation. To monitor the extent of protein aggregation, ultracentrifugation was performed by spinning the samples at 150,000 x *g* for 30 min at 4°C, followed by separation of the supernatant and the pellet fraction. The pellet fraction was resuspended in the same volume of PBS as the starting sample. Both supernatant and pellet fractions were heated with sample buffer in the absence of reducing agent for 5 min at 95°C, run on a 4-12% Bis-Tris gel (Invitrogen) and stained by InstantBlue. Imaging of the gel and the densitometry analysis were performed using the Odyssey DLx Imaging System with Image Studio software (Li-Cor).

Lipofuscin imaging and quantifications for mouse brain tissues

For lipofuscin imaging of mouse brains shown in Figures 6A–6E, S6A, and S6B, brain sections were from a previously published cohort.[10] ~50 μm brain sections were washed in PBS 3x 5 min, permeabilized in PBS containing 0.25% Triton X-100 for 30 min, incubated for 1hr in blocking buffer (2% donkey serum in 0.25% Triton X-100 in PBS) and then stained overnight with anti-NeuN (1:500, ab177487) and anti-Iba1 (1:500, Novus Biologicals, NB100-1028) in blocking buffer. The next day, the sections were washed 3x 5 min in PBS, and incubated with secondary antibodies (donkey anti-Rb 488, Thermofisher A-21206; donkey anti-goat 647, Thermofisher A-21447) at room temperature for 1 h in dark. After 3x 5 min PBS washes, the tissues were stained with DAPI (Invitrogen D1306, 1:10,000 dilution in PBS) for 20 min in dark, washed 3x 5 min in PBS, and mounted on glass with ProlongGold Antifade media and scanned at 20x/0.75 N.A. using Leica Stellaris 8 Confocal microscope. Autofluorescent signals in the entire hippocampus and entorhinal cortex were quantified in the 545 nm excitation/600–650 nm emission channel using ImageJ software with different thresholding of signal over background. The number of DAPI-positive nuclei was calculated using "ImageJ > Analyze Particles > Add to ROI manager"; the levels of autofluorescent lipofuscin signals were then normalized to the number of nuclei. Representative images were acquired using a 40x/1.3 N.A. oil immersion objective.

Lipofuscin induction with *in vivo* injections of CE(20:4)/POPC-lipidated ApoE4

For hippocampal injections of CE(20:4)/POPC-lipidated ApoE4 or PBS into C57BL/6J mice, the surgical area was sterilized with 70% isopropanol and 4% chlorhexidine and surgical instruments were autoclaved. Mice were brought under anesthesia with 4% isoflurane. Hair was shaved from the base of the skull region to between the eyes prior to placement on the stereotaxic surface. 5 mg/kg Meloxicam, 3.25 mg/kg Ethiqa XR and 10 uL of 0.125% Bupivicaine were administered subcutaneously prior to any surgical procedures. With a maintenance level of ~1.5-2% isoflurane, a 1.5-2 mm incision was made in the center of the skull. Following application of 3% hydrogen peroxide, bregma and lambda were used as reference points to set coordinates and drill into the skull. Bilateral

hippocampal injections were conducted for each animal at the following coordinates: AP -2.54, ML -2.98, DV 2.83 (PBS injection) and AP -2.54, ML +2.98, DV 2.83 (CE(20:4)/POPC-lipidated ApoE4 injection), from brain surface. The injection needle (30GA, 15mm, 45 degree angle), attached to a 10μL Hamilton syringe (Hamilton 1701 RN) was inserted into the brain at a rate of 0.2 mm per second. After waiting for one minute to allow for brain sealing around the needle, 2.5 uL of PBS or 4 mg/mL CE(20:4)/POPC-lipidated ApoE4 in 2 mM DTT PBS was administered at 0.4 μL per minute. Five minutes following the injection, the needle was raised slowly out of the brain, the incision was sutured and mice were transferred to a cage placed over a heated recovery pad and observed for full recovery. Mice were monitored daily after surgery to check for weight loss, pain, discomfort, or infections.

Animals were sacrificed 14 d post-surgery. For terminal sample collection, animals were deeply anesthetized via IP injection of 2.5% Avertin (Millipore Sigma T48402-5G) prior to exsanguination. Animals were then transcardially perfused with ice-cold PBS using a peristaltic pump (Gilson Inc. Minipuls Evolution). The brain was removed and post-fixed 24 h in 4% PFA at 4°C, after which tissue was cryoprotected in 30% sucrose and sliced coronally at 25 μm on a freezing microtome at and around the anterior-posterior level of injection. Free-floating sections were first washed in PBS, permeabilized for 15 min in 0.3% TritonX-100 in PBS, and then blocked for 1 h in a PBS based-buffer containing 1% BSA, 0.1% fish gelatin (BioWorld 21761058) and 0.5% TritonX-100. After blocking, sections were incubated with rabbit anti-NeuN antibody (1:500, Abcam, ab177487) and goat anti-Iba1 antibody (1:500, Novus, NB100-1028) in antibody buffer (1% BSA, 0.3% TritonX-100 in PBS) for 3 d at 4°C. Sections were then washed 3x 15 min in PBS + 0.05% Tween 20, before being incubated with secondary antibodies and DAPI in antibody buffer for 3 h at RT. Finally, stained tissues were washed in PBS + 0.05% Tween 20 before being mounted onto slides, dried, and then coverslipped with Prolong Glass mounting medium (Invitrogen, P36984).

For brain-wide lipofuscin imaging and intensity measurement, stained sections were imaged at 20x magnification using a widefield epifluorescence slide scanner (Axio Scan Z1; Carl Zeiss; 20x/0.8 N.A. air objective, with Zeiss Zen 3.7 software). Each channel was acquired independently with the 450-490 nm filter set for NeuN/488, the 625-655 nm filter set for Iba1/647 and the 538-562 nm filter set for capturing the lipofuscin (LF) autofluorescence, corresponding to emission wavelengths of 570-640 nm. Channel excitation intensities and shutter speeds, as well as other parameters, were kept consistent throughout the imaging run. One section per brain hemisphere most closely capturing the site of injection was quantified for LF content, using Zeiss Zen 3.7 software and custom macros written in Zen using Advanced Processing and Macros tools. First, automatic shading correction was applied and images were stitched together in Zen 3.7, then PBS and ApoE-injected hemispheres were denoted manually, and finally custom macros were used to apply consistent local thresholding to images as well as to identify and quantify both the total tissue area and LF-stained areas within each hemisphere. Total areas and total 555nm sum intensities were then quantified for both hemibrain tissue and LF objects, which were then summed across all objects per animal and hemisphere. These numbers were normalized to total hemisphere tissue area and used to compare LF content between conditions.

Confocal imaging was performed for quantifying cell type-specific LF content. For each animal, two regions were selected in both the hippocampus and the cortex to generate mean statistics. Regions were chosen to be close to the site of injection but not in the areas with high microgliosis or visible cell damage. Regions were also specifically selected to contain detectable levels of LF in both neurons and microglia, to facilitate cell-type analyses. Images were captured using a laser scanning confocal microscope (Leica SP8; Leica Microsystems, Inc) with a 40x/1.3 N.A. oil objective, at a pixel size of 126 nm and a format size of 1024x1024. Confocal z-stacks of 0.5 μm were acquired sequentially for each channel, scanned at 1000 Hz between frames with a 2-frame average. Laser excitation and detection intensities and wavelengths were kept consistent throughout experiment, with NeuN-488 excited at 488 nm (0.4%) and collected at 495-555 nm, Iba1-647 excited at 638 nm (0.3%) and collected at 645-749nm, and LF autofluorescence excited at 552 nm (0.5%) and collected at 557-641nm. The representative image is a maximum projection of the three-dimensional image volume, generated using Imaris 10.1.1 (Oxford Instruments, Bitplane). For quantifying the intracellular levels of LF, the signal within the intraneuronal compartment was masked using an intensity-based segmentation of NeuN-positive pixels, and the resulting sum intensities were normalized to the total neuronal volume within a given 3D image volume. Intra-microglial LF was similarly masked using an intensity-based segmentation of Iba1-positive pixels, and the sum intensities of LF signal were quantified and normalized to the total microglial volume. In each case, the mean sum intensity was determined for each region selected (hippocampus and cortex), and these means were then used to calculate the volume-normalized intracellular LF signals.

RNA isolation and qPCR

Cells were lysed with Qiagen RLT-plus buffer containing β-mercaptoethanol, snap frozen and store at -80°C. RNA was extracted from cell lysates using the RNeasy Plus Micro Kit (Qiagen, 74034), followed by cDNA generation by reverse transcription (ThermoFisher VILO Superscript IV #11756050). Transcript levels were evaluated using various TaqMan probes for specific genes of interest (refer to key resources table) with expression of each gene normalized to the housekeeping gene *ACTB* (Thermofisher, Hs01060665_g1). qPCR was performed using a QuantStudio 6 flex system (Applied Biosystems) and average CT values were collected for each probe between technical duplicates. Delta CT values for each sample from the housekeeping gene were calculated, and delta delta CT values relative to samples from the control condition were calculated to generate relative expression for each gene of interest.

Bulk transcriptome sequencing & analysis for *APOE* KO iMg

At the end of each independent *APOE* KO iMg experiment, cells were lysed with Qiagen RLT-plus buffer containing β-mercaptoethanol. Samples were stored at -80°C until cell lysates from all individual experiments were collected, at which point, all lysates

were thawed with technical triplicates for each condition in each experiment pooled into a single sample. ERRC control RNA that was pre-diluted at 1:10,000 was spiked into each sample at a volume directly proportional to the cell count determined during imaging. RNA extraction was performed using the RNeasy Plus Micro Kit (Qiagen, 74034). Quality control of RNA samples was performed using the Agilent 2200 TapeStation System to determine RNA concentration and quality. RNA samples were diluted 1:10 in nuclease-free water prior to loading on the TapeStation System. High sensitivity RNA ScreenTape (Agilent, 5067-5579) was loaded onto the system for sample injections and high sensitivity RNA ScreenTape Ladder (Agilent, 5067-5581) was loaded onto the system as a standard.

For each of the 12 independent biological samples, two replicate sequencing libraries were prepared from 10-60 ng of total RNA each using the QuantSeq 3' mRNA-seq Library Prep Kit FWD for Illumina (Lexogen A01173). The UMI second strand synthesis module was used to introduce unique molecular identifiers (UMIs), following the protocol defined by the manufacturer. Briefly, oligo dT primers were hybridized to total RNA for reverse transcription, followed by RNA removal after first strand synthesis. UMIs were introduced during second strand synthesis and cDNA was purified using magnetic beads, followed by 18 cycles of PCR with dual indexes and PCR purification. Library quantity and quality were assessed with the High Sensitivity D1000 ScreenTape (Agilent 5067-5584). Libraries were pooled in equimolar ratios and the sequencing pool was purified with magnetic beads to remove residual adaptor dimers. Sequencing was performed on an Illumina NovaSeq 6000 instrument (75 bp single end) by SeqMatic (Fremont, CA, USA).

Sequencing data was processed using nf-core/rnaseq (v3.11.2) of the nf-core collection of workflows,[88] using STAR (v2.7.9a)[89] and salmon (v1.10.1)[90] for alignment of reads to the GRCh38 human reference genome and quantification of Gencode (v42) gene models, respectively. One library failed QC (fewer than 1 million total counts) and was excluded from downstream analysis. Differential expression analysis was performed with the edgeR (v3.42.4)[91] and limma (v 3.56.2)[92] Bioconductor packages using R v4.3.1. In brief, count matrices were subset to protein-coding genes, and library size factors were calculated with the TMM method.[93] Gene-wise linear models with experimental group and collection batch as covariates were fit with the voomLmFit function, setting the sample weights argument to TRUE. The relationships of technical and biological replicates were specified via the block argument. Fold changes and p-values for the contrasts of interest were obtained with the eBayes function, and false discovery rates were calculated according to previously published methods.[94] Gene set enrichment analyses were performed with the fgsea R package (v1.26.0)[95] with gene set collections from the msigdbr package (v7.5.1). Volcano plots were generated with the ggplot2 R package (v3.4.3). Raw and processed RNA-seq data are available from the NCBI GEO repository under accession GSE250249.

Cytokine/chemokine assay for *APOE* KO iMg

After 1 d treatment with BODIPY-CE/HDL/ApoE complex described earlier, conditioned media collected from iMg were transferred to a 96-well V-bottom plate (Greiner, 651201), centrifuged at 300 x *g* for 5 min, and the resulting supernatants were transferred to PCR tubes (Thermofisher, AB2005) which were snap frozen and stored at -80 °C. After conditioned media from all four independent experiments were harvested, samples were sent to Eve Technologies (Canada) for cytokine/chemokine measurement using Human Cytokine/Chemokine Panel A 48-Plex Discovery Assay Array. Concentrations of cytokines/chemokines were normalized to cell count of each sample.

Liquid chromatography-mass spectrometry analysis of lipids

For lipid extraction from cells in 96-well plates, cells were washed with cold saline after complete removal of medium, followed by the addition of 100 μL/well of lipid extraction buffer consisting of 3:1 ratio of MS-grade butanol:methanol (v:v) with lipid standards added. After extensive scraping of the wells with pipette tips, the plate was sealed tightly and agitated in a plate shaker at 1,000 rpm for 25 min at 4°C. Next, the plate was incubated at -20°C for at least 1 h to facilitate protein precipitation, following which, the extracts were transferred to a v-bottom 96-well plate and centrifuged at 3,000 x *g* for 20 min at 4°C. The top 70 μL supernatant from each well was collected with samples from technical replicates pooled into one glass vial. Samples were dried under Nitrogen gas and stored at -80°C. Right before liquid chromatography-mass spectrometry (LCMS) analysis, samples were resuspended in methanol and agitated on a shaker for ~15 min. Lipid extraction from CSF was performed as previously described.[96]

Lipid analyses for both cells and CSF samples were performed as descried in the same study[96] by liquid chromatography (UHPLC Nexera X2) coupled to electrospray mass spectrometry (QTRAP 6500+). Lipids were analyzed in both positive and negative mode, using an UPLC BEH C18 column (150 × 2.1 mm, 1.7 μm particle size, Waters Corp.), maintained at 55°C, with gradient elution, and flow rate of 0.25 mL/min over 10 min gradient. The ion pair information, retention time and secondary spectrum data of the detected substances were collected for qualitative analysis. The multiple reaction monitoring mode (MRM) of triple quadrupole mass spectrometry was performed for semi-quantitative analysis. MRM data analysis was performed using Analyst 1.6.3 software (AB SCIEX, Framingham, MA, USA) and MultiQuant (AB SCIEX). For peak area determinations, the data of all metabolites were extracted, and the individual area of the same metabolite was normalized to the integrated area of all peaks (Area Ratio). The quality control (QC) was mixed with all sample extracts.

Evaluation of gene expression levels of APOE receptors in the human brain

Expression of APOE receptors across major brain cell types was explored using human single-nucleus RNA-seq data generated from the postmortem brain tissue provided by the ROSMAP (Religious Orders Study and Rush Memory and Aging Project) cohort at Rush Alzheimer's Disease Center, Rush University Medical Center, Chicago.[97] Results published here are in whole or in part based on data

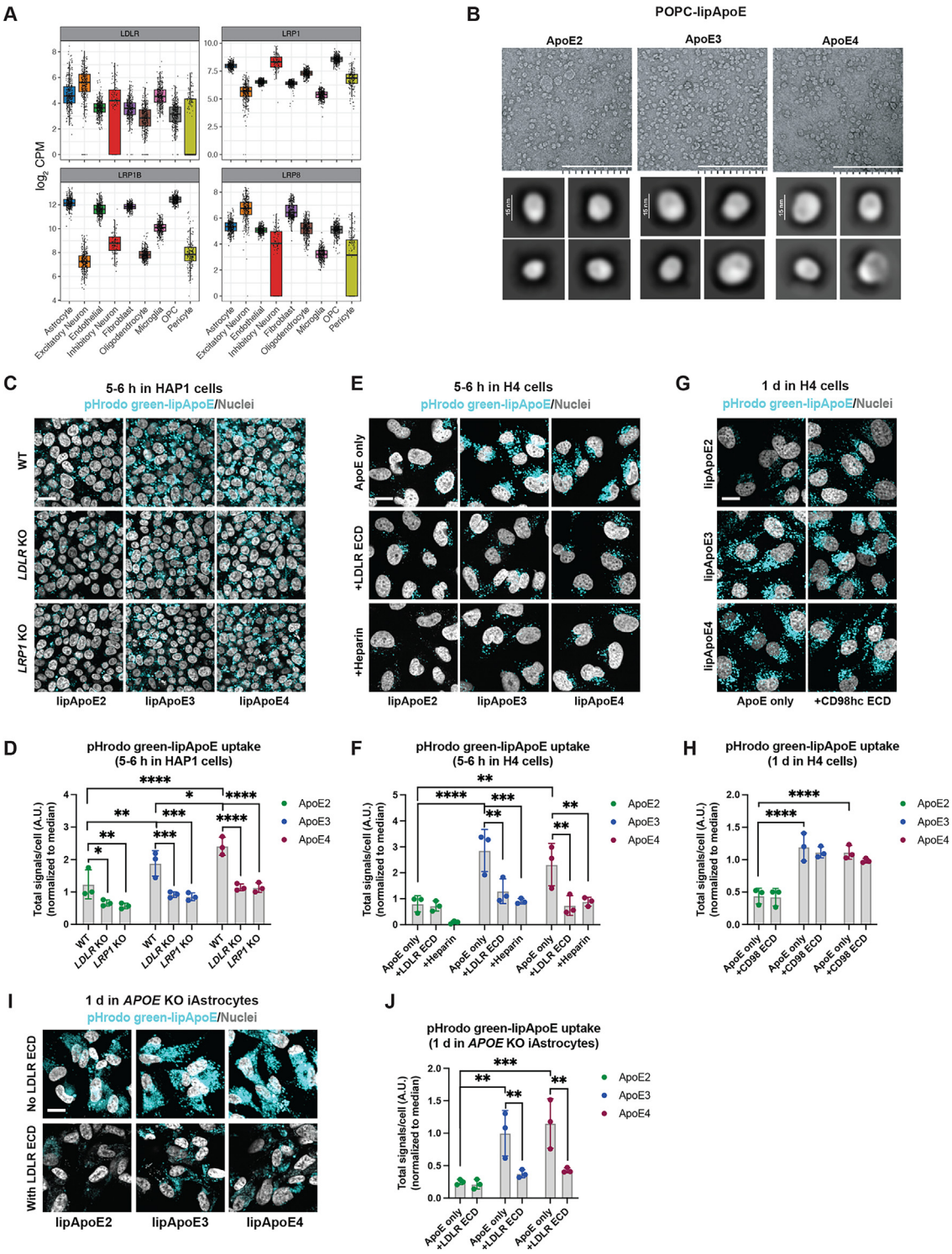
obtained from the AD Knowledge Portal (https://adknowledgeportal.org). Data from the study are available vis Synapse (https://www.synapse.org/Synapse:syn31512863). FASTQ files were mapped to the human genome (GRCh38) using cellranger (v7.0.0). All subsequent analysis was performed using R (v4.3). Resulting single-cell libraries were processed using Seurat v4[98] and the scran suite of tools.[99] Nuclei were retained if they contained expression of at least 200 unique genes and at least 1000 total gene counts. Nuclei were removed if >10% of their gene expression was mitochondrial. Doublets were predicted using the scDblFinder package and subsequently removed.[100] The final dataset consisted of 391 donors. Broad cell type labels were predicted with the SingleR package using the Allen Brain Atlas human brain data set as a reference [101]. Final cell type labels were refined using manual clustering and marker gene analysis in a small subset of donor datasets. These final labels were then projected onto the remaining donor data sets using the label transfer pipeline in Seurat v4. Finally, pseudobulk libraries were generated by aggregating counts across cell types and donor IDs using the scuttle package.[102] Gene expression is represented as $\log_2$ normalized values using a prior pseudocount of 1.

QUANTIFICATION AND STATISTICAL ANALYSIS

For each independent experiment, readouts from 2-4 technical replicates were first averaged for each treatment group and then normalized to the mean or median across treatment groups in that experiment. Normalization to median was the default method, but normalization to mean was used in some experiments to avoid a loss of error bar for one of the experimental conditions that consistently emerges as the median. Normalized data from 3-4 independent experiments were combined for statistical analysis in Prism. In all graphs, data represent mean $\pm$ SEM with each dot in bar graphs representing one independent experiment or one animal. One-way or two-way ANOVA was performed followed by multiple comparisons using specific correction methods stated in the Figure Legends. Significant differences were only plotted for comparisons among ApoE isoforms in graphs unless otherwise indicated. Statistical significance was noted in individual graphs as follows: ns for $p > 0.05$, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$. For analysis of cellular lipidomic data from *APOE* KO iAstrocytes and NGN2 iNeurons, area ratio for each lipid species was first normalized to the sum of all detected lipid species in that sample (*i.e.*, input normalization) before further normalization to the mean of that particular lipid species within each independent experiment followed by log transformation. Global analysis of differences in lipid concentrations among treatment groups was performed using the online version of MetaboAnalyst 5.0.[103] Heatmaps for top lipid species different among treatment conditions were automatically generated by MetaboAnalyst 5.0 with lipid species ranked by fold of changes. The heatmap shown in Figure 3F also included unsupervised clustering of datasets automatically generated by the online tool. Statistical analysis for the most significantly altered lipid species based on MetaboAnalyst analysis was subsequently performed in Prism with one-way ANOVA followed by Holm-Sidak's multiple comparisons. For analysis of lipidomic data from H4 cells with and without Lalistat 2 treatment, unknown variations in data unrelated to biological factors of interest were removed using the ruv package (version 0.9.7.1). Analysis of human CSF lipids was performed using R statistical software. Area ratio values were transformed using a log base 2 transformation. Differences between groups were estimated using a linear mixed-effects model, with subjects treated as random effects to address the repeated measurements. To account for the lack of balance in age and sex across groups, both covariates were adjusted as linear effects across all analytes. A mixed effects model adjusting for age*gender is used here to account for longitudinal measures with the same subject. For microscopy, representative images were included in the figures, based on quantifications.

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Supplemental figures



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Figure S1. ApoE isoform-dependent cellular uptake of POPC-lipidated ApoE, related to Figure 1

(A) Boxplots showing median expression of *LDLR* and other ApoE receptors across major cell types in the human brain based on ROSMAP single-nucleus RNA sequencing dataset. Expression of *LDLR* is comparable to that of *LRP8*, albeit somewhat lower than *LRP1* and *LRP1B*. In excitatory neurons, all receptors are expressed at comparable levels.

(B) Representative micrographs for negative-stain transmission electron microscopy of POPC-lipidated ApoE isoforms (top, scale bar: 200 nm) along with corresponding top 2D classes (bottom, shown are four most populated, scale bar: 15 nm).

(C) Confocal images of pHrodo green-labeled POPC-lipidated ApoE (used at 10 μg/mL) detected in WT, *LDLR* KO, or *LPR1* KO HAP1 cells after 5–6 h incubation. Scale bar: 10 μm.

(D) Quantification of pHrodo green signals in (B) (n = 3 independent experiments, two-way ANOVA, Holm-Sidak's multiple comparisons).

(E) Confocal images of pHrodo green-labeled POPC-lipidated ApoE (used at 10 μg/mL) detected in H4 cells after 5–6 h incubation, with and without co-treatment of 20 μg/mL LDLR extracellular domain (ECD) or 2.5 μg/mL heparin. Scale bar: 10 μm.

(F) Quantification of pHrodo green signals in (D) (n = 3 independent experiments, two-way ANOVA, Holm-Sidak's multiple comparisons).

(G) Confocal images of pHrodo green-labeled POPC-lipidated ApoE (used at 10 μg/mL) detected in H4 cells after 1-day incubation, with and without co-treatment of 20 μg/mL CD98hc ECD. Scale bar: 10 μm.

(H) Quantification of pHrodo green signals in (F) (n = 3 independent experiments, two-way ANOVA, Holm-Sidak's multiple comparisons).

(I) Confocal images of pHrodo green-labeled POPC-lipidated ApoE (used at 10 μg/mL) detected in *APOE* KO iAstrocytes after 1-day incubation, with and without co-treatment of 20 μg/mL LDLR ECD. Scale bar: 10 μm.

(J) Quantification of pHrodo green signals in (H) (n = 3 independent experiments, two-way ANOVA, Holm-Sidak's multiple comparisons).

For (D), (F), (H), and (J), data shown are mean ± SEM; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$.

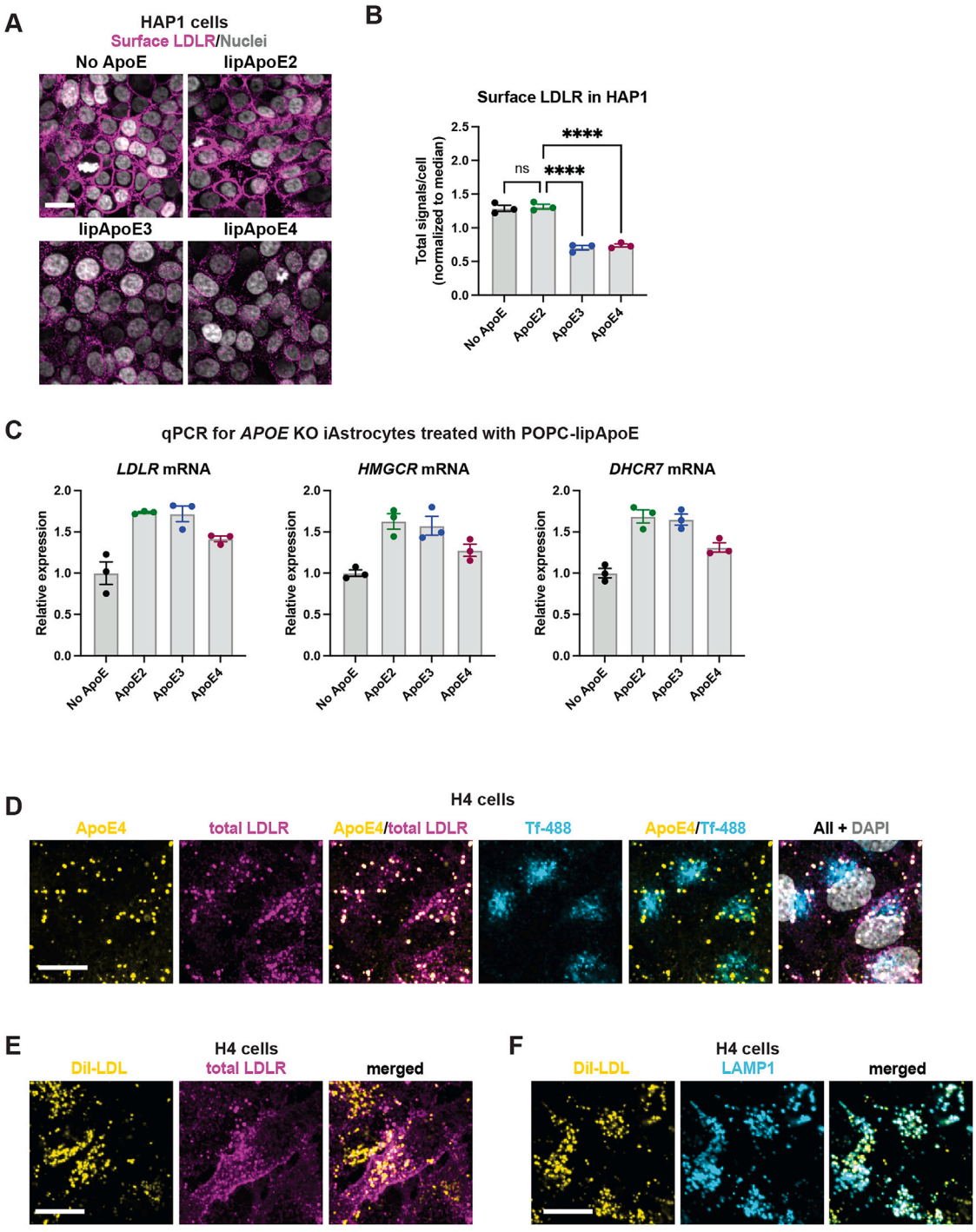


Figure S2. Lipidated ApoE3 and ApoE4 but not LDL sequester LDLR, related to Figure 2

(A) Confocal images of cell surface LDLR staining for HAP1 cells after 1-day incubation with 10 μg/mL POPC-lipidated ApoE. Scale bar: 10 μm.

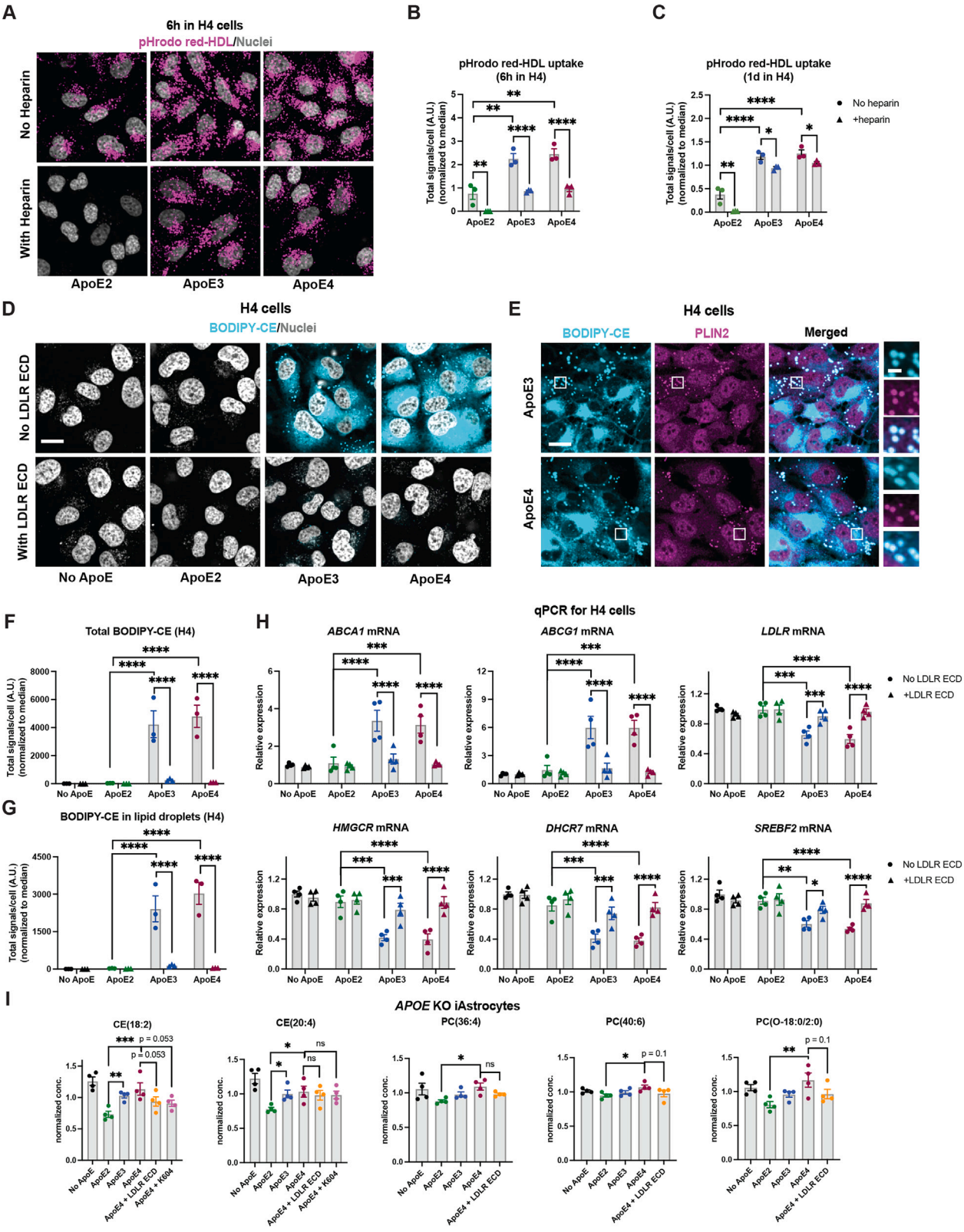
(B) Quantification of surface LDLR signals in (A) ($n = 3$ independent experiments, one-way ANOVA, Sidak's multiple comparisons). Data shown are mean ± SEM; ns: $p > 0.05$, ****$p < 0.0001$.

(C) Relative expression of 3 genes quantified by qPCR for *APOE* KO iAstrocytes after 1-day treatment with 10 μg/mL POPC-lipidated ApoE ($n = 3$ independent experiments). Data shown are mean ± SEM.

(D) Confocal images for co-immunostaining of ApoE and LDLR in H4 cells after 1-day treatment with 10 μg/mL POPC-lipidated ApoE4 followed by 15 min incubation with Alexa 488-labeled transferrin to label recycling endosomes. Scale bar: 10 μm.

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(E) Confocal images showing lack of colocalization between DiI-LDL and total LDLR immunoreactivities in H4 cells after 1-day treatment with 10 μg/mL DiI-LDL. Scale bar: 10 μm.
(F) Confocal images showing partial colocalization between DiI-LDL and LAMP1 immunoreactivities in H4 cells after 1-day treatment with 10 μg/mL DiI-LDL. Scale bar: 10 μm.



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Figure S3. ApoE isoform- and LDLR-dependent lipid uptake impacts lipid homeostasis, related to Figure 3

(A) Confocal images of pHrodo signals detected in H4 cells after 6 h incubation with 10 μg/mL pHrodo red-labeled HDL pre-complexed with 10 μg/mL recombinant ApoE, with and without co-treatment of 2.5 μg/mL heparin. Scale bar: 10 μm.

(B and C) Quantification of pHrodo red signals in (A) and after 1-day incubation, respectively (n = 3 independent experiments, two-way ANOVA, Tukey's multiple comparisons).

(D) Confocal images of BODIPY signals detected in H4 cells after 1-day incubation with 10 μg/mL BODIPY-CE pre-complexed with 10 μg/mL HDL and 10 μg/mL recombinant ApoE, with and without co-treatment of 20 μg/mL LDLR ECD. Scale bar: 10 μm.

(E) Confocal images showing colocalization of punctate BODIPY signals with non-nuclear PLIN2 immunoreactivities in H4 cells after 1-day incubation with 10 μg/mL BODIPY-CE pre-complexed with 10 μg/mL HDL and 10 μg/mL recombinant ApoE3/4. Scale bar: 10 μm. Insets: higher-magnification images for single and merged channels; scale bar: 2 μm.

(F and G) Quantification of total BODIPY signals and punctate BODIPY signals, respectively, in (D) (n = 3 independent experiments, two-way ANOVA, Sidak's multiple comparisons).

(H) Relative expression of 6 genes quantified by qPCR for H4 cells after 1-day treatment with 10 μg/mL CE(18:1) pre-complexed with 10 μg/mL HDL and 10 μg/mL recombinant ApoE, with and without co-treatment of 20 μg/mL LDLR ECD (n = 4 independent experiments, two-way ANOVA, Sidak's multiple comparisons).

(I) Additional lipid species showing ApoE isoform-dependent changes in *APOE* KO iAstrocytes after overnight treatment with 10 μg/mL CE(18:1) pre-complexed with 10 μg/mL HDL and 10 μg/mL recombinant ApoE, with and without co-treatment of 20 μg/mL LDLR ECD or 500 nM ACAT1 inhibitor K604 (n = 4 independent experiments, one-way ANOVA, Sidak's multiple comparisons).

For (B), (C), and (F)–(I), data shown are mean ± SEM; ns: $p > 0.05$, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$.

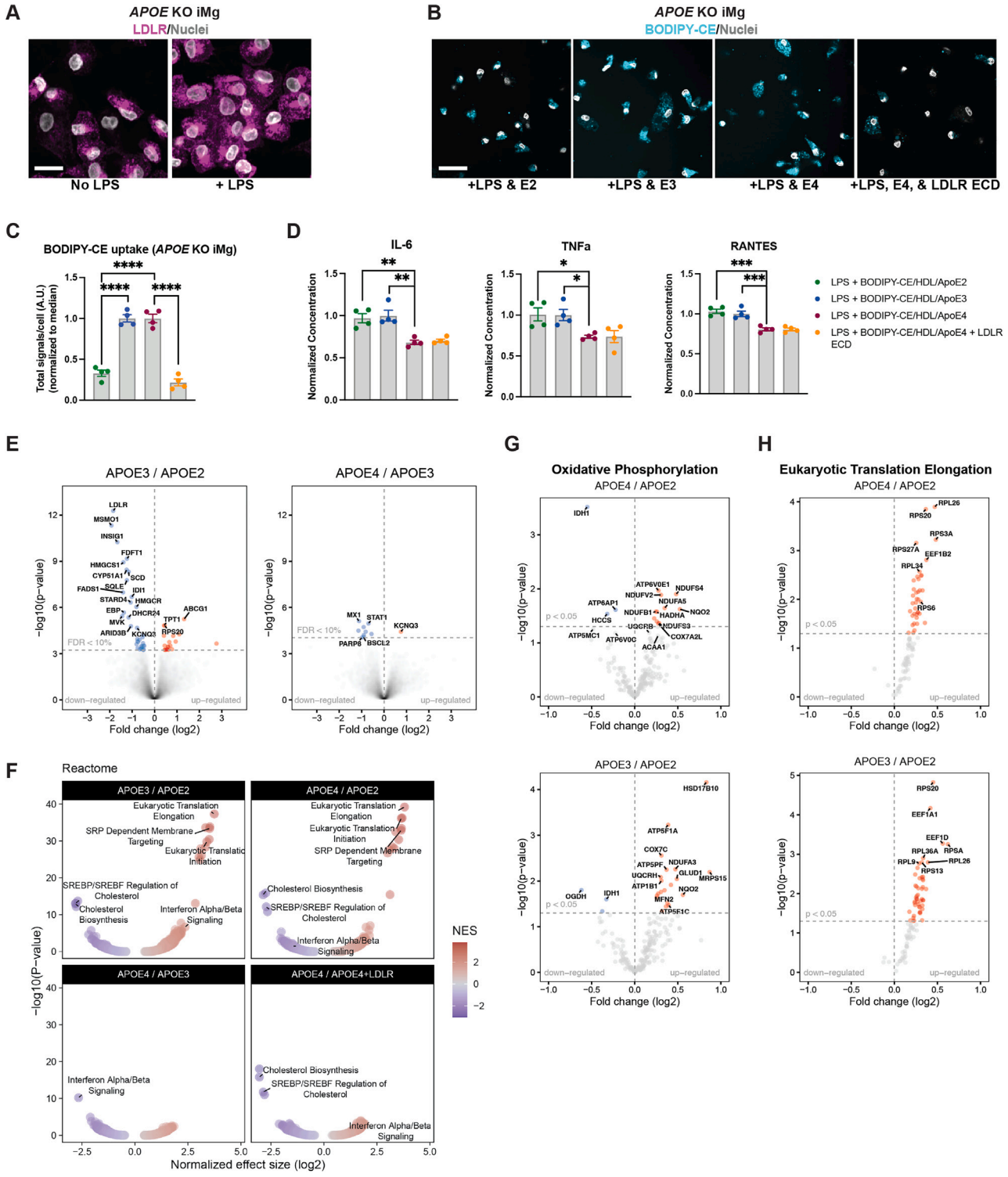


Figure S4. Differential lipid burden modulates transcriptional states of human microglia, related to Figure 4

(A) Confocal images of LDLR immunostaining for *APOE* KO human iPSC-derived microglia (iMg) after 1-day incubation with or without 100 ng/mL LPS. Scale bar: 10 μm.

(B) Confocal images of BODIPY signals detected in *APOE* KO iMg after 1-day incubation with 10 μg/mL BODIPY-CE pre-complexed with 10 μg/mL HDL and 10 μg/mL recombinant ApoE, with and without co-treatment of 20 μg/mL LDLR ECD, in medium containing 100 ng/mL LPS. These experiments were done using iMg from a different round of differentiation from that shown in Figure 4A. Scale bar: 20 μm.

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(C) Quantification of BODIPY signals in (A) (n = 4 independent experiments, one-way ANOVA, Sidak's multiple comparisons).
(D) Normalized concentrations of 3 cytokines/chemokines measured from conditioned media of *APOE* KO iMg after live imaging shown in (A) (n = 4 independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons). These cytokines/chemokines showed ApoE4-specific changes not rescued by LDLR ECD co-treatment.
(E–H) Results from RNA sequencing performed on *APOE* KO iMg after live imaging shown in (A) (n = 3 independent experiments). (E) Volcano plots showing significantly different individual genes with adjusted $p < 0.1$, comparing ApoE3 vs. ApoE2 and ApoE4 vs. ApoE3 treatment conditions. (F) Volcano plots showing Reactome pathways that are the most altered for different pairs of comparisons. (G and H) Volcano plots highlighting genes that are the most significantly different for ApoE4 vs. ApoE2 and ApoE3 vs. ApoE2 comparisons within the oxidation phosphorylation pathway and the eukaryotic translation elongation pathway, respectively.
For (C) and (D), data shown are mean ± SEM; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$.

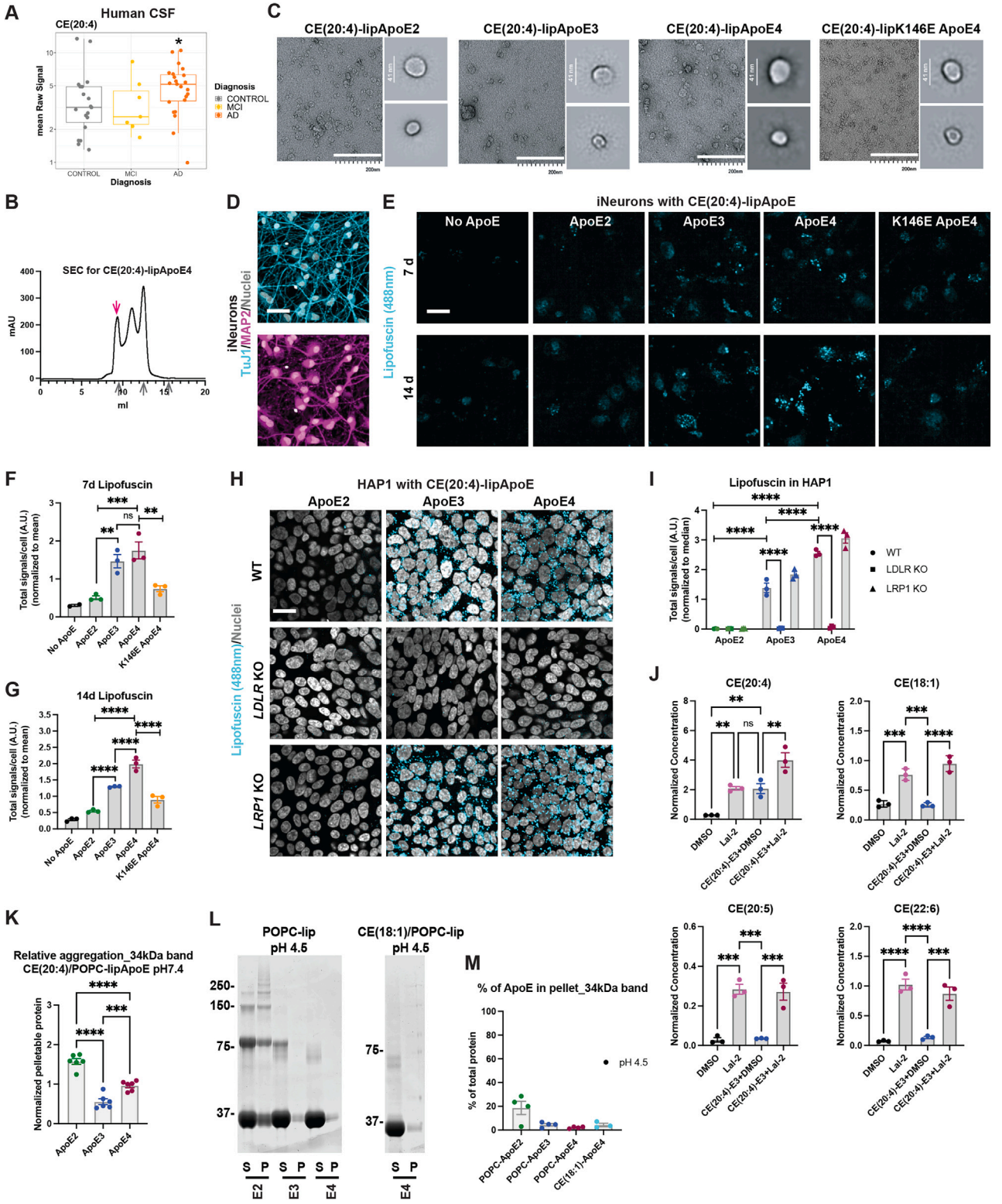
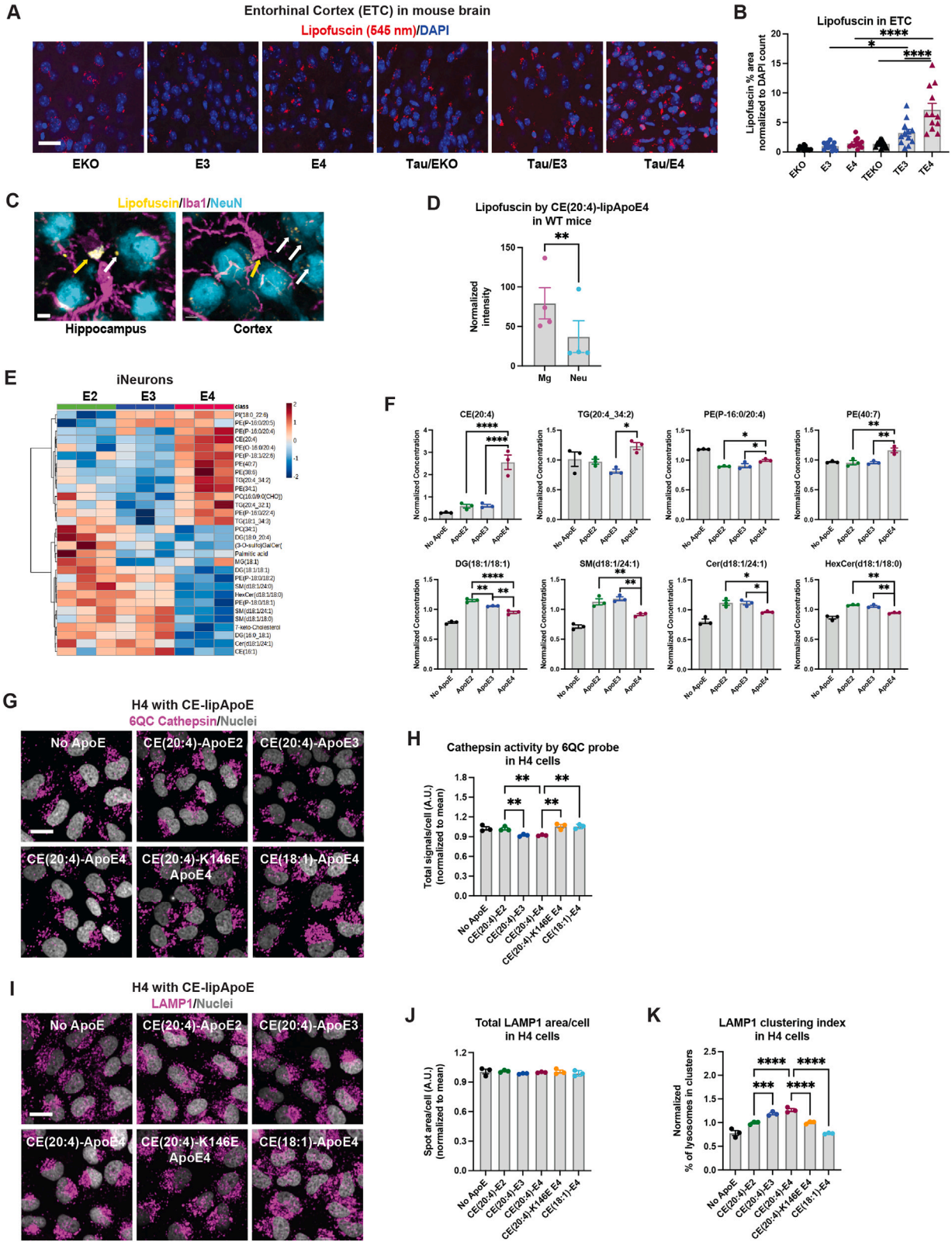


Figure S5. ApoE4 promotes lipofuscin accumulation in cultured cells, related to Figure 5

(A) Levels of CE(20:4) in human CSF samples were stratified by dementia status. Differences between groups were estimated using a linear mixed-effects model with age and sex correction. *$p < 0.05$ compared with control group.

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(B) Representative size exclusion chromatography profile for CE(20:4)/POPC-lipidated ApoE4 with the first peak used for NGN2 human iPSC-derived neurons (iNeuron) treatment indicated by a magenta arrow. The three gray arrows indicate the mean elution volumes for 670, 150, and 44 kDa standards (from left to right) using the same column.
(C) Representative micrographs for negative-stain transmission electron microscopy of CE(20:4)/POPC-lipidated ApoE isoforms from the leftmost peak (left, scale bar: 200 nm) along with selected 2D classes showing size heterogeneity (right, scale bars: 41 nm).
(D) Representative immunostaining of Tuj1 (class III beta-tubulin) and MAP2 (microtubule-associated protein 2) for NGN2 iNeurons at 4 days of differentiation. Scale bar: 20 μm.
(E) Confocal images of lipofuscin (488 nm channel) detected in NGN2 iNeurons after 7 or 14 days incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE isoforms. Scale bar: 10 μm.
(F and G) Quantifications of lipofuscin in (E) (n = 3 independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons). Note that nuclear counts from the endpoint imaging (21-day incubation as shown in Figure 5A) were used for per-cell quantifications.
(G) Confocal images showing partial colocalization of lipofuscin (488 nm) signals with LysoTracker staining in NGN2 iNeurons after 21-day incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE4. Scale bar: 5 μm.
(H) Confocal images of lipofuscin (488 nm channel) detected in WT, *LDLR* KO, and *LRP1* KO HAP1 cells after 3-day incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE. Scale bar: 10 μm.
(I) Quantifications of lipofuscin in (H) (n = 3 independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons).
(J) Normalized levels of different CE species detected by LCMS in H4 cells after 1-day treatment with DMSO, 25 μM Lalistat-2 (Lal-2, in DMSO), 20 μg/mL CE(20:4)/POPC-lipidated ApoE3, or 20 μg/mL CE(20:4)/POPC-lipidated ApoE3 together with 25 μM Lalistat-2 (n = 3 independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons).
(K) Quantification of relative aggregation for CE(20:4)/POPC-lipidated ApoE isoforms after 4-day incubation at 37°C at pH 7.4, with percentage of pelletable ApoE for each isoform normalized by mean pelletable ApoE in each experiment (combined results from 3 independent experiments for 2 batches of preparations).
(L) Non-reduced SDS-PAGE analysis for supernatant (S) and pellet (P) fractions from ultracentrifugation of POPC-lipidated ApoE or CE(18:1)/POPC-lipidated ApoE4 after 4-day incubation at 37°C at pH 4.5.
(M) Quantification of percentage of ApoE recovered in the pellet fraction for experiments shown in (L) (n = 3–4 independent experiments with densitometry of the 34 kDa band shown).
For (F), (G), (I)–(K), and (M), data shown are mean ± SEM; ns: $p > 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$.



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Figure S6. Lipofuscin perturbs lipid metabolism and lysosomes, related to Figure 6

(A) Lipofuscin signals (545 nm channel) in the entorhinal cortex of 9.5-month-old *APOE* KO (EKO), *APOE3* KI (E3), *APOE4* KI (E4), P301S MAPT/APOE KO (Tau/EKO), P301S *MAPT/APOE3* KI (Tau/E3), and P301S *MAPT/APOE4* KI (Tau/E4) mice. Scale bar: 20 μm.

(B) Quantification of lipofuscin in (A) (n = 12 mice/group, one-way ANOVA, Sidak's multiple comparisons).

(C) Images of lipofuscin co-labeled with neuronal (NeuN) and microglial (Iba1) markers for 6-month-old WT mice at 2 weeks after injection with CE(20:4)/POPC-lipidated ApoE4 in the hippocampus and cortex near but not at injection tract. Scale bar: 5 μm. White arrows point to examples of neuronal lipofuscin, and yellow arrows point to examples of microglial lipofuscin.

(D) Quantification for localization of lipofuscin in microglia (Mg) vs. neurons (Neu) shown in (C) (n = 4 mice, two-tailed paired t test).

(E and F) Lipidomic analysis for P301S tau-overexpressing NGN2 iNeurons after 14-day incubation with 20 μg/mL CE(20:4)/POPC-lipidated ApoE isoforms. (E) A heatmap for the top 30 lipids showing the greatest differences among the 3 isoform treatment conditions. (F) A subset of lipid species showing ApoE isoform-dependent changes (n = 3 independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons).

(G) Confocal images for cathepsin activity measured by 6QC probe in live H4 cells after 3-day incubation with CE(20:4)/POPC-lipidated ApoE isoforms or CE(18:1)/POPC-lipidated ApoE. Scale bar: 10 μm.

(H) Quantification of 6QC signals in (G) (n = 3 independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons).

(I) Confocal images of LAMP1 immunostaining in fixed H4 cells after the experiment shown in (G). Scale bar: 10 μm.

(J) Quantification of total LAMP1 spot area for (I) (n = 3 independent experiments).

(K) Quantification of LAMP1 clustering index (percentage of LAMP1 spot area in clusters for each condition normalized to mean across all conditions in each experiment) for (I) (n = 3 independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons).

For (B), (D), (F), (H), (J), and (K), data shown are mean ± SEM; $^{*}p < 0.05$, $^{**}p < 0.01$, $^{****}p < 0.0001$.

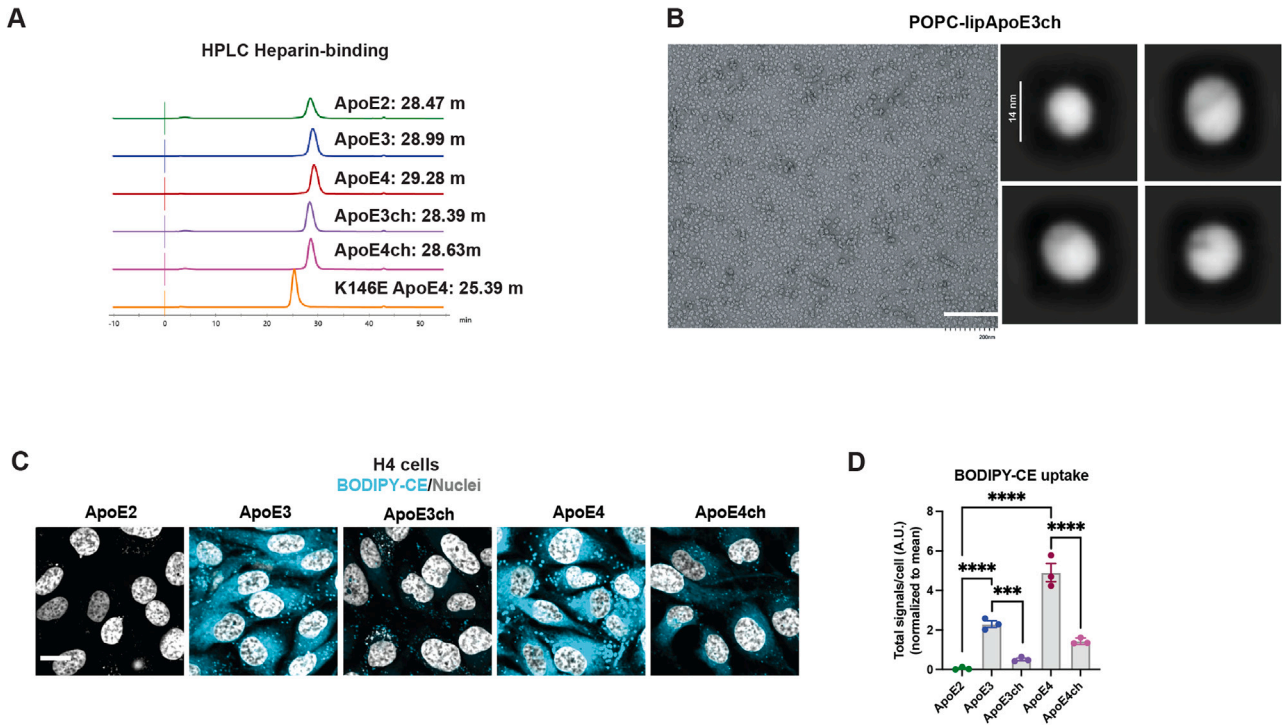


Figure S7. Christchurch mutation reduces receptor binding, lipid uptake, and lipofuscin, related to Figure 7

(A) HPLC profiles for non-lipidated ApoE isoforms eluted from a heparin column in a representative HPLC-based heparin binding experiment, with average retention time from 3 independent experiments indicated for individual isoforms.

(B) Representative micrograph for negative-stain transmission electron microscopy of POPC-lipidated ApoE3ch (left, scale bar: 200 nm) along with top 2D classes (right, shown are four most populated, scale bar: 14 nm).

(C) Confocal images of BODIPY signals detected in H4 cells after 1-day incubation with 10 μg/mL BODIPY-CE pre-complexed with 10 μg/mL HDL and 10 μg/mL recombinant ApoE isoforms. Scale bar: 10 μm.

(D) Quantification of BODIPY signals in (C) ($n = 3$ independent experiments, one-way ANOVA, Holm-Sidak's multiple comparisons). Data shown are mean ± SEM; ***$p < 0.001$, ****$p < 0.0001$.